Formalizing the Reduced Mahler $3/2$ Dynamical System in Lean 4: Exact 1D Slack Reduction, Constructive Coasting Survivors, Conditional Finiteness Theorems, and Certified Transcendence Barriers

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Toolchain: Lean 4.19.0 + Mathlib v4.19.0 (`142` `.lean` files, `0` `sorry`s, `0` warnings, standard Lean axioms `[propext, Classical.choice, Quot.sound]`)

Abstract

Mahler's $3/2$ problem (1968) asks whether there exists any real "Z-number" $\alpha > 0$ such that the fractional part $\{(3/2)^n \alpha\}$ lies in $[0, 1/2)$ for every integer $n \geq 0$. Writing $\alpha = a_0 + \xi$ with $a_0 = \lfloor \alpha \rfloor \in \mathbb{N}$ and $\xi \in [0, 1)$, survival up to step N is equivalent to the discrete ceiling recurrence $g_{n+1} = \lceil 3g_n/2 \rceil = \lfloor (3g_n + 1)/2 \rfloor$ ($g_0 = a_0$) admitting a non-empty intersection of N nested archimedean companion intervals $[\mathrm{lowerBound}(a_0, n),\ \mathrm{upperBound}(a_0, n))$ (`InfiniteZNumberBridgeAndRationalBase.lean`).

In this work, we present a 142-file, `sorry`-free formalization in Lean 4 that investigates both **constructive survival** and **obstructions to infinite orbits**: 1. **Exact 1D Slack Reduction & Infinite-Horizon Z-Number Bridge:** By grouping each odd step $g \mapsto (3g + 1)/2$ with its $s(k) - 1$ subsequent even steps $g \mapsto 3g/2$ via $3b_k + 1 = 2^{s(k)} u(k)$ ($\gcd(u(k), 6) = 1$), we collapse all $\binom{K}{2}$ pairwise companion-interval inequalities along a finite block chain into a **single 1D recurrence** (`minSlack`), and prove its exact two-way equivalence with the existence of a real companion surviving through step $T(K)$ (both in interval form and in direct `Int.fract` form). Normalizing by $2^{T(m)}$ yields the time-homogeneous 1D slack map $\mu_2(s, m + 1) = \min\left(1,\ (3/2)^{s(m)} \mu_2(s, m) - 2\right)$ driven by the arithmetic-controlled block length $s(m) = v_2(3b_m + 1)$ (`CoupledFractionalPartDynamics.lean` , `FiniteSeedCarryAndBackwardDescent.lean`). 2. **Constructive Survivors & 113-Step Record-Holder:** We prove that any block sequence with $s(k) \geq 3$ saturates the ceiling minSlack $s\ m = 2^{T(m)}$ and automatically admits a valid real companion at every finite horizon. Using this sufficiency theorem, we construct closed-form seed families surviving arbitrarily many blocks prior to ceiling exit, and machine-certify a 55-bit integer (`recordHolderA0 = 29871555608238677 < 2^55`) that survives **113 steps** (58 steps past its 55-bit binary length) before dying at step 114 (`MahlerGrandSynthesis.lean` , `InfiniteZNumberBridgeAndRationalBase.lean`). 3. **Unconditional Exclusion of Restricted, Aperiodic, Non-Stationary & Rapid-Growth Classes:** We prove explicit, logarithm-free step cutoffs in Lean 4 ruling out: (i) sub-ceiling bounded-block runs ($s(k) \leq c$), (ii) periodic and uniform-macro-block sequences (`2^(D m) | ((3^D - 2^D)(3a_0+1) + 2^(s 0) R_0)`), (iii) **arbitrary two-letter $\{a, b\}$ ($3 \leq a \leq b$) mechanical/Sturmian block chains** (`no_infinite_mechanical_two_letter_block_chain` , with factor complexity $\leq L + 1$ derived directly from phase thresholds), (iv) **sub-critical liminf-complexity sequences** ($\kappa < \log 2 / \log(3/2) \approx 1.709511$) under a positive Cesàro mean, uniform window balance, or scale-local epoch balance (`no_infinite_tendsto_mean_log_threshold_block_chain` , `no_infinite_uniformWindowMean_global_complexity_block_chain` , `no_infinite_scale_local_balanced_block_chain` , including the non-stationary oscillating-epoch family `nonStationaryEpochSeq a b`), (v) **3-step encoded Thue–Morse and Period-Doubling parity streams** (`no_nonneg_seed_realizes_tmEncodedParity` , `pdEncodedParity_cutoff_and_exclusion` , with quantitative seed-residue lower bounds $\geq 2^{2^k}$) and **Thue–Morse block-length chains** (`no_infinite_tmBlockSeq_block_chain`), and (vi) **sustained rapid-growth chains** ($3 \leq s(m + 1)$ and $3^{s(m)} < 2^{s(m+1)}$, via well-founded odd-quotient descent $u(m + 1) + 2 \leq u(m)$ in `no_infinite_rapid_growth_block_chain`). We also formalize the exact arithmetic setup for Baker–Matveev, Ridout, and Schlickewei Subspace analyses, clearly separating the machine-checked algebraic/analytic bounds from external transcendence theorems not yet in Mathlib. 4. **Exact Coupled Seed-Residue State, 5-Way General Conjecture Equivalence, Certified $B(29) \geq 4096$ Bound & Structural Barriers:** We formalize the denominator-cleared integer feasibility margin $D_{N+1} = \min(2^{N+1},\ 3D_N - 2^{N+1}(\varepsilon_N \bmod 2))$, the coupled quotient–residue–margin transition $2q_{N+1} = 3q_N + 3^{N+1}\eta_N + e_N$, the **5-way general Mahler conjecture equivalence** (`general_mahler_conjecture_iff_positive_stabilized_branch_escape` , `mahler_conjecture_iff_leastPositiveSurvivor_unbounded`), the **Lean-certified global step-29 lower bound $m \geq 4096$ ($B(29) \geq 4096$) and extremal seed `149`** (`positive_survivor_step29_ge_4096` , `seed_149_survives_28_and_fails_29`), and the exact ceiling-recovery theorem (`signedMargin_100_ceiling_recovery`) explaining why $s(m) \geq 3$ (`100`) blocks escape pure scalar-margin contraction and why the general conjecture remains open.

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1. Introduction: From Informal Proof Brainstorming to Machine-Checked Rigor

When attempting to prove Mahler's $3/2$ conjecture or related Collatz-type problems, informal proof sketches typically fall into two traps: 1. **The Probabilistic / Density Fallacy:** Treating 2-adic parity constraints and archimedean interval constraints as statistically independent, or assuming that once the step count T exceeds the bit-length $W = \lfloor \log_2 a_0 \rfloor + 1$ (the "coasting regime"), a trajectory can only survive a handful of steps. 2. **The Hidden Rank-Collapse Trap:** Constructing auxiliary linear combinations, Wronskians, or multi-origin determinants that appear to overdetermine the orbit, without noticing that the underlying 1D affine recurrence forces those combinations to be algebraically dependent.

By formalizing every proposed attack vector in Lean 4—and requiring that every failed approach be diagnosed via a machine-checked counterexample or algebraic identity—we separated genuine number-theoretic structure from mirages.

1.5 Literature Survey, Attribution of Prior Work & Novel Contributions

To place this formalization in its proper mathematical context, we carefully distinguish **prior theorems from the literature that we rediscovered or formalized** from **novel formulations, theorems, and certified barrier analyses originating in this work**.

A. Attribution to Prior Mathematical Literature

1. **Mahler (1968) — Discrete Companion Reduction, Series Formula & Two-Seed Ridout Bound [1]:**
2. Kurt Mahler introduced Z-numbers and proved the foundational reduction: for each integer $g_0 = a_0 \geq 1$, there is *at most one* real companion $\alpha = a_0 + \xi$ ($\xi \in [0, 1)$) such that α is a Z-number, governed by the discrete ceiling recurrence $g_{n+1} = \lceil 3g_n/2 \rceil = \lfloor (3g_n + 1)/2 \rfloor$ and nested archimedean intervals (`MahlerDemandA.lean` , `PureIntegerPinch.lean` , `InfiniteZNumberBridgeAndRationalBase.lean`). As we formally verify in `odd_step_floor_exceeds_half_and_forces_ceil` , when g_n is odd and $0 \leq r_n < 1/2$, the remainder relative to $\lfloor 3g_n/2 \rfloor$ is $1/2 + 3r_n/2 \geq 1/2$, so keeping $\{(3/2)^{n+1}\alpha\} < 1/2$ forces $r_n \geq 1/3$ and $\lfloor (3/2)^{n+1}\alpha \rfloor = \lceil 3g_n/2 \rceil$.
3. Mahler also derived the infinite series representation of the unique candidate companion ξ_∞ (`InfiniteSeries.lean` , `TelescopingSeriesDuality.lean`) and applied **Ridout's Theorem (1957) [2]** to the difference of two hypothetical Z-numbers in the same binary interval $[2^k, 2^{k+1})$ to show that at most finitely many Z-numbers exist per interval (`mahler_two_seed_same_step_ridout_bound` and `two_seed_shared_prefix_divisibility` in `RidoutSubspaceFramework.lean`), yielding $|\{Z\text{-numbers} \leq X\}| = O(X^{0.7})$.
4. **Vijayaraghavan (1940), Pollington (1979) & Choquet (1980) — Distribution of Powers of $3/2$ and the $1/3$ Range Gap [3, 4, 5]:**
5. Vijayaraghavan [3] proved that fractional parts of rational powers $(p/q)^n\alpha$ for $\alpha \in \mathbb{Q}^\times$ have infinitely many limit points and cannot stay confined to $[0, 1/2)$.
6. Pollington [4] and Choquet [5] established lower bounds on the oscillation of $\{(3/2)^n\alpha\}$, leading to the critical gap constant $1/3$.
7. **Flatto, Lagarias & Pollington (1995) — The $1/3$ Range Theorem, Symbolic Dynamics & Forbidden Subwords [6]:**
8. In their landmark paper [6], Flatto, Lagarias, and Pollington proved that for every real $\alpha > 0$,

$$\limsup_{n\to\infty} \{(3/2)^n\alpha\} - \liminf_{n\to\infty} \{(3/2)^n\alpha\} \geq \frac{3-2}{3} = \frac{1}{3}.$$

   This $1/3$ range gap is the continuous foundation of our $\frac{1}{3}$ **Archimedean Drain Budget** ($\sum_{k=1}^{m} (2/3)^{T(k)+1} < 1/3$ in `TelescopingSeriesDuality.lean`) and explains why $s(k) = 1$ (which consumes $(2/3)^2 = 4/9 > 1/3$ in one block) is forbidden (`EscapeBranchTwoCap.lean`).
9. Flatto, Lagarias, and Pollington also developed the **symbolic dynamics of the parity sequence** $g_n \bmod 2$, characterizing admissible finite binary words via an infinite hierarchy of forbidden subwords and deriving average step-rate / topological entropy bounds. Our theorems in `GlobalStepRateBounds.lean` and `MinSlackDynamicalSystem.lean` —including $s(k) \geq 2$, the post-2 recovery requirements $s(k+1) \geq 6, s(k+2) \geq 3, s(k+3) \geq 3$, the $(2, 6, 3, 3, 2, 10, 7)$ cascade, and the $3m - 1$ step-rate bound—rediscover and machine-check these symbolic-dynamical constraints in odd-block length coordinates $s(k)$.
10. **Classical Diophantine & Transcendence Machinery (Ridout [2], Schlickewei [7], Stewart [8], Baker–Wüstholz / Matveev [9, 10], Corvaja–Zannier [11, 12], Adamczewski–Bugeaud [13], Dubickas [14]) & Rational-Base Numeration Systems (Akiyama–Frougny–Sakarovitch [16], Andrieu–Eliahou–Vivion [17]):**
11. Our analyses in Modules 89–102 formalize the exact algebraic and logarithmic inequalities feeding into the classical frameworks of linear forms in logarithms (Baker–Wüstholz [9], Matveev [10]), p-adic Diophantine approximation (Ridout [2], Schlickewei's p-adic Subspace Theorem [7]), simultaneous representation in bases 2 and 3 (Stewart [8]), S-unit GCD bounds (Corvaja–Zannier [11, 12]), and subword-complexity transcendence criteria (Adamczewski–Bugeaud [13]). Because the deep transcendence theorems of Matveev, Ridout, and Schlickewei are not themselves formalized in Mathlib, our Lean modules establish the unconditional algebraic/archimedean upper bounds, GCD

divisibility laws, and hyperplane syzygies directly, while clearly noting where external transcendence theorems are invoked in conditional interpretations.

12. Akiyama, Frougny, and Sakarovitch (2008) [16] introduced rational-base number systems $\mathrm{rep}_{p/q}(n)$, and Andrieu, Eliahou, and Vivion (2025) [17] proved that Borel normality of minimal/maximal rational-base words (or residue equidistribution of $T_{3/2}^n(a_0) = \lceil 3x/2 \rceil \bmod 2^k$) implies the non-existence of Z-numbers. In `InfiniteZNumberBridgeAndRationalBase.lean`, we connect `Mahler.step` directly to $T_{3/2}$ and prove that single deterministic occurrences of forbidden patterns suffice without assuming full normality.

B. Novel Contributions Originating in This Work (and Comparison with Prior Lean Formalizations)

1. **Relationship to Prior Lean Repositories on Mahler's Work:**
2. A direct search of all public Lean repositories on GitHub shows that **Google DeepMind's `formal-conjectures` project [15]** (`FormalConjectures/Wikipedia/Mahler32.lean`, `FormalConjectures/Books/BugeaudDistributionModuloOne/Problem10_9.lean`, and `FormalConjecturesForMathlib/NumberTheory/DiophantineApproximation/ZNumber.lean`, 2025–2026) previously formalized the *definition* `IsZNumber`, a 5-line sanity check `¬ IsZNumber 1`, and the *statements* (with `sorry` placeholders) of Mahler's conjecture, the consequence of $\Omega(3/2) > 1/2$, the Flatto–Lagarias–Pollington bound $\Omega(p/q) \geq 1/p$, Lebesgue measure zero, and Hausdorff dimension < 1.
3. Other Lean 4 repositories bearing Mahler's name address distinct theorems of Kurt Mahler: `giuliocaflisch/mahler` (Mahler's theorem on p-adic continuous functions), `DiegoMarquesMath/MahlerLean` (Mahler's classification of Liouville numbers), and `baobingzhang/jsp-000025-lehmer-schinzel-lean` (polynomial Mahler measure).
4. Thus, while `google-deepmind/formal-conjectures` [15] formalized the top-level conjecture statements for Z-numbers, our 122-module `sorry`-free library is a substantial proved Lean 4 development of the discrete Mahler $3/2$ dynamical system, the finite and infinite companion-interval reductions, the Flatto–Lagarias–Pollington symbolic forbidden-subword constraints in odd-block coordinates, the Sturmian/balanced aperiodic complexity exclusions, the finite-seed vs. 2-adic isolation theorems, and the Diophantine setup identities.
5. **The Reduced Coprime Quotient System & Exact 1D `minSlack` Dynamical Collapse (`CoprimeRemainderSlack.lean`, `MinSlackDynamicalSystem.lean`, `ConverseCompanionReconstruction.lean`, `CoupledFractionalPartDynamics.lean`):**
6. Formulating the dynamics in terms of the coprime odd quotients $2^{s(k+1)}u(k+1) = 3^{s(k)}u(k) + 1$ ($\gcd(u(k), 6) = 1$), the 3-adic block-boundary factorization $b_{k+1} = 3^{s(k)-1}u(k)$, and the polynomial $R(s, m)$.
7. Collapsing all $\binom{K}{2}$ shifted archimedean companion inequalities into the **single 1D recurrence** `minSlack s (m+1) = min (2^(T(m+1))) (3^(s m) * minSlack s m - 2^(T(m+1)+1))` and its normalized 1D update $\mu_2(s, m+1) = \min\left(1,\ (3/2)^{s(m)}\mu_2(s, m) - 2\right)$ driven by the block length $s(m)$ (with uncapped values in $(0, 1]$ partitioned by $(0, 1] = \bigsqcup_{c=2}^{\infty}(2(2/3)^c, 3(2/3)^c] \cap (0, 1]$), together with the two-way **Converse Companion Reconstruction Theorem** (`(∃ ξ, ValidCompanionUpTo a_0 ξ (blockSteps s K)) ↔ 0 < minSlack s (K - 1)`).
8. **Additive Separation of Shifted Slacks & the Three-Origin Subspace Syzygy (`MultiOriginSubspaceSyzygy.lean`):**
9. Proving that the two-parameter triangle of shifted slacks separates additively as $\frac{2^{T(r)}\Delta_{m-r}^{(r)}}{3^{T(m)}} = \frac{\Delta_m^{(0)}}{3^{T(m)}} - \frac{\Delta_r^{(0)} - 2^{T(r)}}{3^{T(r)}}$, eliminating `shiftSeq` completely from `0 < minSlack s m`, and proving the exact **Three-Origin Schlickewei Subspace Hyperplane Syzygy** (Dead End 15) showing that any ≥ 3 shifted origins lie in the 2D plane $\mathrm{span}\{\Delta_m^{(0)}, 3^{T(m)}\}$.
10. **Carryless Base-3 Frozen-Prefix Equivalence & Dual Remainder Complementarity (`SimultaneousBaseDigitRigidity.lean`):**
11. Proving $0 < \Delta_m \iff \lfloor 2^{T(m+1)+1}u(m)/3^{T(m)} \rfloor = 6a_0 + 2$ (and its multi-origin companion equivalence) together with the exact dual remainder law $(P_m \bmod 3^{T(m)}) + (Q_m \bmod 2^{T(m)}) = 3^{T(m)}$.
12. **Exact Three-Way Multi-Step GCD Identities & Two-Sided CRT Quotient Floors (`MultiStepQuotientGCD.lean`, `ThreeAdicAndSeedGCDLocks.lean`, `Mod2304CombinedSieve.lean`, `TwoSidedCRTAndBackwardObstructions.lean`, `Mod216TwoSidedQuotientFloor.lean`):**
13. Proving the exact GCD equalities $\gcd(u(r), u(r+d)) = \gcd(u(r), R_d^{(r)}) = \gcd(u(r+d), R_d^{(r)})$, $\gcd(\Delta_r, \Delta_{r+d}) = \gcd(\Delta_r, R_d^{(r)})$ (with $\gcd(\Delta_r, \Delta_{r+1}) = 1$), and $\gcd(u(m), \Delta_m) = \gcd(u(m), 2a_0 + 1) = \gcd(\Delta_m, 2a_0 + 1)$.
14. Proving the closed-form 3-adic residue lock $u(k+1) \bmod 3^{s(k)} = ((3^{s(k)} + 1)/2)^{s(k+1)} \bmod 3^{s(k)}$ and combining backward 3-adic congruences ($\bmod\ 3, 9, 27$) with forward 2-adic congruences ($\bmod\ 4, 8, 64, 256$) to prove $u(k) \neq 1$ ($k \geq 1$), $1735 \leq u(k)$ at $s(k) = 2$, $65 \leq u(k)$ across $2 \leq s(k) \leq 6$, and $103 \leq u(k)$ on ceiling-saturated runs.
15. **Exact $2^{T(m)}$-Cancellation in the Single-Origin Ridout Product & Explicit Repeated-Subword Cutoffs (`RidoutSubspaceFramework.lean`, `RepeatedSubwordComplexity.lean`):**
16. Proving the single-origin cancellation $2^{s(m)+D+1}u(m)\Delta_m < 3^{T(m)+1}(2a_0 + 1)$ and the exponent identity $(1 - \frac{\log 2}{\log 3}) + \frac{\log 2}{\log 3} + 1 = 2$ (Dead End 14).

17. Proving the parity-enhanced common multiplier $u(r_2) - u(r_1) = 2^{S_{\text{tail}+1}} K_0$, $u(r_2 + d) - u(r_1 + d) = 2 \cdot 3^S K_0$ across any repeated subword and its logarithm-free linear cutoff $\text{blockSteps}^{(r_1)} L \le T(r_2) + W$.
18. **Discovery & Machine Certification of the 55-Bit, 113-Step Coasting Record-Holder (`MahlerGrandSynthesis.lean`):**
19. Using our ceiling-saturation sufficiency theorem (`exists_validCompanionUpTo_of_three_le`) to construct a targeted congruence search that uncovered `recordHolderA0 = 29871555608238677 < 2^55`, surviving 113 steps (58 steps past its 55-bit binary length) and dying at step 114.

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2. The Odd-Block Reduction & Dual 2-Adic / 3-Adic Recurrence

Let `step x = (3 * x + 1) / 2` if `x` is odd and `3 * x / 2` if `x` is even, and let `orbit a_0 t` be the t-th iterate starting from $a_0 \in \mathbb{N}$.

A sequence of block lengths $s : \mathbb{N} \to \mathbb{N}$ and odd quotients $u : \mathbb{N} \to \mathbb{N}$ forms a **`ValidBlockChain a_0 s u K`** (`OddBlockReduction.lean`) if for every $k < K$:

$$1 \le s(k), \qquad u(k) \bmod 2 = 1, \qquad 3 \cdot b_k + 1 = 2^{s(k)} u(k),$$

where $b_k := \text{orbit } a_0\ (T(k))$ and $T(k) := \text{blockSteps } s\ k = \sum_{i=0}^{k-1} s(i)$.

Fundamental Identities Along Any Valid Block Chain

- **One-Step Coprime Quotient Recurrence (`CoprimeRemainderSlack.lean`):**

  $$2^{s(k+1)} u(k+1) = 3^{s(k)} u(k) + 1 \qquad (k+1 < K).$$

- **Exact 3-Adic Factorization of Block-Boundary Orbit Values (`ThreeAdicAndSeedGCDLocks.lean`):**

  $$b_{k+1} = \text{orbit } a_0\ (T(k+1)) = 3^{s(k)-1} \cdot u(k), \qquad \gcd(u(k), 6) = 1.$$

  Thus at every block boundary b_{k+1}, its **3-adic valuation** is $s(k) - 1$ (encoding the *preceding* block length) with 3-free part $u(k)$, while the **2-adic valuation of** $3b_{k+1} + 1$ is $s(k+1)$ (encoding the *upcoming* block length) with 2-free part $u(k+1)$!
- **Multi-Step Coprime Remainder & Intrinsic Slack (`CoprimeRemainderSlack.lean`, `AperiodicSlackCocycle.lean`):** Defining $R(s, 0) = 0$ and $R(s, m+1) = 3^{s(m)} R(s, m) + 2^{\text{tailSteps } s\ m}$ (where $\text{tailSteps } s\ m = \sum_{i=1}^{m} s(i)$), the m-step quotient satisfies:

  $$2^{\text{tailSteps } s\ m} u(m) = 3^{T(m)} u(0) + R(s, m).$$

  Eliminating $u(0)$ via $2^{s(0)} u(0) = 3a_0 + 1$ yields the **Orbit Slack Identity**:

  $$\Delta_m := \text{integerSlack } s\ m := 3^{T(m)} - 2^{s(0)+1} R(s, m) = 3^{T(m)+1}(2a_0 + 1) - 2^{T(m+1)+1} u(m).$$

  Notice that the middle expression $3^{T(m)} - 2^{s(0)+1} R(s, m)$ depends **solely on the block sequence** s and is completely independent of a_0 and u!

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3. Exact 1D Dynamical Reduction & Converse Companion Reconstruction

At first glance, ensuring that all K archimedean companion intervals $[\text{lowerBound}(a_0, T(j)),\ \text{upperBound}(a_0, T(m+1)))$ overlap requires checking $\binom{K}{2}$ pairwise inequalities:

$$\forall\, 0 \le j \le m < K, \qquad 0 < \Delta_{m-j}^{(j)} := \text{integerSlack } (\text{shiftSeq } s\ j)\ (m - j).$$

We proved four equivalent ways to collapse this $\binom{K}{2}$ triangular system into a **1D dynamical condition**:

1. **The 1D `minSlack` Dynamical System (`MinSlackDynamicalSystem.lean`, `ConverseCompanionReconstruction.lean`):** Define:

   $$\text{minSlack } s\ 0 = 1, \qquad \text{minSlack } s\ (m+1) = \min\Big(2^{T(m+1)},\ 3^{s(m)} \cdot \text{minSlack } s\ m - 2^{T(m+1)+1}\Big).$$

   Then for any `ValidBlockChain a_0 s u K` ($K \ge 1$), a real companion $\xi \in \mathbb{R}$ surviving through step $T(K)$ exists **if and only if** `0 < minSlack s (K - 1)`:

   $$\big(\exists \xi : \mathbb{R},\ \text{ValidCompanionUpTo } a_0\ \xi\ (T(K))\big) \iff 0 < \text{minSlack } s\ (K-1).$$

2. **Shift-Free Unshifted Ceiling Comparison (`MultiOriginSubspaceSyzygy.lean`):** Using the **Shifted Slack Affine Collapse Identity**

   $$2^{T(r)} \Delta_d^{(r)} = \Delta_{r+d}^{(0)} + 3^{S_{r,d}} \big(2^{T(r)} - \Delta_r^{(0)}\big),$$

   every `shiftSeq` can be eliminated:

$$0 < \text{minSlack } s\ m \iff \forall j \le m,\quad 3^{T(m)}\left(\Delta_j^{(0)} - 2^{T(j)}\right) < 3^{T(j)}\Delta_m^{(0)}.$$

3. **Carryless Base-3 Shift-and-Add Automaton (`SimultaneousBaseDigitRigidity.lean`):** The scaled quotients $P_m := 2^{T(m+1)+1}u(m)$ satisfy the base-3 shift-and-add recurrence $P_{m+1} = 3^{s(m)}P_m + 2^{T(m+1)+1}$, and unconditionally $P_m = 3^{T(m)}(6a_0+2) + 2^{s(0)+1}R(s,m)$. Thus:

$$0 < \Delta_m^{(0)} \iff \left\lfloor \frac{2^{T(m+1)+1}u(m)}{3^{T(m)}} \right\rfloor = 6a_0 + 2 \quad (\text{no base-3 carry across } 3^{T(m)}!).$$

4. **Time-Homogeneous 1D Slack Map & Lüroth–Markov Partition (`CoupledFractionalPartDynamics.lean` , `FiniteSeedCarryAndBackwardDescent.lean`):** Normalizing by $2^{T(m)}$ via $\mu_2(s,m) := \text{minSlack } s\ m / 2^{T(m)}$ removes the explicit time dependence $T(m)$ from the recurrence:

$$\mu_2(s,0) = 1, \qquad \mu_2(s,m+1) = \min\left(1,\ \left(\frac{3}{2}\right)^{s(m)}\mu_2(s,m) - 2\right).$$

(Scope note: μ_2 still takes the block length $s(m)$ as input—at $\mu_2 = 1$, every $s(m) \ge 3$ preserves $\mu_2 = 1$. Along an integer orbit, $s(m) = v_2(3b_m+1)$ and $u(m) = (3b_m+1)/2^{s(m)}$ are uniquely forced by the arithmetic boundary b_m, as proved in `arith_block_parameters_unique` .) Whenever the uncapped ratio $\theta_2(s,m) = \Delta_m / 2^{T(m)}$ transitions into $(0,1]$, $\theta_2(s,m)$ must lie in the partition cell $I_{s(m)} = (2(2/3)^{s(m)},\ 3(2/3)^{s(m)}]$. Because $3(2/3)^{c+1} = 2(2/3)^c$, the cells $I_c \cap (0,1]$ $(c = 2,3,4,\ldots)$ form a **pairwise-disjoint partition of** $(0,1]$.

5. **Infinite-Horizon Z-Number Bridge & Direct `Int.fract` Equivalence (`InfiniteZNumberBridgeAndRationalBase.lean`):** For any $a_0 \in \mathbb{N}$ and $\alpha \in \mathbb{R}$, `ValidCompanionUpTo a_0 α N` holds **if and only if** $\lfloor \alpha \rfloor = a_0$ and $\forall n \le N,\ \text{Int.fract}(\alpha(3/2)^n) < 1/2$ (`validCompanionUpTo_iff_floor_and_fract_lt_half`), and `ValidCompanionForever a_0 α` holds **if and only if** $\lfloor \alpha \rfloor = a_0$ and $\forall n \in \mathbb{N},\ \text{Int.fract}(\alpha(3/2)^n) < 1/2$ (`validCompanionForever_iff_floor_and_forall_fract_lt_half`). Along an infinite block chain, any α_∞ satisfying the strict block-boundary separation $\forall m,\ \text{lowerBound } a_0\ (T(m)) \le \alpha_\infty < \text{upperBound } a_0\ (T(m))$ is a valid companion across all steps (`validCompanionForever_of_block_boundary_separation`).

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4. Ceiling Saturation & the 113-Step 55-Bit Record-Holder

Whenever $s(k) \ge 3$ for all $k < m$, `minSlack` remains pinned at its ceiling (`CeilingSaturationSynthesis.lean`):

$$\left(\forall k < m,\ 3 \le s(k)\right) \implies \text{minSlack } s\ m = 2^{T(m)} > 0.$$

Consequently, **every** block chain with $s(k) \ge 3$ automatically satisfies all archimedean companion constraints (`exists_validCompanionUpTo_of_three_le`).

- **Arbitrarily Long Closed-Form Survivors (`ExplicitHorizonCutoffs.lean`):** The closed-form integer family

$$a_0(r) = \frac{92 \cdot 2^{12r} - 27}{65}$$

executes r consecutive $(3,3,3,3)$ cycles ($12r$ steps) with $s(k) = 3$, surviving arbitrarily many steps before its first $s = 2$ block.
- **The 113-Step Coasting Record-Holder (`MahlerGrandSynthesis.lean`):** Searching the ceiling-saturated congruence tree uncovered the 55-bit seed:

$$\text{recordHolderA0} = 29{,}871{,}555{,}608{,}238{,}677 < 2^{55}.$$

In `recordHolderA0_exact_113_survival_and_114_death` , Lean machine-checks that `recordHolderA0` executes 31 blocks with $s(k) \ge 3$ through $T(31) = 109$, hits its first $s(31) = 2$ at step 109, takes $s(32) = 1$ at step 111, **survives 113 steps** (58 steps past its 55-bit binary length!), and dies at step 114.

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5. Unconditional Finiteness Theorems & Two-Sided CRT Sieves

While ceiling-saturated chains can be constructed up to $T(m) \approx W$ for $a_0 < 2^W$, once $T(m) > W$ (the **coasting regime**), we proved unconditional finiteness across the following regimes:

1. **Global Step-Rate & Even-Step Domination (`GlobalStepRateBounds.lean` , `ThreeAdicValuationConservation.lean`):** Every surviving orbit satisfies $s(k) \ge 2$, $s(k) = 2 \implies s(k+1) \ge 6, s(k+2) \ge 3, s(k+3) \ge 3$, and the global step-rate bound:

$$3m \le T(m) + 1 \iff 2m \le \text{evenStepSum } s\ m + 1.$$

Moreover, the product of the first m block-boundary orbit values has exact 3-adic valuation $T(m) - m$:

$$\prod_{k=0}^{m-1} b_{k+1} = 3^{T(m)-m} \prod_{k=0}^{m-1} u(k), \qquad \gcd\left(\prod_{k=0}^{m-1} u(k),\ 6\right) = 1.$$

2. **Time-Independent Bounded-Block Escape Ceilings (`AperiodicBoundedBlockSqueeze.lean`, `TimeIndependentDeficiencySqueeze.lean`):** During any run of blocks bounded by $s(k+i) \le c$ after $s(k-1) = 2$, the $2^{T(k)}$ factor cancels identically from `minSlackDeficiency`, bounding the run to at most **6 steps** for $c = 3$ and **4 blocks** for $c = 5$ regardless of how large $T(k)$ is!

3. **Repeated-Subword & Aperiodic Sturmian / Balanced Complexity Cutoffs (`RepeatedSubwordComplexity.lean`, `SturmianAndBalancedComplexityExclusion.lean`):** Whenever an L-block subword repeats at two origins $r_1 < r_2$ ($\forall i < L,\ s(r_1 + i) = s(r_2 + i)$), subtracting the multi-step quotient equations cancels the unknown remainder R_{win} identically and yields a common positive integer $K_0 \ge 1$ such that:

$$u(r_2) - u(r_1) = 2^{\text{tailSteps}^{(r_1)}(L-1)+1} K_0, \qquad u(r_2 + L - 1) - u(r_1 + L - 1) = 2 \cdot 3^{\text{blockSteps}^{(r_1)}(L-1)} K_0.$$

Raising the resulting exponential bound $2^{T(r_2)+S+2} < 3^{T(r_2)+1}(2a_0 + 1)$ to the 5th ($3^5 = 243 < 2^8 = 256$) and 17th ($3^{17} < 2^{27}$) powers yields the **weighted logarithm-free cutoffs**:

$$5S \le 3T(r_2) + 5W + 2, \qquad 17S \le 10T(r_2) + 17W + 9, \qquad 51L \le 10T(r_2) + 17W + 26.$$

Combining these weighted inequalities with the **Finite Pigeonhole Principle on subword complexity** (`exists_repeated_subword_of_pattern_classifier`) unconditionally excludes substantial **aperiodic** classes—including ceiling-saturated classes ($s(k) \ge 3$) that never trigger any archimedean or deficiency pinch:

4. **Aperiodic Ceiling-Saturated $\{3,4\}$ & $[3,5]$ Sturmian Classes (`aperiodic_sturmian_three_four_exclusion`, `aperiodic_sturmian_three_five_exclusion`):** Any chain in $\{3,4\}$ (resp. $[3,5]$) of complexity $p(L) \le L + C_0$ satisfies $3L \le 12C_0 + 5W + 2$ (resp. $L \le 50C_0 + 17W + 9$).

5. **General Sub-Critical Balanced Complexity (`balanced_subcritical_complexity_exclusion`):** For *any* average block length λ, any surviving chain with $10T(r_2) \le (17 - \delta)S + D_0$ ($\delta \ge 1$, corresponding to complexity factor $\kappa \le (17 - \delta)/10 < 1.7$) satisfies $3\delta L \le D_0 + 17W + 9 + \delta$.

6. **Two-Sided Chinese Remainder Sieves & Quotient Floors (`Mod256CompleteSieve.lean`, `Mod2304CombinedSieve.lean`, `TwoSidedCRTAndBackwardObstructions.lean`, `Mod216TwoSidedQuotientFloor.lean`):**

7. **Mod-2304 (256×9) Two-Block Sieve:** Eliminates 2283 of 2304 residue classes of $(3^{s(k)}u(k) + 1) \bmod 2304$ (99.09%), locking the post-2 recovery branch to the single class $Y = 28 \pmod{2304}$.
8. **Unconditional Impossibility of $u(k) = 1$ ($k \ge 1$):** Looking one step backward ($k - 1 \to k$) mod 3 forces $u(k) = 1 \implies s(k)$ even, and looking one step forward ($k \to k + 1$) mod 4 forces $3^{s(k)} + 1 \equiv 2 \pmod 4 \implies s(k+1) = 1$ (immediate death).
9. **Mod-54, Mod-216 & Mod-1728 Quotient Floors:** Combining backward Mod-27 locks ($3 \le s(k-1)$) with forward 2-adic locks forces $1735 \le u(k)$ at $s(k) = 2$, $65 \le u(k)$ across $2 \le s(k) \le 6$, and $103 \le u(k)$ on ceiling-saturated runs ($3 \le s(k) \le 6$ with $3 \le s(k+1)$).

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6. Certified Transcendence & Diophantine Barriers (Why the General Case Remains Open)

A central contribution of this formalization is proving **exact algebraic conservation laws (Dead Ends 1–18)** that explain why every standard transcendence and Diophantine method falls just short on a general aperiodic sequence s:

Barrier 1: Why Infinite Solutions Exist in $\mathbb{Z}_2 \times \mathbb{R}$ vs. Finite Positive Seeds (`TwoAdicCauchyConvergence.lean`, `MahlerSeriesHankelFramework.lean`, `FiniteSeedCarryAndBackwardDescent.lean`, `SubstitutionMahlerMethodAndGrandSynthesis.lean`)

The single power series $\Sigma_m(s) = \sum_{k=1}^{m} (2/3)^{T(k)+1} = \frac{2^{s(0)+1} R(s,m)}{3^{T(m)+1}}$ converges in **both** $\mathbb{R}$ (since $|2/3|_\infty = 2/3 < 1$) and $\mathbb{Q}_2$ (since $|2/3|_2 = 1/2 < 1$). Along any valid block chain (`dual_place_rational_limit_sign_split`):

$$0 < \Sigma_m(s) < \frac{1}{3}, \qquad -\frac{6a_0 + 2}{3} \le -\frac{2}{3} < 0, \qquad \Sigma_m(s) - \left(-\frac{6a_0 + 2}{3}\right) = \frac{2^{T(m+1)+1} u(m)}{3^{T(m)+1}}.$$

In $\mathbb{Z}_2 \times \mathbb{R}$, **infinite solutions genuinely exist** for every sequence s with minSlack $s\ m > 0$: there is a unique 2-adic integer $\alpha_2 \in \mathbb{Z}_2$ and real companion α_∞ satisfying all congruences and archimedean bounds.

For a **positive integer seed** $a_0 \in \mathbb{N}$, the rational partial sums $\Sigma_m(s) \in (0, 1/3)$ converge in $\mathbb{Q}_2$ to the negative rational $-\frac{6a_0+2}{3} < 0$. *(Scope note: because $\mathbb{R}$ and $\mathbb{Q}_2$ are distinct completions of $\mathbb{Q}$, opposite real signs of the archimedean and 2-adic limits are not by themselves a contradiction for an arbitrary aperiodic series; they yield an immediate algebraic contradiction only when the series is forced to represent a common rational or algebraic relation in both places, as in the periodic macro-block case below.)* - **Periodic / Uniform Macro-Block Sign Obstruction (`periodic_macro_block_seed_divisibility`):** Whenever s has uniform macro-block step sum D and remainder R_0, the geometric series sums to the *same* positive rational $\frac{2^{s(0)+1}R_0}{3(3^D-2^D)} > 0$ in both $\mathbb{R}$ and $\mathbb{Q}_2$, forcing $2^{Dm} \mid \big((3^D - 2^D)(3a_0+1) + 2^{s(0)}R_0\big)$. - **Concrete Resolution of the $G_{3,6}$ Product-Graph False Positive (`g3_6_false_positive_cycle_and_real_orbit`, `constant_three_block_chain_eight_pow_dvd`):** In `G_{3,6}`, the 3-cycle $(5,5) \to (4,2) \to (6,3) \to (5,5)$ ($s(k) = 3$) has valid fractional cycle $9/19 \to 4/19 \to 6/19 \to 9/19$ and 2-adic seed $a_0 = -9/19 \in \mathbb{Z}_2$, but any natural seed $a_0 \in \mathbb{N}$ must satisfy $8^m(19b_m + 9) = 27^m(19a_0 + 9)$, hence $8^m \mid (19a_0 + 9)$ and $3m \le W + 4$ for $a_0 < 2^W$. - **Backward-Lifting Descent Dichotomy (`predecessor_lt_boundary`, `fracPart_backward_lift_identity`, `backward_archimedean_survival_dichotomy`):** Every predecessor block boundary $a = (2^c u - 1)/3$ satisfies $a < b = 3^{c-1}u$ and $r_a = 1/3 + (2/3)^c r_b$, with $1/3 \le r_a < 13/27 < 1/2$ holding **unconditionally** for every $c \ge 3$.

Barrier 2: Rank-1 Determinant & Padé–Hankel Valuation Collapse (Dead Ends 12 & 18, `DeterminantFactorization.lean`, `MahlerSeriesHankelFramework.lean`)

When one forms the 2×2 determinant between steps r and $r+d$ to eliminate $2a_0 + 1$ (either from the archimedean slacks Δ_k or from the 2-adic numerators $P_k = 2^{T(k+1)+1}u(k)$), the determinant factors identically as:

$$3^{T(r+d)}\Delta_r - 3^{T(r)}\Delta_{r+d} = 3^{T(r)} \cdot 2^{T(r+1)+1} \cdot R(\text{shiftSeq } s\ r,\ d),$$

$$3^{T(r)+1}P_{r+d} - 3^{T(r+d)+1}P_r = 3^{T(r)+1} \cdot 2^{T(r+1)+1} \cdot R(\text{shiftSeq } s\ r,\ d).$$

Because $\gcd(R_d^{(r)}, 6) = 1$, the 2-adic valuation of the determinant drops from $T(r+d+1)+1$ all the way back to $T(r+1)+1$, preventing Padé approximation from gaining a 2-adic advantage over a single step.

Barrier 3: Baker–Matveev Linear Forms in Logarithms & Multi-Step GCDs (Dead Ends 13 & 17, `BakerMatveevFramework.lean`, `MultiStepQuotientGCD.lean`)

In `BakerMatveevFramework.lean`, we proved the exponentially small 3-logarithm linear form bounds:

$$0 < s(m+1)\log 2 - s(m)\log 3 + \log\left(\frac{u(m+1)}{u(m)}\right) < \frac{1}{3a_0+1}\left(\frac{2}{3}\right)^{T(m+1)},$$

$$0 < (T(m)+1)\log 3 - (T(m+1)+1)\log 2 + \log\left(\frac{2a_0+1}{u(m)}\right) \le \frac{1}{6a_0+2}\left(\frac{2}{3}\right)^{T(m)}.$$

When $u(m)$ is supported on a fixed finite set of primes (S-units), all algebraic bases are fixed, so invoking Matveev's external lower bound $\log|\Lambda| \ge -C(S, a_0)\log T(m)$ beats $-T(m)\log(3/2)$ and implies finiteness conditional on Matveev's theorem. However, in the general case: - Consecutive quotients are always coprime ($\gcd(u(m), u(m+1)) = 1$, `consecutive_u_coprime`). - Across d steps (`multi_step_u_gcd_eq_gcd_coprimeRemainder`), $\gcd(u(r), u(r+d)) = \gcd(u(r), R_d^{(r)}) < 3^{S_{r,d}}/2^{s(r)+1}$. - Even after canceling $\gcd(u(r), u(r+d))$, the reduced denominator satisfies $2^{T(r)}3^{S_{r,d}}\big(u(r)/\gcd(u(r), u(r+d))\big) > 3^{T(r)}(3a_0+1)$ (`multi_step_reduced_denominator_exponential_lower_bound`). Thus the logarithmic Weil height of the variable rational term grows linearly as $h \sim T(m)\log(3/2)$, causing Matveev's lower bound $-C_3 \cdot T(m)\log T(m)$ to lose to $-T(m)\log(3/2)$ by a factor of $\log T(m)$.

Barrier 4: Ridout's Theorem & the Critical Q_m^{-2} Exponent Identity (Dead End 14, `RidoutSubspaceFramework.lean`)

In the orbit equation $Q_m - P_m = \Delta_m$ with $Q_m = 3^{T(m)+1}(2a_0+1)$ and $P_m = 2^{T(m+1)+1}u(m)$, the $\{2,3\}$-free part of P_m is $u(m) \sim (3/2)^{T(m)}$ (with $2^{T(m)}$ in the denominator), while in any sub-ceiling phase $\Delta_m \le 2^{T(m)}$ (with $2^{T(m)}$ in the numerator). Multiplying $u(m)$ and Δ_m **cancels $2^{T(m)}$ identically** (`ridout_non_s_part_cancellation`):

$$2^{s(m)+D+1} \cdot u(m) \cdot \Delta_m < 3^{T(m)+1}(2a_0+1) = Q_m.$$

Consequently, the 3-place $\{\infty, 2, 3\}$ Ridout product satisfies (`ridout_three_place_quadratic_bound`):

$$\left(1 - \frac{P_m}{Q_m}\right) \cdot \frac{u(m)}{P_m} \cdot \frac{2a_0+1}{Q_m} < \frac{(6a_0+3)(2a_0+1)}{2^{s(m)+1}(6a_0+2)} \cdot \frac{1}{Q_m^2}.$$

Because the three place exponents sum to **exact** 2:

$$\underbrace{\left(1-\frac{\log 2}{\log 3}\right)}_{|\cdot|_\infty}+\underbrace{\left(\frac{\log 2}{\log 3}\right)}_{|\cdot|_2}+\underbrace{1}_{|\cdot|_3}=2,$$

the general bounded-block orbit sits **right on Ridout's critical Q_m^{-2} boundary**, crossing below $Q_m^{-2-\varepsilon}$ only when $s(m)\geq\delta T(m)$ or $\Delta_m\leq 2^{(1-\delta)T(m)}$.

Barrier 5: Multi-Origin Schlickewei Subspace Hyperplane Syzygy (Dead End 15, `MultiOriginSubspaceSyzygy.lean`)

Attempting to break the single-origin Q_m^{-2} tangency by combining $k\geq 3$ shifted origins $(0,r_1,r_2,\ldots)$ in Schlickewei's higher-dimensional p-adic Subspace Theorem fails because every scaled shifted slack $2^{T(r)}\Delta_{m-r}^{(r)}$ lies in the **same 2-dimensional plane** $\operatorname{span}\{\Delta_m^{(0)},3^{T(m)}\}$ as $m\to\infty$. Any three origins $(0,r_1,r_2)$ satisfy the exact m-independent hyperplane syzygy (`three_origin_subspace_syzygy`):

$$D_0(r_2)\cdot 3^{T(r_1)}\left(2^{T(r_1)}\Delta_{m-r_1}^{(r_1)}-\Delta_m^{(0)}\right)=D_0(r_1)\cdot 3^{T(r_2)}\left(2^{T(r_2)}\Delta_{m-r_2}^{(r_2)}-\Delta_m^{(0)}\right),$$

landing identically inside the degenerate exceptional subspace of Schlickewei's theorem.

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7. Quantitative Extensions of Dubickas (2008 / 2009): Shifted Carry Words, General Rational Bases p/q, and Explicit $O(\log\xi)$ Confinement Escape Bounds up to Total Length $25/36$

While Mahler's Z-number question concerns the specific half-interval $[0,1/2)$ at base $3/2$, A. Dubickas (2008, *Math. Nachr.* **281**, 951–958; 2009, *Acta Arith.* **137**, 233–239) introduced a powerful relaxation based on the **shifted integer-part / carry recurrence**: for a coprime rational base $p/q>1$ ($1\leq q<p$, $\gcd(p,q)=1$) and a real shift $\theta\in\mathbb{R}$, decomposing $\xi(p/q)^n+\theta=x_n+y_n$ with $x_n=\lfloor\xi(p/q)^n+\theta\rfloor\in\mathbb{Z}$ and $y_n=\{\xi(p/q)^n+\theta\}\in[0,1)$ gives the exact dual identity (`dubBaseCarry_eq_fract_linear`):

$$s_n:=qx_{n+1}-px_n=py_n-qy_{n+1}-(p-q)\theta\in\mathbb{Z}.$$

In `DubickasConfinementAndComplexityExtension.lean`, we formalized and substantially extended Dubickas's framework in five directions:

7.1 Multi-Step Carry Cancellation & the Sharp $\log q/\log(p/q)$ Sub-Critical Complexity Threshold

Iterating $qx_{n+1}=px_n+s_n$ over m steps yields the dual integer and fractional summation identities (`dubBaseX_multi_step_carry_identity` , `dubBaseY_multi_step_carry_identity`):

$$q^m x_{n+m}=p^m x_n+\sum_{t=0}^{m-1}p^{m-1-t}q^t s_{n+t},\qquad q^m y_{n+m}=p^m y_n-\sum_{t=0}^{m-1}p^{m-1-t}q^t s_{n+t}-(p^m-q^m)\theta.$$

Whenever a length-L factor of s_n repeats at two indices $i<j$ ($\forall t<L,\ s_{i+t}=s_{j+t}$), subtracting the L-step identities cancels the carry sum identically (`dubBaseX_repeat_factor_cancellation`):

$$q^L(x_{j+L}-x_{i+L})=p^L(x_j-x_i)\implies q^L\mid(x_j-x_i)\implies q^{L+j}<p^jq^W$$

for $\xi<q^W$ (`dubBaseX_repeated_window_exp_and_power_cutoff`). Raising $q^{L+j}<p^jq^W$ to the b-th power for any (a,b) satisfying $p^b<q^{a+b}$ yields the universal logarithm-free cutoff $bL+1\leq aj+bW$. Combining this with our general-base real logarithmic selection bridge (`exists_ab_of_lt_general_log_threshold`) proves (`no_subcritical_general_log_threshold_dubBaseCarry`): - **General-Base Sub-Critical Carry Complexity Theorem:** For **every** coprime pair $2\leq q<p$ (including the open regime $p>q^2$), every $\xi\geq 2+\frac{2q}{p-q}$, every $\theta\in[-1,0]$, and **every** slope $\kappa<\frac{\log q}{\log(p/q)}$ (in particular, $\kappa<\frac{\log 2}{\log(3/2)}\approx 1.709511$ at base $3/2$ in `no_subcritical_log_threshold_dubCarry`), the shifted carry word cannot satisfy $\operatorname{prefixFactorCount}(s,L,C(L))\leq C(L)\leq\kappa L+C_0$ at arbitrarily large lengths L. This strictly upgrades Dubickas's Sturmian carry-word contradiction ($\kappa=1$) to the exact critical slope $\log q/\log(p/q)$.

7.2 Universal Two-Letter Alphabet, Finite Bispecial-Free Complexity $\leq L+1$ & Linear-in-W Dubickas (2009) Escape

When $\{\xi(p/q)^n\}\in[s,s+1/p]$ ($s\in[0,1-1/p]$), setting $\theta=-s$ places $y_n=\{\xi(p/q)^n-s\}\in[0,1/p]$, so $(p-q)s-q/p\leq s_n\leq(p-q)s+1$. Since $0<q/p<1$, s_n is confined to the two-letter alphabet $\{\lfloor(p-q)s\rfloor,\ \lfloor(p-q)s\rfloor+1\}$ (`dubBaseCarry_mem_two_letters`). Furthermore, if two occurrences of a length-$(m-1)$ factor at i,j branched in both directions at their left and right endpoints, the m-step fractional identity would force $p^m+q^m\leq p^{m-1}+q^m$ ($p^m\leq p^{m-1}$, impossible for $p\geq 2$, `dubBaseCarry_no_bispecial_pair`). - **Finite Bispecial-Free Binary Complexity Theorem (`prefixFactorCount_le_add_one_of_no_bispecial`):** By partitioning the length-$(k+1)$ factors in $[0,k+1+C)$ by their final bit and bounding the overlap of their length-k prefixes via the maximum differing index of any two distinct right-special prefixes, any binary word with no bispecial pair on $[0,L+C]$ satisfies $\operatorname{prefixFactorCount}(w,L,C)\leq L+1$. - **Full General-Base Dubickas (2009) Theorem 1 &**

Strictly Linear $O(W)$ Escape (`dubickas2009_general_base_explicit_and_infinite_escape`, `dubickas2009_general_base_linear_in_W_escape`, `shifted_third_unconditional_explicit_escape`): For **every** coprime $2 \le q < p < q^2$ and **every** $s \in [0, 1 - 1/p]$, choosing (a, b) with $a < b$ and $p^b < q^{a+b}$ forces $\{\xi(p/q)^n\}$ to escape $[s, s + 1/p]$ within $n \le 2bW + 2a + 2$ steps for all $\xi \in [2 + \frac{2q}{p-q}, q^W)$ (and within $n \le W^2 + 3W + 2$ using $(a, b) = (1, 1)$ directly), and infinitely often for all $\xi > 0$. At base $3/2$ ($a = 3, b = 5, 3^5 < 2^8$), every $\xi \in [1, 2^W)$ escapes $[s, s + 1/3]$ by step $n \le 5W + 14$. - **Floor-Residue Non-Stabilization & Open-Regime ($p > q^2$) Single-Cell Escape (`general_base_floor_mod_non_stabilization`, `general_base_non_straddling_cell_escape`):** Using $\lfloor x \rfloor \bmod m = r \iff \{x/m\} \in [r/m, (r+1)/m)$, we prove that $\lfloor \xi(p/q)^n \rfloor \bmod m$ never stabilizes for any coprime $2 \le q < p < q^2$ and $m \ge p$. Moreover, in the open regime $p > q^2$ ($p \ge 2q$, e.g. $5/2, 7/2, 7/3$), we prove that for *every* cell $[r/m, (r+1)/m)$ that avoids the finitely many fixed points $\frac{k}{p-q}$ ($0 \le k < p - q$), s_n is locked to a single constant k, forcing escape within $n \le W + 4$ steps.

7.3 Human-Readable Inequality Proofs for Beyond-$1/p$ Single Windows at Bases $3/2$ and $4/3$

Dubickas (2008) showed that for certain shifts s, one can exceed the $1/p$ barrier on a single interval by forbidding short carry words: - **Base $3/2$ on $[1/6, 13/24)$ of Length $3/8 > 1/3$ (`shifted_three_eighths_carry_period_three`, `shifted_three_eighths_explicit_and_infinite_escape`):** Five linear fractional inequalities rule out `11` (2 steps), `000` (3 steps), `10101` (5 steps), `101001010` (9 steps), and `101001001` (9 steps), forcing $s_{k+3} = s_k$ (period 3) after step 2 and yielding an explicit $n \le W + 15$ escape bound. - **Base $4/3$ on $[1/3, 5/8)$ of Length $7/24 > 1/4 = 1/p$ (`base_four_three_carry_period_two`, `base_four_three_beyond_quarter_explicit_and_infinite_escape`):** Four linear fractional inequalities rule out `11` (2 steps: $16y_n - 9y_{n+2} = 7$), `000` (3 steps: $64y_n - 27y_{n+3} = 0$), `00100` (5 steps: $1024y_n - 243y_{n+5} = 144$), and `001010` (6 steps: $4096y_n - 729y_{n+6} = 891$), forcing $s_{k+2} = s_k$ (period 2) from step 2 onward and proving $Z_{4/3}(1/3, 5/8) = \emptyset$ with an explicit $n \le W + 8$ escape bound for $\xi \in [8, 3^W)$.

7.4 Verified Backward–Forward Integer Funnel Engine & Escape Bounds up to Total Length $25/36 > 2/3 > 1/2$

To certify explicit $O(W)$ escape bounds on multi-interval unions of total length well above $1/2$, we designed a **verified two-phase integer interval-arithmetic engine** (`inRatIntervalList_prev_of_funnelStepOk`, `inRatIntervalList_next_of_forwardStepOk`): 1. **Phase 1 (Backward Coarsened Funnel):** Starting from a scaled integer interval list T_0 at denominator D, `funnelStepOk D Tnext Tprev` checks that for every $(a, b) \in T_{\text{prev}}$, every carry $s \in \{-1, 0, 1, 2\}$, and every $(c, d) \in T_{\text{next}}$, the preimage $([a, b) \cap [(2c + sD)/3, (2d + sD)/3))$ is either empty or contained in a single interval of T_{prev}. Thus $y_0, \ldots, y_K \in T_0 \implies y_n \in T_K$ for all $n \le N - K$. 2. **Phase 2 (Forward Transient Pruning):** Because backward preimages retain transient intervals with in-degree 0 that eventually enter a cycle, `forwardStepOk D Tcur Tamb Tnext` pushes $y_n \in T_{\text{cur}}$ forward one step under $y_{n+1} = (3y_n - s)/2 \in T_{\text{amb}}$, eliminating all in-degree-0 transient components in 1–2 steps without any graph cycle search and collapsing the survivor set to a single fixed-carry or period-1 component!

Confinement Set / Family	Base p/q	Total Length	Carry Lock Mechanism	Explicit Finite Escape Bound	Lean Theorem
$[s,\ s + 1/p]$ ($s \in [0, 1 - 1/p]$)	$2 \le q < p < q^2$	$1/p$	No bispecial pair $\implies P(L) \le L + 1$	$n \le 2bW + 2a + 2$ ($W^2 + 3W + 2$)	`dubickas2009_general_ba`
$[s,\ s + 1/3]$ ($s \in [0, 2/3]$)	$3/2$	$1/3 \approx 0.3333$	No bispecial pair + $3^5 < 2^8$	$n \le 5W + 14$	`shifted_third_unconditi`
$[r/m,\ (r+1)/m)$ non-straddling	$p \ge 2q \ge 4$	$1/m \le \frac{1}{2(p-q)}$	Single-letter carry lock $s_n = k$	$n \le W + 4$	`general_base_non_stradd`
$[1/3,\ 5/8)$	$4/3$	$7/24 \approx 0.2917 > 1/4$	4-inequality period-2 lock ($s_{k+2} = s_k$)	$n \le W + 8$	`base_four_three_beyond_`
$[4/27,\ 1/2)$	$3/2$	$19/54 \approx 0.3519 > 1/3$	`11`, `101`, `000` ban ($s_{k+3} = s_k$)	$n \le W + 8$	`punctured_four_twenty_s`
$[8/57,\ 1/2)$	$3/2$	$41/114 \approx 0.3596 > 1/3$	Homoclinic catapult into $[4/19, 1/4)$	$n \le 3W + 20$	`punctured_eight_fifty_s`
$[\frac{280}{2187}, \frac{648200}{4782969}) \cup [\frac{8}{57}, \frac{1}{2})$	$3/2$	$\frac{66729163}{181752822} \approx 0.3671$ (73.4% of $[0, \frac{1}{2})$)	7-step death band + $[8/57, 1/2)$ catapult	$n \le 3W + 27$	`two_island_punctured_ha`
$[1/6,\ 13/24)$	$3/2$	$3/8 = 0.3750 > 1/3$	5-inequality period-3 lock ($s_{k+3} = s_k$)	$n \le W + 15$	`shifted_three_eighths_e`
$[0,\ 1/5) \cup [4/5,\ 1)$	$3/2$	$2/5 = 0.4000$	1-step carry lock $s_{n+1} = s_n$	$n \le W + 5$	`nearest_int_fifth_expli`
$(8/39, 18/39) \cup (21/39, 31/39)$	$3/2$	$20/39 \approx 0.5128 > 1/2$	2-step `00` / `11` ban ($s_{k+2} = s_k$)	$n \le W + 8$	`dubickas2008_open_union`

Confinement Set / Family	Base p/q	Total Length	Carry Lock Mechanism	Explicit Finite Escape Bound	Lean Theorem
$[8/39, 18/39] \cup [21/39, 31/39]$	$3/2$	$20/39 \approx 0.5128 > 1/2$	Open interior + 4-cycle $2^m \mid (13x_r + v_r)$	$n \le 3W + 19$	`dubickas2008_closed_uni`
4-interval union $U_{7/12}$	$3/2$	$7/12 \approx 0.5833 > 1/2$	7-step backward funnel ($D = 7296$)	$n \le W + 12$	`union_seven_twelfths_ex`
5-interval union $U_{2/3}$	$3/2$	$2/3 \approx 0.6667 > 1/2$	9-step backward + 2-step forward ($D = 118098$)	$n \le W + 14$	`union_two_thirds_explic`
6-interval union $U_{25/36}$	$3/2$	$25/36 \approx 0.6944 > 2/3$	11-step backward + 1-step forward ($D = 708588$)	$n \le W + 16$	`union_twenty_five_thirt`
7-interval union $U_{17/24}$	$3/2$	$17/24 \approx 0.7083 > 70\%$	12-step self-refining + 1-step forward ($D = 472392$)	$n \le W + 17$	`union_seventeen_twenty_`

7.5 The Thue–Morse Period-Doubling Cantor Cascade on $[c, 1-c]$, Fermat-Denominator Reduction & the $[2/7, 5/7]$ Cutoff

For centered windows $[c, 1-c]$ ($c \in (1/4, 2/5]$), the real hold set under $2y_{n+1} = 3y_n - s_n$ ($s_n \in \{0, 1\}$) is governed by a **Thue–Morse period-doubling Cantor cascade**: - **Fermat-Denominator Reduction & Mahler Product Identity (`tmCascadeCarryAcc_succ_factored`, `tmCascadeCarryAcc_mahlerProd_identity`):** Because the length-2^{k+1} Thue–Morse word decomposes as $T_{k+1} = T_k \overline{T_k}$, its carry accumulator $A(T_{k+1}) = \sum_{t=0}^{2^{k+1}-1} 3^{2^{k+1}-1-t} 2^t \, \mathrm{tmBit}(t)$ factors as

$$A(T_{k+1}) = \left(3^{2^k} - 2^{2^k}\right)\left(A(T_k) + 2^{2^k}\right), \qquad \left(3^{2^k} - 2^{2^k}\right) - 2A(T_k) = \prod_{n=0}^{k-1} \left(3^{2^n} - 2^{2^n}\right).$$

Canceling $3^{2^{k-1}} - 2^{2^{k-1}}$ from $3^{2^k} - 2^{2^k}$ reduces the denominator of the period-2^k Thue–Morse cycle to the Fermat-like denominator $F_{k-1} := 3^{2^{k-1}} + 2^{2^{k-1}} \in \{5, 13, 97, 6817, 43112257, \ldots\}$. - **Closed-Form Entry Thresholds c_k, Difference-of-Squares Ratio & $[2/7, 5/7]$ Cutoff (`tmCascadeEntry_closed_form`, `tmCascadeEntry_step_ratio_and_strictAnti`, `tmCascadeEntry_explicit_values_and_two_sevenths_cutoff`, `tmCascade_periodic_orbit_two_pow_dvd`):** The minimum coordinate c_k of the period-2^k Thue–Morse cycle occurs at cyclic shift $2^{k-1} + 1$ and satisfies the exact closed-form product identity and step ratio:

$$\left(3^{2^{k-1}} + 2^{2^{k-1}}\right)(4c_k - 1) = 3\prod_{n=0}^{k-2}\left(3^{2^n} - 2^{2^n}\right), \qquad \frac{4c_{k+1} - 1}{4c_k - 1} = \frac{3^{2^k} - 2^{2^k}}{3^{2^k} + 2^{2^k}} \in (0, 1).$$

Hence c_k strictly descends toward Dubickas's (2006) constant $c_\infty = \frac{1 + \prod_{n \ge 0}(1 - (2/3)^{2^n})}{4} \approx 0.28564732$, with first five rational values:

$$c_1 = \frac{2}{5} > c_2 = \frac{4}{13} > c_3 = \frac{28}{97} > c_4 = \frac{1948}{6817} > \frac{2}{7} > c_5 = \frac{12314908}{43112257},$$

where $c_4 - \frac{2}{7} = \frac{2}{47719} > 0$ and $\frac{2}{7} - c_5 = \frac{20158}{301785799} > 0$. Moreover, any real ξ landing on the level-k cycle at step r satisfies the 2-adic divisibility trap $2^{2^k m} \mid (F_{k-1} x_r + v)$ for all $m \ge 0$. - **Exact Match with Mahler's Binary Partition Sequence through Depth 15 (`cantorTwoSevenths_matches_mahlerBinaryPartition_up_to_15_and_splits_at_16`):** Because $c_5 < 2/7 < c_4$, the Cantor cylinder tree of $(2/7, 5/7)$ admits the first 4 levels of the Thue–Morse cascade ($L = 2, 4, 8, 16$) and matches Mahler's binary partition sequence $B(d) = B(d-1) + B(\lfloor d/2 \rfloor)$ identically for all $0 \le d \le 15$ (`[1, 2, 4, 6, 10, 14, 20, 26, 36, 46, 60, 74, 94, 114, 140, 166]`), before clipping the two mirror $k = 5$ (period-32) branches at depth $d = 2^4 = 16$ (`200` vs. $B(16) = 202$).

7.6 The General Gap-g Window-to-Residue Ladder ($g \ge 3$) & General g-Rung First-Defect Valuation Lock

For bottom-anchored windows $[0, a)$ ($g \ge 3$), defining $\alpha_g = \frac{3^{g-1}}{3^g - 2^g}$ and $\beta_g = \frac{1 + (2/3)^{g-1}}{3} = \frac{3^{g-1} + 2^{g-1}}{3^g}$, we formalized the general g-rung ladder and its 2-adic valuation lock: - **Strict Rung Ordering & Short-Gap Fractional Cost (`rungAlpha_lt_rungBeta_lt_half`, `gap_ladder_short_gap_cost_and_cap`):** Since $2^g < 3^{g-1}$ for all $g \ge 3$, every rung satisfies $\frac{1}{3} < \alpha_g < \beta_g < \frac{1}{2}$ with $\beta_g - \alpha_g = \frac{2^{g-1}(3^{g-1} - 2^g)}{3^g(3^g - 2^g)} > 0$. Any odd step followed by another odd step at distance $1 \le d \le g - 1$ has fractional part $\ge \frac{1}{3} + (2/3)^d \cdot \frac{1}{3} \ge \beta_g$, whereas any block of length $c \ge g$ maps $[0, \alpha_g]$ into $[0, \alpha_g]$. - **Universal Rung-g Affine Invariant $I_g(b) = (3^g - 2^g)b + 3^{g-1}$ & First-Defect Valuation Lock (`rungAffInv_single_block_identity`, `rungAffInv_pure_prefix_identity`, `general_rung_first_defect_valuation_lock`):** Across any block $3b + 1 = 2^c u$, $I_g(b)$ satisfies the universal identity $3I_g(b) = (3^g - 2^g)2^c u + 2^g$. Consequently, m consecutive blocks of length g force $2^{gm} I_g(b_m) = 3^{gm} I_g(a_0)$, and a first defect $s(j_0) \ge g + 1$ after j_0 pure blocks of length g permanently locks the 2-adic valuation of $(3^g - 2^g)a_0 + 3^{g-1}$ to **exactly** $g(j_0 + 1)$:

$$2^{g(j_0+1)} \,\Big|\, \big((3^g-2^g)a_0+3^{g-1}\big) \quad \text{and} \quad 2^{g(j_0+1)+1} \nmid \big((3^g-2^g)a_0+3^{g-1}\big).$$

7.7 The Punctured $[0,1/2)$ Homoclinic Catapult Trap on $[8/57,1/2)$ & the $\varepsilon^* = 8/57$ Phase Transition

When Mahler's interval $[0,1/2)$ is punctured at the origin to $[\varepsilon,1/2)$, the unbounded `0^g` staircase near 0 is cut off and the Cantor cylinder tree undergoes an exact homoclinic phase transition at $\varepsilon^* = \frac{8}{57} = \frac{2}{3}\cdot\frac{4}{19} \approx 0.140351$: - **Pure Period-3 Lock on** $[4/27,1/2)$ **(`punctured_half_step_dichotomy_and_forbidden_short_gaps`, `punctured_four_twenty_sevenths_period_three_lock_and_escape`):** For any $\varepsilon \geq 1/16$, `11` and `101` are forbidden on $[\varepsilon,1/2)$. When $\varepsilon \geq 4/27 \approx 0.14815$, three consecutive zeros (`000`) are also forbidden (`8 y_{k+3} = 27 y_k \ge 4`), forcing $s_k + s_{k+1} + s_{k+2} = 1$ and $s_{k+3} = s_k$ for all $k \geq 0$, with explicit escape in $n \leq W + 8$ steps. - **The Homoclinic Catapult Trap on** $[8/57,1/2)$ **(`punctured_eight_fifty_sevenths_catapult_and_absorbing_trap`, `punctured_eight_fifty_sevenths_explicit_and_infinite_escape`):** On $[8/57,1/2)$ (length $41/114 \approx 0.3596 > 1/3$), four consecutive zeros (`0000`) are forbidden, and any three-zero block (`000`) at step k forces $s_{k+3} = 1$ and **catapults** y_{k+4} into the absorbing period-3 trap $[4/19,1/4)$:

$$16y_{k+4} = 81y_k - 8 \geq 81\cdot\frac{8}{57} - 8 = \frac{64}{19} \implies \frac{4}{19} \leq y_{k+4} < \frac{1}{4}.$$

Once $y_m \in [4/19,1/4)$, every subsequent 3-step block is forced to be `(001)` with $y_{m+3} \in [4/19,1/4)$, locking $s_{t+3} = s_t$ for all $t \geq m-1$. Thus at most **one** `000` block can occur in an entire orbit, and every $\xi \in [1,2^W)$ escapes $[8/57,1/2)$ in at most $n \leq 3W+20$ steps ($Z_{3/2}([8/57,1/2)) = \emptyset$). - **Sub-**$8/57$ **Homoclinic Periodic Cascade & Kernel-Certified Cylinder Counts (`homoclinicCycleMin_closed_form_and_strictMono`, `cantorPuncturedHalf_phase_transition_counts`):** Below $\varepsilon^* = 8/57$, the periodic cycles $\mathcal{O}_m = \texttt{(001)}^m\texttt{(0001)}$ of period $3m+4$ appear at minimum coordinates

$$\varepsilon_m = \frac{8}{57} - \frac{4\cdot 2^{3m+4}}{57\,(3^{3m+4}-2^{3m+4})} \quad \left(\varepsilon_1 = \frac{280}{2059} < \varepsilon_2 = \frac{8072}{58025} < \varepsilon_3 = \frac{222040}{1586131} < \cdots \uparrow \frac{8}{57}\right).$$

Kernel evaluation (`by decide`) certifies the Cantor cylinder counts $|C_d([\varepsilon,1/2))|$ for $0 \leq d \leq 10$ across all four regimes: constant 3 at $\varepsilon = 4/27$ (`[1, 2, 3, 3, ..., 3]`), constant 6 at $\varepsilon = 1/7$ (`[1, 2, 3, 4, 5, 6, 6, ..., 6]`), exact $d+1$ linear growth at $\varepsilon^* = 8/57$ (`[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`), and bifurcation above $d+1$ at depth $d = 7$ (period 7 of $\mathcal{O}_1$) for $\varepsilon = 1/8 < \varepsilon_1$ (`[1, 2, 3, 4, 5, 6, 7, 9, 11, 13, 16]`).

7.8 The Universal Rung-g Catapult Staircase ($\varepsilon_g^* \to 0$) and the Leak-Free Inductive Elevator

Section 7.7's catapult trap at $g = 3$ ($\varepsilon_3^* = 8/57$) extends to an infinite inductive staircase across **all rungs** $g \geq 3$ whose left boundary ε_g^* converges to 0: - **Universal Four-Term Rung Ordering, Corridor Overlap & Geometric Decay (`rungCatapult_ordering_and_corridor_overlap`):** For every $g \geq 3$, define the period-g post-reset fixed point $z_g^* = \frac{2^{g-1}}{3^g-2^g}$, the $(g+1)$-defect boundary $b_g = \frac{2^{g-1}}{3^g}$, and the Rung-$g$ catapult floor $\varepsilon_g^* = \frac{2^g}{3(3^g-2^g)} = \frac{2}{3}z_g^*$. Because $2^g < 3^{g-1}$ for all $g \geq 3$, these satisfy the strict inequality chain

$$0 < \varepsilon_{g+1}^* < z_{g+1}^* < \varepsilon_g^* < b_g < z_g^* < b_{g-1}, \qquad \varepsilon_g^* < \left(\frac{2}{3}\right)^g \xrightarrow{g\to\infty} 0,$$

with explicit rational triples $(\varepsilon_3^*, b_3, z_3^*) = (8/57, 4/27, 4/19)$, $(\varepsilon_4^*, b_4, z_4^*) = (16/195, 8/81, 8/65)$, and $(\varepsilon_5^*, b_5, z_5^*) = (32/633, 16/243, 16/211)$. Because $\varepsilon_g^* < b_g$, adjacent Rung-g corridors $J_g = [\varepsilon_g^*, b_{g-1})$ overlap on $[\varepsilon_g^*, b_g)$ and tile $(0, 2/9)$ all the way down to 0. - **Universal Rung-g Catapult & Absorbing Upward Expansion (`universal_rung_catapult_and_upward_expansion`):** Under the k-step post-reset return map $T_k(z) = (3/2)^k z - 1/2$, every rung $g \geq 3$ satisfies

$$T_g(z) - z_g^* = \left(\frac{3}{2}\right)^g (z - z_g^*), \qquad T_{g+1}(z) - z_g^* = \left(\frac{3}{2}\right)^{g+1} (z - \varepsilon_g^*).$$

Hence any $(g+1)$-step block from $z \geq \varepsilon_g^*$ **catapults** $T_{g+1}(z) \geq z_g^* > b_g$, and any g-step block from $z \geq z_g^*$ expands upward ($T_g(z) \geq z \geq z_g^*$). - **The Leak-Free Inductive Elevator Across Rungs** $3 \leq g \leq G$ **(`leak_free_inductive_elevator_monotone_rung_descent`):** Each interval $[b_g, b_{g-1})$ splits into the **Rung-g leak window** $L_g = [b_g, z_g^*)$ of width $\mathrm{rungLeakWidth}(g) = \frac{2^{2g-1}}{3^g(3^g-2^g)} \leq \frac{32}{513}\left(\frac{4}{9}\right)^{g-3}$ and the **Rung-g upward trap** $\mathrm{Trap}_g = [z_g^*, b_{g-1})$. Whenever a post-reset trajectory evolves in the leak-free union $\mathcal{U}_G = [\varepsilon_G^*, b_G) \cup \bigcup_{g=3}^{G} [z_g^*, b_{g-1})$ (`ValidLeakFreeElevator G z s`), at most one initial catapult step ($s_0 = G+1$) occurs at $m = 0$, after which $z_1 \leq z_2 \leq \cdots$ is monotonically non-decreasing and the block lengths satisfy $G+1 \geq s_0 \geq s_1 \geq s_2 \geq \cdots \geq 3$—forcing one-way upward promotion toward Rung 3 regardless of how close ε_G^* is to 0.

7.9 Self-Cleaning Death-Band Decomposition of Leak Windows $L_g = [b_g, z_g^*)$ & the $280/2059$ Seven-Step Catapult-Death Trap

Even without manually puncturing $L_g = [b_g, z_g^*)$, almost the entire leak window is **automatically self-cleaning** on $[\varepsilon, 1/2)$: - **The m-Step Automatic Death-Band Ladder** $D_{g,m}(\varepsilon) \subset [b_g, z_g^*)$

(`rungDeathBand_step_and_measure_decomposition`): For every rung $g \geq 3$ and depth $m \geq 1$, the m-step preimage of $[0, \varepsilon)$ under $T_g(w) = (3/2)^g w - 1/2$ is the interval $D_{g,m}(\varepsilon) = [\mathrm{rungDeathBandLo}(g,m),\ \mathrm{rungDeathBandHi}(g,m,\varepsilon))$ with

$$\mathrm{rungDeathBandLo}(g,m) = z_g^* \left(1 - \left(\frac{2}{3}\right)^{gm}\right), \qquad \mathrm{rungDeathBandHi}(g,m,\varepsilon) = \mathrm{rungDeathBandLo}(g,m) + \left(\frac{2}{3}\right)^{gm} \varepsilon.$$

At $m = 1$, $\mathrm{rungDeathBandLo}(g,1) = b_g$, and every $w \in D_{g,1}(\varepsilon)$ falls into $[0, \varepsilon)$ in a single g-step block ($0 \leq T_g(w) < \varepsilon$). For all $m \geq 1$, T_g maps $D_{g,m+1}(\varepsilon)$ onto $D_{g,m}(\varepsilon)$, and for $0 \leq \varepsilon \leq b_g$ these bands are pairwise disjoint inside $[b_g, z_g^*)$ with total geometric-series measure $\mathrm{rungCleanedMeasure}(g,\varepsilon) = \frac{2^g}{3^g - 2^g}\varepsilon$ and residual uncleaned measure

$$\mathrm{rungResidualLeakMeasure}(g,\varepsilon) = \mathrm{rungLeakWidth}(g) - \frac{2^g}{3^g - 2^g}\varepsilon = \frac{2^g}{3^g - 2^g}(b_g - \varepsilon).$$

At $\varepsilon = \varepsilon_3^* = 8/57$ ($g = 3$), the automatic death bands clean out $\frac{64}{1083}$ out of $|L_3| = \frac{32}{513}$—an exact fraction $\frac{18}{19} \approx 94.74\%$ of the Rung-3 leak window—leaving a residual uncleaned measure of only $\frac{32}{9747} \approx 0.003283$. - **The 7-Step Catapult-into-Death-Band Trap Below $8/57$**
(`catapult_into_death_band_seven_step_escape`): For $\varepsilon \geq 1/8$, any point $y_k \in \left[\frac{280}{2187},\ \frac{280+128\varepsilon}{2187}\right)$ is forced to take the 4-step block `0001` into the 1-step death band $D_{3,1}(\varepsilon) = \left[\frac{4}{27},\ \frac{4+8\varepsilon}{27}\right)$, followed by the 3-step block `001` into $[0, \varepsilon)$—dying in $4 + 3 = 7$ steps (`0001001`)! Moreover, $\varepsilon < \frac{280+128\varepsilon}{2187} \iff \varepsilon < \frac{280}{2059} = \varepsilon_1$, explaining mechanically why $\varepsilon_1 = \frac{280}{2059}$ is the first homoclinic threshold below $\frac{8}{57}$.

7.10 Higher-Order Gap-to-Trigger Bijection ($\mathrm{Gap}_{g,m}(\varepsilon) \xrightarrow{T_g^m} [\varepsilon, b_g)$), K-Pass Decay to 0, and the $2^{7m} \mid (2059x_r + 280)$ Leak-Threading Survivor Lock

What happens to the remaining uncleaned gaps $\mathrm{Gap}_{g,m}(\varepsilon) := [\mathrm{rungDeathBandHi}(g,m,\varepsilon),\ \mathrm{rungDeathBandLo}(g,m+1))$ between consecutive death bands $D_{g,m}(\varepsilon)$ and $D_{g,m+1}(\varepsilon)$? - **Universal Gap-to-Trigger Bijection & K-Pass Measure Decay**
(`rungDeathGap_maps_onto_catapult_zone`): Under the m-fold return map $T_g^{[m]} = \mathrm{rungReturnMapIter}(g,m,\cdot)$, every uncleaned gap $\mathrm{Gap}_{g,m}(\varepsilon)$ ($g \geq 3, m \geq 1$) satisfies the displacement law $T_g^{[m]}(w) - z_g^* = (3/2)^{gm}(w - z_g^*)$, has exact width $(2/3)^{gm}(b_g - \varepsilon)$, and maps **bijectively onto the catapult trigger zone** $[\varepsilon, b_g)$:

$$T_g^{[m]}\big(\mathrm{rungDeathBandHi}(g,m,\varepsilon)\big) = \varepsilon, \qquad T_g^{[m]}\big(\mathrm{rungDeathBandLo}(g,m+1)\big) = b_g,$$

$$w \in \mathrm{Gap}_{g,m}(\varepsilon) \iff T_g^{[m]}(w) \in [\varepsilon, b_g).$$

Consequently, every point surviving the first pass through L_g is driven back into $[\varepsilon, b_g)$ and stretched by T_{g+1} across the death bands $D_{g,k}(\varepsilon)$ a second, third, ..., K-th time (at $g = 3, \varepsilon = 8/57$, the single-pass survival ratio $\frac{1}{19}$ decays as $(1/19)^2 = \frac{1}{361}$ and $(1/19)^3 = \frac{1}{6859} \to 0$, eliminating 100% of the width in the $K \to \infty$ limit). - **Real 7-Cycle Inside $\mathrm{Gap}_{3,1}(\varepsilon_1)$, $2^{7m} \mid (2059x_r + 280)$ Lock & Constructive Leak-Threading Survivors**
(`homoclinic_seven_cycle_in_gap_and_divisibility_lock`): At $\varepsilon_1 = \frac{280}{2059}$, the post-`0001` point $w = \frac{388}{2059} = T_4(\varepsilon_1) = \mathrm{rungDeathBandHi}(3,1,\varepsilon_1) \in \mathrm{Gap}_{3,1}(\varepsilon_1) = \left[\frac{388}{2059},\ \frac{140}{729}\right)$ returns to $T_3(w) = \frac{280}{2059} = \varepsilon_1$, generating the real 7-cycle $\mathcal{O}_1 = \texttt{(0001)(001)}$ inside $\left[\frac{280}{2059}, \frac{1}{2}\right)$. However, any Mahler trajectory $\xi(3/2)^n = x_n + y_n$ with $y_r = y_{r+7m} = \frac{280}{2059}$ satisfies the 2-adic divisibility lock

$$\left(2^{7m} : \mathbb{Z}\right) \,\Big|\, (2059x_r + 280),$$

ruling out $m = \infty$ for all positive seeds while constructively producing explicit finite-horizon leak-threading survivors: $x_0 = 56 \pmod{2^7}$, $x_0 = 1464 \pmod{2^{14}}$, and $x_0 = 1443256 \pmod{2^{21}}$.

7.11 Universal Homoclinic Lock Family $2^{(3k+4)m} \mid (D_k x_r + N_k)$ & Mutually Incompatible Multi-Defect Valuation Locks

We generalize our 2-adic locks across the entire infinite homoclinic cascade and across consecutive multi-defect transitions: - **Universal Homoclinic Cycle Family Lock**
(`homoclinic_family_cycle_divisibility_lock`): For every $k \in \mathbb{N}$ in the homoclinic cascade $\mathcal{O}_k = \texttt{(001)}^k\texttt{(0001)}$ approaching $\frac{8}{57}$ from below, the integer numerator $N_k = \mathrm{homoclinicCycleNum}(k)$ and denominator $D_k = \mathrm{homoclinicCycleDen}(k) = 3^{3k+4} - 2^{3k+4}$ satisfy

$$19N_k = 8\left(27^{k+1} - 8^{k+1}\right), \qquad \varepsilon_k = \mathrm{homoclinicCycleMin}(k) = \frac{N_k}{D_k},$$

with $(N_1, D_1) = (280, 2059)$, $(N_2, D_2) = (8072, 58025)$, and $(N_3, D_3) = (222040, 1586131)$. Whenever $\mathrm{dubY}\ \xi\ 0\ r = \mathrm{dubY}\ \xi\ 0\ (r + (3k+4)m) = \varepsilon_k$, the integer part $x_r = \mathrm{dubX}\ \xi\ 0\ r$ satisfies:

$$\left(2^{(3k+4)m} : \mathbb{Z}\right) \,\Big|\, \left(D_k x_r + N_k\right).$$

- **Universal Rung-g Defect Cocycle & Incompatible Multi-Defect Valuation Locks**
(`rungAffInv_general_defect_step_identity`, `exactTwoPowLock_incompatible_of_ne`,

`general_rung_two_defect_simultaneous_lock`**):** Across any defect block $3b+1 = 2^c u$ ($c \ge g \ge 1$) with $b' = \text{orbit } b\ c = 3^{c-1}u$, the Rung-$g$ affine invariant $I_g(b) = (3^g - 2^g)b + 3^{g-1}$ satisfies:

$$2^c I_g(b') + 2^g 3^{g-1}\left(3^{c-g} - 2^{c-g}\right) = 3^c I_g(b).$$

Consequently, if a trajectory has j_0 pure blocks of length g, a first defect $c = s(j_0) \ge g+1$, j_1 pure blocks of length g, and a second defect $s(j_0+1+j_1) \ge g+1$, then a_0 must **simultaneously** satisfy: 1. $\mathrm{ExactTwoPowLock}\big(g(j_0+1),\ I_g(a_0)\big)$, and 2. $\mathrm{ExactTwoPowLock}\Big(gj_0 + c + g(j_1+1),\ 3^{gj_0+c}I_g(a_0) - 2^{g(j_0+1)}3^{g-1}\big(3^{c-g} - 2^{c-g}\big)\Big)$, where $\mathrm{ExactTwoPowLock}(e, I) :\iff 2^e \mid I \wedge 2^{e+1} \nmid I$. Because $\mathrm{ExactTwoPowLock}(e_1, I)$ and $\mathrm{ExactTwoPowLock}(e_2, I)$ are **mutually incompatible** whenever $e_1 \ne e_2$, no starting seed a_0 can be compatible with two distinct first-defect horizons $j_0 \ne j_0'$ OR two distinct second-defect horizons $j_1 \ne j_1'$.

7.12 The Two-Island Punctured $[0, 1/2)$ Theorem Reaching $\frac{280}{2187}$ (73.43% of $[0,1/2)$), the $17/24 > 70\%$ Seven-Interval Union, and the Arbitrary K-Defect Valuation Tower

- **Two-Island Punctured $[0, 1/2)$ Theorem Reaching Down to $\frac{280}{2187} \approx 0.12803$ (`two_island_punctured_half_explicit_and_infinite_escape`):** Combining Section 7.7's catapult trap on $[8/57, 1/2)$ with Section 7.9's 7-step catapult-into-death-band trap at $\varepsilon = \frac{280}{2187} \approx 0.12803$ ($\frac{280+128\varepsilon}{2187} = \frac{648200}{4782969} \approx 0.13552$), we prove that on the two-island subset of $[0, 1/2)$:

  $$\mathcal{W}_{280/2187} = \left[\frac{280}{2187},\ \frac{648200}{4782969}\right) \ \cup\ \left[\frac{8}{57},\ \frac{1}{2}\right)$$

  (which removes only a gap of width $\frac{438784}{90876411} \approx 0.004828 < \frac{1}{200}$ below $\frac{8}{57}$ and covers $\frac{66729163}{181752822} \approx 0.367142$, or 73.43% **of** $[0,1/2)$), every $\xi \in [1, 2^W)$ escapes in at most $n \le 3W + 27$ **steps** ($Z_{3/2}(\mathcal{W}_{280/2187}) = \emptyset$).
- **Self-Refining Funnel Engine & the $17/24 = 51/72 \approx 0.70833 > 70\%$ Seven-Interval Union (`inRatIntervalList_terminal_of_selfFunnelChainOk`, `union_seventeen_twenty_fourths_explicit_and_infinite_escape`):** By refining both endpoints against the current funnel level T_k (`selfFunnelChainOk` checking `funnelStepOk D Tk Tk Tk1`), we certify in 12 backward steps and 1 forward step over $D = 472392$ (at most 8 intervals per level) that the 7-interval union

  $$U_{17/24} = \left[0, \frac{1}{12}\right) \cup \left[\frac{1}{9}, \frac{11}{36}\right) \cup \left[\frac{4}{9}, \frac{2}{3}\right) \cup \left[\frac{25}{36}, \frac{3}{4}\right) \cup \left[\frac{7}{9}, \frac{19}{24}\right) \cup \left[\frac{5}{6}, \frac{8}{9}\right) \cup \left[\frac{11}{12}, 1\right)$$

  of total length $17/24 = 51/72 \approx 0.708333 > 70\%$ forces every $\xi \in [1, 2^W)$ to escape in at most $n \le W + 17$ **steps**.
- **Arbitrary K-Defect Simultaneous Valuation Lock Tower (`rungDefectCocycle_tower_identity`, `general_rung_multi_defect_valuation_tower`):** Along any `ValidBlockChain a_0 s u K` with $s(k) \ge g \ge 1$ for all $k < K$, defining $\mathrm{rungDefectCocycle}(g, s, m)$ recursively via $\mathrm{rungDefectCocycle}(g, s, m+1) = 3^{s(m)}\mathrm{rungDefectCocycle}(g, s, m) + 2^{\mathrm{blockSteps}(s,m)+g}3^{g-1}(3^{s(m)-g} - 2^{s(m)-g})$, every horizon $m \le K$ satisfies

  $$2^{\mathrm{blockSteps}(s,m)} \cdot I_g(b_m) + \mathrm{rungDefectCocycle}(g, s, m) = 3^{\mathrm{blockSteps}(s,m)} \cdot I_g(a_0),$$

  and at **every** defect index $m < K$ with $s(m) \ge g+1$ simultaneously, the seed a_0 satisfies $\mathrm{ExactTwoPowLock}\big(\mathrm{blockSteps}(s,m) + g,\ 3^{\mathrm{blockSteps}(s,m)}I_g(a_0) - \mathrm{rungDefectCocycle}(g,s,m)\big)$.

7.13 Pushing Into the Cantor Dust Below $8/57$: The k-Block Minimum Defect Spacing Law, Forced `0001001` Macro-Blocks on $[280/2059, 1/2)$, and the $(3, 3k+4)$ Perron–Fibonacci Hierarchy

What is the exact combinatorial and dynamical structure of the zero-width Cantor dust when ε drops below $\varepsilon^* = \frac{8}{57}$ through the homoclinic cascade $\varepsilon_k = \frac{N_k}{D_k}$ ($k \ge 1$)? - **Universal k-Block Minimum Defect Spacing Law (`cantor_dust_minimum_defect_spacing_law`):** For every $k \ge 1$ and every post-reset point $z \ge \varepsilon_k = \mathrm{homoclinicCycleMin}(k)$, after a 4-step defect block `0001` ($z_1 = T_4(z)$), every intermediate 3-step return $0 \le r < k$ stays strictly above the Rung-3 defect boundary $b_3 = \frac{4}{27}$:

$$\mathrm{rungReturnMapIter}\big(3, r, \mathrm{rungReturnMap}(4, z)\big) \ge \frac{388}{2059} > \frac{4}{27} = \mathrm{rungBoundary}(3),$$

and the k-th return satisfies $\mathrm{rungReturnMapIter}\big(3, k, \mathrm{rungReturnMap}(4, z)\big) \ge \varepsilon_k$. Consequently, on $[\varepsilon_k, 1/2)$, **every `0001` defect block is forced to be followed by at least k pure `001` blocks** before another `0001` block can occur—forming a composite macro-block $\texttt{0001(001)}^k$ of length $3k+4$. - **Direct 7-Step Macro-Block Lock on $[280/2059, 1/2)$ (`punctured_eps1_forced_seven_step_macro_block`):** On $[\varepsilon_1, 1/2) = [\frac{280}{2059}, \frac{1}{2})$, any `000` at step n forces the full 7-step carry word $(s_n, \ldots, s_{n+6}) = (0,0,0,1,0,0,1)$ (`0001001`), with $\frac{388}{2059} \le y_{n+4} < \frac{1}{4}$ and $\frac{280}{2059} \le y_{n+7} < \frac{1}{4}$—forbidding two consecutive 4-blocks (`0001000`). - **Kernel-Certified $(3, 3k+4)$ Perron–Fibonacci Cantor Cylinder Hierarchy & Characteristic Polynomial Root Squeeze (`cantorDust_perron_fibonacci_counts`, `perronCharPoly_sign_change_and_root_squeeze`):** Because the Cantor dust on $(\varepsilon_{k-1}, \varepsilon_k]$ is generated by

macro-blocks of lengths 3 (`001`) and $3k+4$ (`0001(001)`^{k}), its cylinder first-differences $\Delta C_k(d) = |C_{d+1}| - |C_d|$ satisfy the $(3, 3k+4)$ Perron–Fibonacci recurrence $\Delta C_k(d) = \Delta C_k(d-3) + \Delta C_k(d-(3k+4))$, kernel-certified (`by decide`) through depth $d = 14$ at $k = 1$ ($L_1 = 7$, `[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 14, 16, 19, 22]`) and at $k = 2$ ($L_2 = 10$, `[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 15, 17]`). Moreover, its characteristic polynomial $P_k(\lambda) = \lambda^{3k+4} - \lambda^{3k+1} - 1 = \lambda^{3k+1}(\lambda^3 - 1) - 1$ satisfies $P_k(1) = -1 < 0$ and $P_k(1 + 1/m) > 0$ whenever $m^2 \le 3(3k+1)$, squeezing the Perron growth rate λ_k into $(1, 1 + 1/m)$ and proving $\lambda_k \to 1^+$ (topological entropy $\log \lambda_k \to 0$) as $k \to \infty$. - **Universal Catapult-Death Band Ladder $C_m(\varepsilon) = \left[\frac{N_m}{3^{3m+4}}, \frac{N_m + 2^{3m+4}\varepsilon}{3^{3m+4}}\right)$ & Homoclinic Tangency (`universal_catapult_death_band_ladder`):** For every $m \ge 1$, the `0001` preimage $C_m(\varepsilon) := T_4^{-1}(D_{3,m}(\varepsilon))$ of the m-th Rung-3 death band has lower endpoint $\mathrm{catapultDeathLo}(m) = \frac{N_m}{3^{3m+4}}$ and upper endpoint $\mathrm{catapultDeathHi}(m, \varepsilon) = \frac{N_m + 2^{3m+4}\varepsilon}{3^{3m+4}}$, sharing the exact numerator $N_m = \mathrm{homoclinicCycleNum}(m)$ of $\varepsilon_m = \frac{N_m}{3^{3m+4} - 2^{3m+4}}$. Every point in $C_m(\varepsilon)$ falls into $[0, \varepsilon)$ in $3m+4$ steps (`0001 (001)^m`), $\mathrm{catapultDeathHi}(m, \varepsilon_m) = \varepsilon_m$ (with $\varepsilon < \mathrm{catapultDeathHi}(m, \varepsilon) \iff \varepsilon < \varepsilon_m$), and for $0 \le \varepsilon \le \frac{4}{27}$ the bands $C_m(\varepsilon)$ are pairwise disjoint inside $[\frac{280}{2187}, \frac{8}{57})$ with total measure $\frac{128}{1539}\varepsilon$—bringing the combined measure of $[8/57, 1/2) \cup \bigcup_{m=1}^{\infty} C_m(\frac{280}{2187})$ to $\frac{2492689}{6731586} \approx 0.370297$, or 74.06% **of** $[0, 1/2)$. - **Hidden Arithmetic Structure in the Cantor Dust: $\{3, 4\}$ Block Confinement on $[8/81, 1/2)$, Defect-to-Defect Recurrence $2^{3j_1+4}U_1 + 9 = 3^{3j_0+4}U_0$, Mod-8 Lock & Sparse-Defect Descent (`punctured_eight_eighty_firsts_forbids_four_zeros_and_short_gaps`, `cantor_dust_defect_to_defect_quotient_recurrence`, `cantor_dust_mod_eight_lock_and_sparse_descent`, `no_infinite_sparse_defect_cantor_dust_chain`):** On $[\frac{8}{81}, \frac{1}{2}) \approx [0.098765, 0.5)$ (80.25% of $[0, 1/2)$), `11`, `101`, and `0000` are forbidden, so every block has length $s(m) \in \{3, 4\}$. Across consecutive `0001` defects separated by j_0 and j_1 blocks of 3, the odd defect quotients U_0, U_1 ($19a_0 + 9 = 2^{3j_0+3}U_0$, $19b_{j_0+1} + 9 = 2^{3j_1+3}U_1$) satisfy:

$$2^{3j_1+4} \cdot U_1 + 9 = 3^{3j_0+4} \cdot U_0.$$

This forces $16 \mid (3^{3j_0+4}U_0 - 9)$ and $(3^{j_0}U_0) \bmod 8 = 1$—excluding $U_0 \bmod 8 \in \{5, 7\}$ and locking ($j_0 \bmod 2 = 0 \leftrightarrow U_0 \bmod 8 = 1$) and ($j_0 \bmod 2 = 1 \leftrightarrow U_0 \bmod 8 = 3$). Moreover, whenever defects are sparse ($3^{3j_r+4} \le 2^{3j_{r+1}+4}$, e.g. $2j_r + 2 \le j_{r+1}$), $U_{r+1} + 2 \le U_r$, unconditionally excluding all infinite sparse-defect chaotic chains in the Cantor dust! - **Complete Categorization of Cantor-Dust Families: Exact 3-Adic Valuation $v_3(U_r) = 2$, Non-Stationarity $V_{r+1} \ne V_r$ & Cumulative-Epoch Exclusion (`three_pow_ne_two_pow_add_one_of_four_le`, `cantor_dust_three_adic_valuation_and_normalized_recurrence`, `cantor_dust_multi_step_telescoping_and_epoch_descent`, `no_infinite_cumulative_sparse_epoch_cantor_dust_chain`):** Across consecutive defect steps, every U_r ($r \ge 1$) satisfies $U_r = 9V_r$ with $\gcd(V_r, 6) = 1$, reducing the transition to the coprime recurrence $2^{3j_2+4}V_2 + 1 = 3^{3j_1+4}V_1$ with ($j_2 \bmod 2 = 0 \leftrightarrow V_2 \bmod 3 = 2$) and ($j_2 \bmod 2 = 1 \leftrightarrow V_2 \bmod 3 = 1$). Reducing $3^A = 2^B + 1$ ($B \ge 4$) modulo 16 and 5 proves $V_{r+1} \ne V_r$ ($U_{r+1} \ne U_r$) for all $r \ge 1$. Furthermore, telescoping across $d \ge 1$ defects proves $2^{\sum_{t=1}^{d}(3j(r_0+t)+4)}U(r_0+d) < 3^{\sum_{t=0}^{d-1}(3j(r_0+t)+4)}U(r_0)$, so any oscillating chaotic path with infinitely many cumulative sparse epochs $[R_k, R_{k+1}]$ undergoes strict descent $U(R_{k+1}) + 2 \le U(R_k)$ and terminates! - **Exact 2-adic Valuation Determinism, Universal Mod-16 & Mod-128 Spacing Bijections, Zero-Branching Uniqueness & Dual $(2, 3)$-Adic CRT Renormalization (`defectNormalizedQuotient_exact_two_adic_valuation_and_forward_uniqueness`, `defectNormalizedQuotient_mod_sixteen_spacing_bijection`, `defectNormalizedQuotient_orbit_unique_of_V0`, `defectNormalizedQuotient_binary_crt_parametrization`, `defectNormalizedQuotient_cooldown_mod_128_and_three_adic_lock`, `defectNormalizedQuotient_renormalized_k_recurrence`):** Because V_{r+1} is odd, $2^{3j_{r+1}+4}$ is the exact power of 2 dividing $3^{3j_r+4}V_r - 1$, so (V_r, j_r) uniquely forces (j_{r+1}, V_{r+1}). Moreover, $3^{3j_r+4}V_r \equiv 1 \pmod{16}$ restricts $V_r \bmod 16 \in \{1, 3, 9, 11\}$ and bijectively determines $j_r \bmod 4$ ($0 \leftrightarrow 1$, $1 \leftrightarrow 3$, $2 \leftrightarrow 9$, $3 \leftrightarrow 11$); hence any two trajectories in $\{0, 1, 2, 3\}^{\mathbb{N}}$ (and in particular in the binary Cantor dust $\{1, 2\}^{\mathbb{N}}$) with $V(0) = V'(0)$ are identically equal at every step (**zero branching**). When $j_{r+1} \ge 1$ (on $[280/2059, 1/2)$), $3^{3j_r+4}V_r \equiv 1 \pmod{128}$ forces $V_r \bmod 128 = 35$ ($j_r = 1$) or 25 ($j_r = 2$), eliminating 126 of 128 residue classes; combining this 2-adic lock with the 3-adic lock $V_{r+1} \bmod 3^{3j_r+4} \in \{598, 1715\} \pmod{2187}$ or $\{11533, 23585\} \pmod{59049}$ yields the exact single-parameter CRT representation $(V_r, V_{r+1}) = (2^{3j_{r+1}+11}k_r + R_0,\ 128 \cdot 3^{3j_r+4}k_r + R_1)$ with composite V_{r+1}-moduli 279,936 and 7,558,272, and the exact two-step renormalized recurrence $3^{3j_r+4}k_r - 2^{3j_{r+2}+4}k_{r+1} = C(j_r, j_{r+1}, j_{r+2})$ across all 8 triples in $\{1, 2\}^3$. - **Level-2 and Level-3 CRT Renormalizations, Extinction of All $\ell_1 = 0$ Base Witnesses ($\ell_1 \ge 7$, $V_1 \ge 94{,}389{,}155$), and the General Multi-Step Homogeneous $(2, 3)$-Adic Tower Law (`defectNormalizedQuotient_level2_crt_parametrization`, `defectNormalizedQuotient_level3_renormalized_l_recurrence_j1_eq_one`, `defectNormalizedQuotient_level3_renormalized_l_recurrence_j1_eq_two`, `defectNormalizedQuotient_multi_step_homogeneous_tower`):** Solving the 8 Level-1 renormalized equations locks $(k_1, k_2, V_1, V_2, V_3)$ across $\{1, 2\}^3$ in terms of $\ell_1 = (V_1/2^{3j_2+11})/2^{3j_3+4}$, with pure 2-adic modulus $2^{3j_2+3j_3+15}$ on V_1, mixed $(2, 3)$-adic modulus $2^{3j_3+11} \cdot 3^{3j_1+4}$ on V_2, and 3-adic modulus $128 \cdot 3^{3j_1+3j_2+8}$ on V_3. Chaining three transitions $(j_1, j_2, j_3, j_4) \in \{1, 2\}^4$ yields the 16 Level-3 renormalized recurrences $3^{3j_1+4}\ell_1 - 2^{3j_4+4}\ell_2 = D(j_1, j_2, j_3, j_4)$, forcing $\ell_1 \pmod{2^{3j_4+4}} \in \{45, 756, 73, 910, 9, 78, 65, 864, 105, 558, 125, 412, 28, 571, 7, 980\}$ ($\ell_1 \ge 7 > 0$ in every case) so that every $\ell_1 = 0$ base witness dies at Step 4 and $V_1 \ge 94{,}389{,}155$ across all 16 4-tuples. Finally, for any spacing sequence $j : \mathbb{N} \to \mathbb{N}$ and any two solutions V, V', every origin r_0 and depth d satisfy

$2^{\text{defectStepSum}(\text{fun } t \Rightarrow j(r_0+1+t),d)}V(r_0+d)+3^{\text{defectStepSum}(\text{fun } t \Rightarrow j(r_0+t),d)}V'(r_0)=2^{\text{defectStepSum}(\text{fun } t \Rightarrow j(r_0+1+t),d)}V'(r_0+d)+3^{\text{defectStepSum}(\text{fun } t$
. - **Base-3^{3j_1+4} Carry-Digit Identity, Verified Backward 2-Adic Renormalization Tree Engine & Kernel-Certified Horizon Bounds Through Depth 10 ($V_0 \geq 7.729 \times 10^{21} > 2^{72}$ Across All $2{,}048$ Words in $\{1,2\}^{11}$) (`renormalized_three_adic_carry_digit_identity`, `backward_two_adic_residue_lift_of_stepOk`, `binary_cantor_dust_min_seed_of_certifyBinaryDustDepth`, `binary_cantor_dust_certified_horizon_bounds_up_to_depth_ten`):** Across any step $2^{3j_2+4}V_2+1=3^{3j_1+4}V_1$ with $V_1 \bmod 2^{e_2+(3j_2+4)}=R_1$ and $V_2 \bmod 2^{e_2}=R_2$, the renormalized parameters $\ell_1=V_1/2^{e_2+(3j_2+4)}$ and $\ell_2=V_2/2^{e_2}$ satisfy $\ell_2=3^{3j_1+4}\ell_1+K_2$ where $K_2=\lfloor 3^{3j_1+4}R_1/2^{e_2+(3j_2+4)}\rfloor < 3^{3j_1+4}$ is the exact base-3^{3j_1+4} carry digit ($3^{3j_1+4}R_1=2^{e_2+(3j_2+4)}K_2+2^{3j_2+4}R_2+1$), and $\ell_1=0 \wedge \ell_2=0 \iff 3^{3j_1+4}V_1<2^{e_2+(3j_2+4)}$. Combining `Nat.ModEq.cancel_left_of_coprime` on $\gcd(2^{e'},3^{3j_1+4})=1$ with a 6-step Hensel/Newton quadratic-doubling inverse `oddInvModTwoPow` and a recursive tree verifier `certifyBinaryDustDepth`, the Lean 4 kernel (`by decide`) certifies all $4{,}092$ binary words from $d=1$ through $d=10$, proving $V_0 \geq 590{,}121{,}613{,}202{,}017{,}059 > 2^{59}$ across all 512 words in $\{1,2\}^9$ ($d=8$) and $V_0 \geq 7{,}729{,}594{,}727{,}432{,}150{,}074{,}275 > 2^{72}$ across all $2{,}048$ words in $\{1,2\}^{11}$ ($d=10$, over $258{,}000\times$ larger than `recordHolderA0`). - **Universal $(2,3)$-Adic Quotient-Remainder Conservation Law, Multi-Step Quotient Duality, Depth-12 Binary Bounds ($V_0 > 1.13 \times 10^{27} > 2^{89}$), and Ternary $\{1,2,3\}$ Tree Certification (`one_step_quotient_and_remainder_conservation`, `universal_two_three_adic_quotient_duality`, `binary_cantor_dust_certified_depth_11_and_12`, `defectNormalizedQuotient_ternary_cooldown_mod_128_sieve`, `ternary_cantor_dust_min_seed_of_certifyTernaryDustDepth`, `ternary_cantor_dust_certified_horizon_bounds_up_to_depth_six`):** For any $A \geq 1, B \geq 2, M \geq 1$ and $BV_2+1=AV_1$, we prove $1 \leq V_1 \bmod (MB) < MB$, the exact quotient conservation $V_2/(MA)=V_1/(MB)$, and the exact remainder recurrence invariance $B(V_2 \bmod (MA))+1=A(V_1 \bmod (MB))$. Iterating across d defect steps proves that for every $r_0, d \in \mathbb{N}$ and $M \geq 1$, the 3-adic quotient of $V(r_0+d)$ equals the 2-adic quotient of $V(r_0)$:

$$\frac{V(r_0+d)}{M \cdot 3^{\text{defectStepSum}(\text{fun } t \Rightarrow j(r_0+t),\, d)}} \;=\; \frac{V(r_0)}{M \cdot 2^{\text{defectStepSum}(\text{fun } t \Rightarrow j(r_0+1+t),\, d)}}.$$

Pushing our binary tree verifier to $d=11$ ($4{,}096$ words) and $d=12$ ($8{,}192$ words) certifies $V(0) \geq 2{,}425{,}581{,}233{,}956{,}690{,}499{,}486{,}627 > 2^{81}$ ($d=11$) and $V(0) \geq 1{,}130{,}319{,}219{,}940{,}026{,}671{,}593{,}178{,}019 > 2^{89}$ ($d=12$); and extending the Mod-128 Cooldown Sieve to $j_r=3$ ($V_r \bmod 128 = 91$) certifies across all $2{,}187$ ternary words in $\{1,2,3\}^7$ ($d=6$) that $V(0) \geq 12{,}576{,}832{,}373{,}667 > 2^{43}$. - **Single-Exponent Homogeneous Reduction, Universal Archimedean Floor Invariant $\lfloor 2^{E_d}V(d)/3^{S_d}\rfloor = V(0)-1$, Cyclotomic Mod-7/13/19 Automata & Crossing the 100-Bit Barrier (`defectStepSum_shift_cancellation_and_single_exponent_identity`, `defectNormalizedQuotient_archimedean_floor_invariant`, `defectNormalizedQuotient_cyclotomic_mod_7_13_19_locks`, `mahler_min_seed_of_mahlerLeafOk`, `mahler_binary_cantor_dust_min_seed_of_certifyMahlerBinaryDustDepth`, `mahler_binary_cantor_dust_certified_bounds_up_to_depth_12`):** Telescoping $E_d=S_{d+1}-(3j(0)+4)=S_d+3(j(d)-j(0))$ collapses the d-step system to the single-exponent homogeneous equation $2^{S_{d+1}}V(d)+\Delta(j,d)=2^{3j(0)+4}\cdot 3^{S_d}V(0)$, where $4\Delta(j,d)+2^{S_d} \leq 3^{S_d}$. This pins $2^{E_d}V(d)$ into the narrow archimedean corridor $64 \cdot 3^{S_d}V(0) < 64 \cdot 2^{E_d}V(d)+3^{S_d}$ and makes $\lfloor 2^{E_d}V(d)/3^{S_d}\rfloor = V(0)-1$ a global constant of motion for all $d \geq 1$. Moreover, cyclotomic reduction modulo $7, 13, 19$ proves the Mod-7 forward automaton ($2^{3j_2+4} \equiv 2 \pmod 7$), the Mod-13 backward automaton ($3^{3j_1+4} \equiv 3 \pmod{13}$), and the Mod-19 Mahler integrality conservation law $(2^{3j_r+3}V_r) \bmod 19 = 1$ ($V_r \bmod 19 \in \{11,18,7\}$ for $j_r=1,2,3$). Combining the Mod-19 lock with our backward 2-adic tree via coprime cancellation lifts each leaf modulo $19 \cdot 2^{e_0}$ and certifies across all $8{,}192$ words in $\{1,2\}^{13}$ ($d=12$) that $V(0) \geq 6{,}644{,}210{,}551{,}001{,}288{,}379{,}891{,}372{,}825 > 2^{92}$ and $9 \cdot 2^{3j(0)+3}V(0) > 2^{101}$—crossing the **100-bit barrier** in the Lean 4 kernel. - **Advancing on the Dense Core $\{1,2\}^{\mathbb{N}}$ ($d=\infty$): Strict Doubling Growth, Repeated Defect-Subword Cancellation $2^E \mid (V(r_2)-V(r_1))$, and Unconditional Exclusion of Low-Complexity Dense-Core Trajectories (`dense_core_strict_doubling_and_monotonicity`, `dense_core_repeated_subword_divisibility_and_exp_bound`, `dense_core_repeated_subword_cutoff_and_complexity_exclusion`):** In the Dense Core $j(r) \in \{1,2\}$, $3^{3j(r)+4} \geq 2187 > 2048 \geq 2 \cdot 2^{3j(r+1)+4}$ forces $V(r+1) \geq 2V(r)+1 > V(r)$ ($r_1<r_2 \implies V(r_1)<V(r_2)$). Whenever a defect-spacing subword of length $L+1$ repeats at $r_1<r_2$ ($j(r_1+t)=j(r_2+t)$ for $0 \leq t \leq L$), subtracting the homogeneous tower identities cancels the inhomogeneous accumulator completely:

$$2^E\big(V(r_2+L)-V(r_1+L)\big) \;=\; 3^S\big(V(r_2)-V(r_1)\big),$$

forcing $2^E \mid (V(r_2)-V(r_1))$, $3^S \mid (V(r_2+L)-V(r_1+L))$, and $2^{7L} \leq 2^E < V(r_2)$. Combining this with Section 28's archimedean upper bound $2^{E_{r_2}}V(r_2) < 3^{S_{r_2}}V(0)$, the term-by-term bound $3^{S_{r_2}} \leq 2^{S_{r_2}+6r_2}$, and endpoint cancellation $S_{r_2} \leq E_{r_2}+3$ proves that for $V(0)<2^W$, any repeated subword of length $L+1$ at $r_1<r_2$ must satisfy $7L<6r_2+W+3$—unconditionally excluding all Dense-Core trajectories satisfying `prefixFactorCount j (L + 1) C ≤ C` with $6C+W+3 \leq 7L$! - **Further Advancing the Dense Core $\{1,2\}^{\mathbb{N}}$: Right-Special 2-Adic Valuation Locks, the 7-Bit Spectral Gap, Symmetric Bispecial $(2^S,3^S)$ Resonance, Parameterized (p,q) Cutoffs & Transcendental $\frac{7\log 2}{10\log(3/2)}$ Complexity Exclusion (`dense_core_all_V_odd_and_mod_four_divergence`, `dense_core_right_special_exact_two_adic_valuation_and_spectral_gap`, `dense_core_symmetric_bispecial_resonance`, `dense_core_parameterized_pq_cutoff_and_convergents`,**

`no_subcritical_log_threshold_dense_core`, `no_density_weighted_subcritical_log_threshold_dense_core`**):** Because the $+1$ in $2^{3j(r+1)+4}V(r+1)+1 = 3^{3j(r)+4}V(r)$ does not depend on $j(r+1)$, matching only $K+1$ symbols ($j(r_1+t) = j(r_2+t)$ for $0 \le t \le K$, with no match assumed at $K+1$) already cancels $+1$ at step $K \to K+1$:

$$2^{E_K}\Big(2^{3j(r_2+K+1)+4}V(r_2+K+1) - 2^{3j(r_1+K+1)+4}V(r_1+K+1)\Big) \;=\; 3^{S_{K+1}}\big(V(r_2)-V(r_1)\big),$$

yielding $2^{7(K+1)} \le 2^{E_K+7} < V(r_2)$ (a $+7$-bit upgrade at every length). When $j(r_1+K+1) \ne j(r_2+K+1)$ (a **right-special factor**), dividing the left difference by 128 leaves an odd integer (`8 V - V'` or `V - 8 V'`), locking $\nu_2(V(r_2)-V(r_1)) = E_K + 7$ (`((V(r_2) - V(r_1)) / 2^(E_K + 7)) % 2 = 1`) whereas $j(r_1+K+1) = j(r_2+K+1)$ forces $2^{E_K+14} \mid (V(r_2)-V(r_1))$—creating a **7-bit 2-adic spectral gap**:

$$j(r_1+K+1) = j(r_2+K+1) \iff 2^{E_K+8} \mid \big(V(r_2)-V(r_1)\big) \iff 2^{E_K+14} \mid \big(V(r_2)-V(r_1)\big).$$

For a middle match of length $K+1$ at (r_1+1, r_2+1) (a **bispecial** configuration), canceling $+1$ at both boundaries equalizes the 2-adic and 3-adic tower exponents to $S_{\text{mid}} = \sum_{t=0}^{K}(3j(r_1+1+t)+4)$:

$$2^{S_{\text{mid}}}\Big(2^{3j(r_2+K+2)+4}V(r_2+K+2) - 2^{3j(r_1+K+2)+4}V(r_1+K+2)\Big) \;=\; 3^{S_{\text{mid}}}\Big(3^{3j(r_2)+4}V(r_2) - 3^{3j(r_1)+4}V(r_1)\Big).$$

Finally, applying arbitrary rational power bounds $3^q \le 2^{p+q}$ ($q \ge 1$) and canceling qE_{r_2} via Section 28's shift identity proves $7qL < pS_{r_2} + q(W+3) \le 10pr_2 + q(W+3)$ for any length-L match, yielding concrete convergent cutoffs $119L < 100C + 17(W+3)$ at $(10, 17)$ and $287L < 240C + 41(W+3)$ at $(24, 41)$, proving strict suffix aperiodicity ($\forall r_1 < r_2, \exists t \le 6r_2 + W + 3, j(r_1+t) \ne j(r_2+t)$), and excluding all complexity slopes $10\kappa\log(3/2) < 7\log 2$ ($\kappa < \frac{7\log 2}{10\log(3/2)} \approx 1.196658$) and density-weighted slopes $\mu\kappa\log(3/2) < 7\log 2$ ($\kappa < \frac{7\log 2}{\mu\log(3/2)} \in [1.1967, 1.7095]$ for $\mu \in [7, 10]$). - **Top-Part & Overflow Isolation, Bottom-Part Constant of Motion, Universal Shifted-Horizon 98-Bit Corridor & Three-Occurrence Ultrametric Pigeonhole (`bottom_part_remainder_invariance_and_floor_conservation`, `dense_core_shifted_horizon_floor_and_ninety_eight_bit_corridor`, `dense_core_repeated_window_top_part_isolation_and_squeeze`, `dense_core_three_occurrence_ultrametric_pigeonhole`):** Splitting $V(r_0) = (M \cdot 2^{E_d})\ell + R_0$ and $V(r_0+d) = (M \cdot 3^{S_d})\ell + R_d$ into their **Top Part** $\ell = V(r_0)/(M \cdot 2^{E_d}) = V(r_0+d)/(M \cdot 3^{S_d})$ and **Bottom/Overflow Parts** (R_0, R_d) cancels ℓ identically from the single-exponent homogeneous equation, proving that the Top Part is 100% drift-free ($2^{E_d}(V(r_0+d) - R_d) = 3^{S_d}(V(r_0) - R_0)$) while the Bottom Part absorbs all $+1$ drift ($2^{S_{d+1}}R_d + \Delta_d = 2^{3j(r_0)+4} \cdot 3^{S_d}R_0$), satisfies $R_0 \ge 1$, and obeys its own Constant of Motion $\lfloor 2^{E_d}R_d / 3^{S_d} \rfloor = R_0 - 1$. Applying our Depth-12 tree certificates to every shifted origin $r_0 \in \mathbb{N}$ proves $V(r_0) > 2^{89}$ universally (and $V(r_0) \ge 6{,}644{,}210{,}551{,}001{,}288{,}379{,}891{,}372{,}825 > 2^{92}$ whenever $(2^{3j(0)+3}V(0)) \bmod 19 = 1$), locking the **Top 98 binary bits** of $V(r_0+d)$ to $\frac{3^{S_d}}{2^{E_d}}V(r_0)$ via $2^{98}(3^{S_d}V(r_0) - 2^{E_d}V(r_0+d)) < 3^{S_d}V(r_0)$ for all $r_0 \in \mathbb{N}$ and $d \ge 1$. Moreover, on any repeated window of length $K+1$ at $r_1 < r_2$, the Bottom Parts match identically ($V(r_1) \bmod 2^{E_K+7} = V(r_2) \bmod 2^{E_K+7}$), isolating $V(r_2) - V(r_1) = 2^{E_K+7}(\ell_2 - \ell_1)$ where m matching bits in the Top Parts ($2^m \mid (\ell_2 - \ell_1)$) tighten the cutoff to $q(7(K+1)+m) < pS_{r_2} + q(W+3)$; and whenever a length-$(K+1)$ factor appears at **three** indices $r_1 < r_2 < r_3$, binary Pigeonhole forces at least one pair across the 7-bit spectral gap ($128 \mid (\ell_b - \ell_a)$ and $2^{E_K+14} \mid (V(r_b) - V(r_a))$), gaining $+7$ bits and forcing $7q(K+2) < pS_{r_3} + q(W+3) \le 10pr_3 + q(W+3)$. - **The Infinite $(128^d, 2187^d)$ Sieve Tower, Explicit "4 in 16,384" and "8 in 2,097,152" Bijections & the Exact Prefix-Collision Biconditional (`normalizedDefectAcc_master_identity_and_infinite_sieve_tower`, `dense_core_depth2_four_in_16384_and_depth3_eight_in_2097152_sieves`, `dense_core_exact_prefix_collision_biconditional_and_mod_cutoff`):** Factoring $2^{3j(0)+4}$ out of `singleExpDefectAcc j d` defines `normalizedDefectAcc j d` ($\tilde{\Delta}(j,d)$) and establishes the factor-free master d-step identity $2^{E_d}V(d) + \tilde{\Delta}(j,d) = 3^{S_d}V(0)$. In the Dense Core, for every horizon $d = K+1 \ge 1$, this yields both the **forward "2^d in 128^d" power-of-2 sieve** $(3^{S_{K+1}}V(0)) \bmod 2^{7(K+1)} = \tilde{\Delta}(j, K+1) \bmod 2^{7(K+1)}$ (explicitly restricting $V(r) \bmod 16{,}384 \in \{931, 2851, 4889, 8601\}$ at $d = 2$ and $V(r) \bmod 2{,}097{,}152 \in \{17315, 230307, 1035043, 1952547, 2020121, 1872665, 582041, 1237401\}$ at $d = 3$) and the **backward "2^d in 2187^d" power-of-3 sieve** $(2^{E_{K+1}}V(K+1) + \tilde{\Delta}(j, K+1)) \bmod 3^{7(K+1)} = 0$. Finally, induction over the 7-bit spectral gap proves the **exact prefix-collision biconditional for all horizons** $K \in \mathbb{N}$:

$$(\forall t \le K,\ j(r_1+t) = j(r_2+t)) \iff 2^{E_K+7} \mid \big(V(r_2)-V(r_1)\big) \iff V(r_1) \bmod 2^{E_K+7} = V(r_2) \bmod 2^{E_K+7},$$

proving that any two steps $r_1 < r_2$ colliding modulo 2^{10K+7} must share their entire $(K+1)$-step prefix and obey $7q(K+1) < pS_{r_2} + q(W+3) \le 10pr_2 + q(W+3)$. - **Attacking the Conspiracy Barrier: Exact $\tilde{\Delta}$ Cocycle Composition, Frozen-Remainder Stabilization Beyond Bit W, and the Return-Loop Homoclinic Denominator Syzygy (`normalizedDefectAcc_cocycle_composition_law`, `dense_core_frozen_remainder_and_exact_valuation_conspiracy_law`, `dense_core_return_loop_homoclinic_denominator_syzygy`):** Across any split horizon $d_1 + m$, $\tilde{\Delta}(j, d_1+m)$ obeys the $(2,3)$-adic cocycle law $\tilde{\Delta}(j, d_1+m) = 3^{S_{\text{mid}}}\tilde{\Delta}(j, d_1) + 2^{E_{d_1}}\tilde{\Delta}(\text{shift}_{d_1}j, m)$. For any finite seed $1 \le V(0) < 2^W$, $3^{S_d}V(0) - \tilde{\Delta}(j,d) = 2^{E_d}V(d)$ has **exact 2-adic valuation** E_d at every step $d \in \mathbb{N}$ (

$\neg(2^{E_d+1} \mid (3^{S_d}V(0) - \tilde{\Delta}(j,d)))$), and once $W \le 7(K+1)$, $V(0)$ has exhausted all of its binary bits: its Top Part vanishes permanently ($V(0)/2^{E_K+7} = 0$), its sieve remainder freezes at $V(0)$ itself ($V(0) \bmod 2^{E_K+7} = V(0)$), and $V(K+1) < 3^{S_{K+1}}$. Moreover, whenever a length-$(K+1)$ subword at r_1 repeats at $r_1 + d$ ($d \ge 1$), combining the shifted master equation with Top-Part isolation eliminates $V(r_1+d)$ and yields the **Exact Homoclinic Return Syzygy**:

$$\left(3^{S_{\text{ret}}} - 2^{E_{\text{ret}}}\right)V(r_1) \;=\; \tilde{\Delta}_{\text{ret}} \;+\; 2^{E_{\text{ret}}+E_K+7}(\ell_2 - \ell_1),$$

which locks $V(r_1)$ modulo the super-modulus $2^{E_{\text{ret}}+E_K+7} \ge 2^{7(d+K+1)}$ via $((3^{S_{\text{ret}}} - 2^{E_{\text{ret}}})V(r_1)) \bmod 2^{E_{\text{ret}}+E_K+7} = \tilde{\Delta}_{\text{ret}} \bmod 2^{E_{\text{ret}}+E_K+7}$ and forces the coprime homoclinic return denominator $3^{S_{\text{ret}}} - 2^{E_{\text{ret}}}$ to divide $\tilde{\Delta}_{\text{ret}} + 2^{E_{\text{ret}}+E_K+7}(\ell_2 - \ell_1)$. - **Eliminating the Seed $V(0)$ Completely: Global Origin-to-Return Syzygy & the Seed-Free Two-Return-Loop Compatibility Obstruction (`dense_core_global_origin_return_loop_syzygy`, `dense_core_seed_free_two_return_loop_syzygy_and_congruence`):** Multiplying the return-loop syzygy by $2^{E_{r_1}}$ and substituting $2^{E_{r_1}}V(r_1) = 3^{S_{r_1}}V(0) - \tilde{\Delta}(j,r_1)$ together with the cocycle law eliminates $V(r_1)$ and yields the **Global Origin-to-Return Syzygy**:

$$C_A \cdot V(0) + D_A \;=\; \tilde{\Delta}(j, r_1+d) + 2^{M_A}(\ell_2 - \ell_1),$$

where $C_A = 3^{S_{r_1}}(3^{S_{\text{ret}}} - 2^{E_{\text{ret}}})$ is odd, $D_A = 2^{E_{\text{ret}}}\tilde{\Delta}(j,r_1)$, and $M_A = E_{r_1+d} + E_K + 7 \ge 7(r_1+d+K+1)$, locking $V(0)$ modulo 2^{M_A} directly via $(C_A \cdot V(0) + D_A) \bmod 2^{M_A} = \tilde{\Delta}(j, r_1+d) \bmod 2^{M_A}$. Finally, across any **two** return loops $A = (r_1, d, K)$ and $B = (r_1', d', K')$ in a Dense-Core trajectory, cross-multiplying by C_B and C_A eliminates $V(0)$ identically, and reducing modulo 2^{M_A} (for $M_A \le M_B$) eliminates both Top-Part quotients Q_A, Q_B as well—leaving a **purely combinatorial, V-free and W-free 2-adic congruence on the binary sequence $j \in \{1,2\}^{\mathbb{N}}$ alone**:

$$\left(C_B \cdot \tilde{\Delta}(j, r_1+d) + C_A \cdot D_B\right) \bmod 2^{M_A} \;=\; \left(C_A \cdot \tilde{\Delta}(j, r_1'+d') + C_B \cdot D_A\right) \bmod 2^{M_A}.$$

- Universal 3-Step Maximal Branching Loop Spawning, the Seed-Free Bit-M_A Flip Law & the Seed-Free Two-Loop 7-Bit Spectral Gap Dichotomy (`dense_core_every_three_step_window_spawns_maximal_branching_return_loop`, `dense_core_seed_free_two_loop_bit_flip_and_spectral_gap`): By binary Pigeonhole on $\{j(r_0), j(r_0+1), j(r_0+2)\} \subseteq \{1,2\}$ and strict suffix aperiodicity, **every** 3-step window $[r_0, r_0+2]$ in any Dense-Core trajectory automatically spawns a maximal branching return loop (r_1, d, K) with $r_1 \in \{r_0, r_0+1\}$, $d \in \{1,2\}$, $K \le 6(r_1+d) + W + 3$, $(\forall t \le K, j(r_1+t) = j(r_1+d+t)) \wedge j(r_1+K+1) \ne j(r_1+d+K+1)$, and odd Top-Part jump $Q_A \bmod 2 = 1$. Pitting any return loop $A = (r_1, d, K)$ against any later return loop $B = (r_1', d', K')$ at the bits above M_A proves that when Loop A branches at $K+1$ ($j(r_1+K+1) \ne j(r_1+d+K+1)$) and $M_A < M_B$, $\text{LHS}_{A,B}$ and $\text{RHS}_{A,B}$ are **forced to disagree at bit M_A** ($(\text{LHS}_{A,B} + 2^{M_A}) \bmod 2^{M_A+1} = \text{RHS}_{A,B} \bmod 2^{M_A+1}$ and $\text{LHS}_{A,B} \bmod 2^{M_A+1} \ne \text{RHS}_{A,B} \bmod 2^{M_A+1}$), whereas when Loop A stays together at $K+1$ ($j(r_1+K+1) = j(r_1+d+K+1)$) and $M_A + 7 \le M_B$, they agree for 7 more bits ($\text{LHS}_{A,B} \bmod 2^{M_A+7} = \text{RHS}_{A,B} \bmod 2^{M_A+7}$), establishing the **Seed-Free Two-Loop 7-Bit Spectral Gap Dichotomy**:

$$j(r_1+K+1) = j(r_1+d+K+1) \iff \text{LHS}_{A,B} \bmod 2^{M_A+1} = \text{RHS}_{A,B} \bmod 2^{M_A+1} \iff \text{LHS}_{A,B} \bmod 2^{M_A+7} = \text{RHS}_{A,B} \bmod 2^{M_A}$$

- Exact Constant-Run ($2059V-1$, $58025V-1$) & Alternating-Run ($129{,}009{,}091V - \{60073, 2315\}$) Valuation Locks Across Every 3-Step Window (`dense_core_constant_run_length_2059_and_58025_valuation_locks`, `dense_core_alternating_run_129009091_valuation_locks`): Evaluating Section 33's return-loop syzygy at the local periods $d=1$ and $d=2$ proves that at **every single step** r_1 of a Dense-Core trajectory, the duration of the current constant or alternating run is 100% locked to the exact 2-adic valuation of one of four explicit linear forms: - **Run of 1s ($2059 = 3^7 - 2^7$):** $\forall t \le K+1,\ j(r_1+t) = 1 \implies 2059V(r_1) = 1 + 2^{7(K+2)}(\ell_2 - \ell_1)$, with $\nu_2(2059V(r_1) - 1) = 7(K+2)$ when the run ends at $j(r_1+K+2) = 2$. - **Run of 2s ($58025 = 3^{10} - 2^{10}$):** $\forall t \le K+1,\ j(r_1+t) = 2 \implies 58025V(r_1) = 1 + 2^{10(K+1)+7}(\ell_2 - \ell_1)$, with $\nu_2(58025V(r_1) - 1) = 10(K+1) + 7$ when the run ends at $j(r_1+K+2) = 1$. - **Alternating `1, 2, 1, ...` Run ($129{,}009{,}091 = 3^{17} - 2^{17},\ \tilde{\Delta}_{\text{ret}} = 60{,}073$):** $129009091V(r_1) = 60073 + 2^{E_K+24}(\ell_2 - \ell_1)$, with $\nu_2(129009091V(r_1) - 60073) = E_K + 24$ when the alternation breaks at $j(r_1+K+1) \ne j(r_1+2+K+1)$. - **Alternating `2, 1, 2, ...` Run ($129{,}009{,}091 = 3^{17} - 2^{17},\ \tilde{\Delta}_{\text{ret}} = 2{,}315$):** $129009091V(r_1) = 2315 + 2^{E_K+24}(\ell_2 - \ell_1)$, with $\nu_2(129009091V(r_1) - 2315) = E_K + 24$ when the alternation breaks at $j(r_1+K+1) \ne j(r_1+2+K+1)$. - **Exact Run-Compression & Macro-Transition Laws ($8 \cdot 2059V = 3^{7L}U_1 + 1$ and $58025V = 3^{10L}U_2 + 8$) Across All Run Lengths $L = m+1 \ge 1$ (`dense_core_one_run_compression_and_macro_transition`, `dense_core_two_run_compression_and_macro_transition`):** Multiplying the one-step recurrence by the homoclinic denominators $2059 = 3^7 - 2^7 = 29 \times 71$ and $58025 = 3^{10} - 2^{10} = 25 \times 11 \times 211$ homogenizes every interior run transition ($2^{7m}(2059V(r_1+m) - 1) = 3^{7m}(2059V(r_1) - 1)$ on runs of 1s; $2^{10m}(58025V(r_1+m) - 1) = 3^{10m}(58025V(r_1) - 1)$ on runs of 2s) and compresses **any maximal run of**

length $L = m+1 \geq 1$ (including singletons $L=1$) into a single macro-step connecting the start of the run $V(r_1)$ to the start of the next run $V(r_1+L)$: - **Run of $L \geq 1$ ones switching to 2:** $\nu_2(2059V(r_1)-1) = 7L$ ($U_1 = (2059V(r_1)-1)/2^{7L}$ is odd), and

$$8 \cdot 2059 \cdot V(r_1+L) \;=\; 3^{7L} \cdot U_1 + 1,$$

forcing $(3^{7L}U_1+1) \bmod 8 = 0$, $(3^{7L}U_1+1) \bmod 2059 = 0$, and $(8 \cdot 2059V(r_1+L)) \bmod 3^{7L} = 1$. - **Run of $L \geq 1$ twos switching to 1:** $\nu_2(58025V(r_1)-1) = 10m+7 = 10L-3$ ($U_2 = (58025V(r_1)-1)/2^{10m+7}$ is odd), and

$$58025 \cdot V(r_1+L) \;=\; 3^{10L} \cdot U_2 + 8,$$

forcing $(3^{10L}U_2+8) \bmod 58025 = 0$ and $(58025V(r_1+L)) \bmod 3^{10L} = 8$. - **Bypassing Exponential Word Complexity (2^L) via the Universal $4(L+1)$ Anagram Pigeonhole Theorem & Partial Drift-Cancellation Cutoff (`dense_core_anagram_weight_decomposition_and_universal_pigeonhole`, `dense_core_anagram_collision_syzygy_and_drift_cancellation_cutoff`):** Although a sequence $j \in \{1,2\}^{\mathbb{N}}$ can have up to 2^L distinct subwords of length L, its step-sum $S_L(r) = 7L + 3 \cdot \mathrm{defectWindowTwoCount}(j,r,L)$ depends only on the count of 2s in $\{0,\dots,L\}$ ($L+1$ values), and by the endpoint shift identity $E_L(r) + 3j(r) = S_L(r) + 3j(r+L)$, the combined weight-and-boundary classifier $\mathrm{defectAnagramClass}(j,L,r) = 4 \cdot \mathrm{defectWindowTwoCount}(j,r,L) + 2(j(r)-1) + (j(r+L)-1)$ takes at most $4(L+1)$ values ($< 4(L+1)$). By the Finite Pigeonhole Principle, **every** Dense-Core sequence $j \in \{1,2\}^{\mathbb{N}}$ (with zero complexity assumptions!) has an anagram pair $r_1 < r_2 \leq 4(L+1)$ with $j(r_1) = j(r_2)$, $j(r_1+L) = j(r_2+L)$, $S_L(r_1) = S_L(r_2) = S$, and $E_L(r_1) = E_L(r_2) = E$, satisfying:

$$2^E\big(V(r_2+L) - V(r_1+L)\big) + \tilde{\Delta}_2 \;=\; 3^S\big(V(r_2) - V(r_1)\big) + \tilde{\Delta}_1.$$

Whenever $\tilde{\Delta}_1 \bmod 2^m = \tilde{\Delta}_2 \bmod 2^m$ for $m \leq E$, coprimality forces $2^m \mid (V(r_2) - V(r_1))$ and $qm < pS_{r_2} + q(W+3) \leq 40p(L+1) + q(W+3)$ ($17m < 400(L+1) + 17(W+3)$ at $(p,q) = (10,17)$). - **The $58025 - 8 \cdot 2059 = 3^7 \cdot 19$ Homoclinic Determinant Identity, Macro-Run Interface Renormalization, Three-Run Wave Elimination & Strict Well-Founded Descent $W_4 + 2 \leq W_2$ Inside the Dense Core (`dense_core_one_to_two_run_interface_syzygy_and_ten_mul_valuation_lock`, `dense_core_two_to_one_run_interface_and_wave_elimination`, `dense_core_two_one_two_wave_elimination_and_strict_W_descent`, `dense_core_multi_wave_telescoping_and_epoch_W_descent`):** Eliminating the boundary state $V(r_2)$ between a run of $L_1 = m_1 + 1$ ones and a run of $L_2 = m_2 + 1$ twos via the identity $58025 - 8 \cdot 2059 = (3^{10} - 2^{10}) - 2^3(3^7 - 2^7) = 3^7(3^3 - 2^3) = 2187 \cdot 19 = 41{,}553$ proves that every post-initial run quotient is divisible by $3^7 = 2187$ ($U_k = 2187W_k$, $W_k \bmod 2 = 1$) and collapses the Dense Core to the macro-run interface system:

$$\begin{aligned} 58025 \cdot 3^{7m_1}U_1 + 19 \;&=\; 2059 \cdot 2^{10L_2}W_2 \qquad \big(\nu_2(58025 \cdot 3^{7m_1}U_1 + 19) = 10L_2\big), \\ 58025 \cdot 2^{7L_3}W_3 + 19 \;&=\; 2059 \cdot 3^{10L_2}W_2 \qquad \big(\nu_2(2059 \cdot 3^{10L_2}W_2 - 19) = 7L_3,\ 3^{10L_2} \mid (58025 \cdot 2^{7L_3}W_3 + 19)\big). \end{aligned}$$

Eliminating W_2 across a $1 \to 2 \to 1$ wave and dividing by $58025 = 3^{10} - 2^{10}$ yields $3^{10L_2+7m_1}U_1 + 19\frac{3^{10L_2}-2^{10L_2}}{58025} = 2^{10L_2+7L_3}W_3$, while eliminating W_3 across a $2 \to 1 \to 2$ wave and dividing by $2059 = 3^7 - 2^7$ flips the $+19$ term to the 2-adic side:

$$2^{7L_3+10L_4}W_4 \;+\; 19 \cdot \left(\frac{3^{7L_3} - 2^{7L_3}}{2059}\right) \;=\; 3^{10L_2+7L_3}W_2 \;\;\Longrightarrow\;\; 2^{7L_3+10L_4}W_4 \;<\; 3^{10L_2+7L_3}W_2.$$

Consequently, even though $V(r)$ strictly grows at every step of the Dense Core, **the renormalized 2-run odd quotient W undergoes strict well-founded descent $W_4 + 2 \leq W_2$ inside the Dense Core** whenever $3^{10L_2+7L_3} \leq 2^{7L_3+10L_4}$ (such as on the 4-step pattern `2, 1, 2, 2` where $L_2 = 1, L_3 = 1, L_4 \geq 2$ and $3^{17} < 2^{27}$). Chaining $N \geq 1$ consecutive waves telescopes all interior 2-run lengths out of the exponent difference ($2^{S_{\mathrm{wave}}(N)+10L_2(N)}W(N) < 2^{10L_2(0)}3^{S_{\mathrm{wave}}(N)}W(0)$), unconditionally excluding any Dense-Core trajectory with infinitely many wave-descent epochs! - **Same-Middle-Run Wave Cancellation, Non-Descent Run-Growth Speed Limit ($17L_4 < 27L_2 + 7L_3$) & the $(L_2 = 1, L_3 = 1)$ Alternation-vs.-Descent Dichotomy (`dense_core_same_middle_run_wave_cancellation_and_divisibility`, `dense_core_non_descent_run_growth_speed_limit_and_212_dichotomy`):** Because the inhomogeneous term $19 \cdot (3^{7L_3} - 2^{7L_3})/2059$ in the $2 \to 1 \to 2$ wave equation depends **only on the middle 1-run length L_3** (and $19 \cdot (3^{10L_2} - 2^{10L_2})/58025$ in the $1 \to 2 \to 1$ wave equation depends **only on L_2**), any two waves sharing the same middle run length cancel their $+19Q$ drift identically:

$$3^{7L_3} \cdot (3^{10L_2}W_2) + 2^{7L_3} \cdot (2^{10L_4'}W_4') \;=\; 3^{7L_3} \cdot (3^{10L_2'}W_2') + 2^{7L_3} \cdot (2^{10L_4}W_4),$$

forcing $2^{7L_3} \mid (W_2' - W_2)$ when $L_2 = L_2'$ and both $2^{7L_3+10L_4} \mid (W_2' - W_2)$ and $3^{10L_2+7L_3} \mid (W_4' - W_4)$ when $(L_2, L_4) = (L_2', L_4')$. Contrapositively, whenever a wave **avoids** strict descent ($W_2 \leq W_4$), every rational power bound $3^q \leq 2^{p+q}$ ($q \geq 1$) forces the run-growth speed limit:

$$10qL_4 \;<\; 10(p+q)L_2 + 7pL_3 \qquad \big(17L_4 < 27L_2 + 7L_3 \text{ at } (p,q) = (10,17)\big),$$

and across $N \geq 1$ waves with $W(0) \leq W(N)$, $10qL_2(N) < 10qL_2(0) + p \cdot \mathrm{waveStepSum}(L_2, L_1, N)$. Specializing $17L_4 < 27L_2 + 7L_3$ to $L_2 = 1, L_3 = 1$ ($m_2 = 0, m_3 = 0$) proves the $(L_2 = 1, L_3 = 1)$

Alternation-vs.-Descent Dichotomy: either $m_4 = 0$ ($L_4 = 1$, extending the alternating `2, 1, 2` run) or $W_4 + 2 \le W_2$ (strict descent of W)! - **Universal Macro-Run Mod-16/128/1024 Bijections, Disjoint Mod-8 Sectors ($W_{\text{two}} \bmod 8 = 1$ vs. $W_{\text{one}} \bmod 8 \in \{5, 7\}$), 4x/8x Accelerated Wave Descent ($W_4 + 8 \le W_2$) & Singleton-Bridge Cross-Wave $(2, 3)$-Power Balance** (`dense_core_macro_run_mod_16_128_1024_bijections_and_step_8_descent`, `dense_core_singleton_bridge_wave_identities_and_cross_wave_power_balance`): Reducing the two macro-run interface equations modulo $8, 16, 128$, and 1024 proves that every 2-run quotient satisfies $(2059 \cdot 3^{10L_2} W_2) \bmod 128 = 19$ and $W_2 \bmod 8 = 1$ **universally** ($W_2 \bmod 16 \in \{1, 9\}$ bijectively determining $L_2 \bmod 2 \in \{1, 0\}$), whereas every 1-run quotient satisfies $(58025 \cdot 3^{7L_3} W_3 + 19) \bmod 1024 = 0$ and $W_3 \bmod 8 \in \{5, 7\}$ ($W_3 \bmod 16 \in \{5, 7, 13, 15\}$ bijectively determining $L_3 \bmod 4 \in \{0, 3, 2, 1\}$). Because the residue sectors of 2-run and 1-run quotients modulo 8 are completely disjoint, $W_2 \neq W_3$ always, and any descending wave $W_4 < W_2$ drops by **at least** 8 ($W_4 + 8 \le W_2$, or $W_4 + 16 \le W_2$ when $L_2 \equiv L_4 \pmod 2$). Finally, on singleton-bridge waves ($L_2 = 1$ in $1^{L_1} \to 2^1 \to 1^{L_3}$ and $L_3 = 1$ in $2^{L_2} \to 1^1 \to 2^{L_4}$), both inhomogeneous quotients equal 1, reducing both wave equations to $+19$ ($3^{7L_1+10} W_1 + 19 = 2^{7L_3+10} W_3$ and $2^{10L_4+7} W_4 + 19 = 3^{10L_2+7} W_2$) and yielding the pure $(2, 3)$-power balance:

$$2^{7L_3+10} W_3 \;+\; 2^{10L_4+7} W_4 \;=\; 3^{7L_1+10} W_1 \;+\; 3^{10L_2+7} W_2,$$

forcing $2^{17} \mid (3^{7L_1+10} W_1 + 3^{10L_2+7} W_2)$ and $3^{17} \mid (2^{7L_3+10} W_3 + 2^{10L_4+7} W_4)$. - **Pure Power-of-3 Macro-Run Sieves ($W_2 \bmod 128 \in \{113, 9\}$, $W_3 \bmod 1024 \in \{143, 141\}$), the $1/6{,}377{,}292$ Macro-Wave Corridor & the Macro-Wave Constant of Motion** $\lfloor 2^{7L_3+10L_4} W_4 / 3^{10L_2+7L_3} \rfloor = W_2 - 1$ (`dense_core_macro_run_pure_power_three_sieves_and_exact_residues`, `dense_core_macro_wave_archimedean_floor_invariant_and_corridor`): Substituting $2059 \equiv 3^7 \pmod{128}$ and $58025 \equiv 3^{10} \pmod{1024}$ into the two macro-run interface equations eliminates all composite factors and yields **pure power-of-3 sieves**:

$$(3^{10L_2+7} \cdot W_2) \bmod 128 \;=\; 19, \qquad (3^{7L_3+10} \cdot W_3 + 19) \bmod 1024 \;=\; 0,$$

which lock $W_2 \bmod 128 = 113$ ($L_2 = 1$) vs. 9 ($L_2 = 2$) and $W_3 \bmod 1024 = 143$ ($L_3 = 1$) vs. 141 ($L_3 = 2$). Simultaneously, in the $2^{L_2} \to 1^{L_3} \to 2^{L_4}$ wave equation $2^{7L_3+10L_4} W_4 + 19 Q_{L_3} = 3^{10L_2+7L_3} W_2$, the inhomogeneous drift $19 Q_{L_3}$ is bounded by $108 \cdot (19 Q_{L_3}) < 3^{7L_3} \le 3^{10L_2+7L_3} / 59049$, squeezing the wave transition into a microscopic archimedean corridor of relative width $< \frac{1}{108 \cdot 59049} = \frac{1}{6{,}377{,}292}$:

$$6{,}377{,}292 \cdot 3^{10L_2+7L_3} \cdot W_2 \;<\; 6{,}377{,}292 \cdot 2^{7L_3+10L_4} \cdot W_4 \;+\; 3^{10L_2+7L_3},$$

and proving that for **all** $W_2 \ge 1$ of arbitrary size, the macro-wave integer floor satisfies the exact **Macro-Wave Constant of Motion**:

$$\left\lfloor \frac{2^{7L_3+10L_4} \cdot W_4}{3^{10L_2+7L_3}} \right\rfloor \;=\; W_2 - 1.$$

- **3-Adic Mirror Sieves ($\bmod 3^7 = 2187$ and $\bmod 3^{10} = 59049$), Fermat-Prime Mod-11 & Mod-25 Homoclinic Sieves ($W_2 \bmod 11 = 4$, $W_2 \bmod 4400 \in \{609, 4041\}$), Mod-39600 6-Way Bijection, Full-Power 3-Adic Remainders, $+19$-Cancellation Homogeneous Difference Identities & the 4-Run W_2-Free Cross-Wave Syzygy** (`dense_core_three_adic_mirror_sieves_and_mod_24_144_bijections`, `dense_core_dual_wave_full_three_adic_remainders_and_one_run_floor_invariant`, `dense_core_plus_19_cancellation_homogeneous_difference_and_run_expansion_laws`, `dense_core_fermat_homoclinic_sieves_and_mod_4400_39600_bijections`, `dense_core_four_run_W2_free_cross_wave_syzygy`): 1. **3-Adic Mirror Pure Power-of-2 Sieves & Fermat Homoclinic Sieves:** Using $2059 + 128 = 2187 = 3^7$ and $58025 + 1024 = 59049 = 3^{10}$ in the macro-run interface equations yields the 3-adic mirror sieves $(2^{10L_2+7} W_2 + 19) \bmod 2187 = 0$ ($W_2 \bmod 2187 \in \{340, 829\}$ for $L_2 = 1, 2$) and $(2^{7L_3+10} W_3) \bmod 59049 = 19$ ($W_3 \bmod 59049 \in \{23717, 45856\}$ for $L_3 = 1, 2$), locking $W_2 \bmod 24 = 1$ universally, a 6-way bijection between $L_2 \bmod 6$ and $W_2 \bmod 144 \in \{97, 73, 49, 25, 1, 121\}$, and a 6-way bijection between $L_3 \bmod 6$ and $W_3 \bmod 9 \in \{2, 1, 5, 7, 8, 4\}$. Simultaneously, reducing modulo $58025 = 25 \times 11 \times 211$ and $2059 = 29 \times 71$ locks $W_2 \bmod 58025 \in \{52034, 51616\}$ ($L_2 = 1, 2$) and $W_3 \bmod 2059 \in \{117, 17\}$ ($L_3 = 1, 2$), and by Fermat's Little Theorem ($3^{10} \equiv 1 \pmod{11}$ and $3^{10} \equiv -1 \pmod{25}$) locks $W_2 \bmod 11 = 4$ **and** $W_2 \bmod 88 = 81$ **universally across all** $L_2 \ge 1$, restricts $W_2 \bmod 4400 \in \{609, 4041\}$ ($L_2 \bmod 2 = 1 \leftrightarrow 609$, $L_2 \bmod 2 = 0 \leftrightarrow 4041$), locks $W_2 \bmod 264 = 169$ and $W_2 \bmod 13200 \in \{9409, 12841\}$, and establishes a 6-way bijection between $L_2 \bmod 6$ and $W_2 \bmod 39600 \in \{35809, 39241, 9409, 12841, 22609, 26041\}$—accelerating wave descent to $W_4 + 88 \le W_2$ (unconditional), $W_4 + 264 \le W_2$ (post-initial), $W_4 + 4400 \le W_2$ ($L_2 \equiv L_4 \pmod 2$), $W_4 + 13200 \le W_2$ ($L_2 \equiv L_4 \pmod 2$, post-initial), and $W_4 + 39600 \le W_2$ ($L_2 \equiv L_4 \pmod 6$, a 19,800× **speedup**)! 2. **Unconditional Full-Power 3-Adic Remainders & Dual $1 \to 2 \to 1$ Floor Invariant:** For all $U_1, W_2, W_3, W_4 \in \mathbb{N}$, Euclidean division yields the quotient-independent full-power 3-adic remainders $(2^{7L_3+10L_4} W_4) \bmod 3^{10L_2+7L_3} = 3^{10L_2+7L_3} - 19 Q_{L_3}$ and $(2^{10L_2+7L_3} W_3) \bmod 3^{10L_2+7m_1} = 19 R_{L_2}$, together with the $1/3053$ dual corridor and exact floor invariant $\lfloor 2^{10L_2+7L_3} W_3 / 3^{10L_2+7m_1} \rfloor = U_1$. 3. **$+19$-Cancellation Homogeneous Difference Identities, Two-Sided $(2, 3)$-Power Wave Corridors, Mandatory Run Expansion & 4-Run W_2-Free Cross-Wave Syzygy:** Directly subtracting consecutive interface equations cancels $+19$ identically and yields:

$$2^{10L_4} \cdot W_4 = 3^{10L_2} \cdot W_2 + 58025 \cdot Q_{L_3} \cdot W_3, \qquad 2^{7L_3} \cdot W_3 = 3^{7m_1} \cdot U_1 + 2059 \cdot R_{L_2} \cdot W_2,$$

proving $(58025Q_{L_3}) \mid (2^{10L_4}W_4 - 3^{10L_2}W_2)$, $(2059R_{L_2}) \mid (2^{7L_3}W_3 - 3^{7m_1}U_1)$, and the two-sided $(2,3)$-power wave corridor $3^{10L_2}W_2 < 2^{10L_4}W_4$ and $2^{7L_3+10L_4}W_4 < 3^{10L_2+7L_3}W_2$. Consequently, whenever $2^{10L_4} \le 3^{10L_2}$ (including $L_4 \le L_2$), $W_2 < W_4$ strictly ($W_2 + 8 \le W_4$), and whenever $W_4 \le W_2$, the run length must expand via $3^{10L_2} < 2^{10L_4}$ ($3L_2 < 2L_4$, and $(p'+q')L_2 < q'L_4$ for all $2^{p'+q'} \le 3^{q'}$). Finally, eliminating W_2 between the two homogeneous difference identities yields the **4-Run W_2-Free Cross-Wave Syzygy**:

$$2059 \cdot R_{L_2} \cdot 2^{10L_4} \cdot W_4 + 3^{10L_2+7m_1} \cdot U_1 \;=\; \Big(3^{10L_2+7L_3} + 2^{10L_2+7L_3} - 2^{10L_2} \cdot 3^{7L_3}\Big) \cdot W_3.$$

- Exact 2-Adic Wave Valuations & Short-Run Euclidean Laws, the Symmetric $(2^{S_N}, 3^{S_N})$ Multi-Wave Master Identity, the $275 \cdot 2^{S_N+10L_{\text{end}}+4}$ Super-Divisibility Law & the $+12$-Bit Macro-Wave Archimedean Repetition Cutoff

(`dense_core_wave_exact_two_adic_valuations_and_short_run_euclidean_laws`, `dense_core_symmetric_multi_wave_master_identity_and_boundary_independence`, `dense_core_multi_wave_interior_match_cancellation_and_275_super_divisibility`, `dense_core_macro_wave_archimedean_repetition_cutoff`): 1. **Exact 2-Adic Wave Valuations & Short-Run 2-Adic Euclidean Quotients/Remainders:** Because $W_4 \bmod 2 = 1$ and $W_3 \bmod 2 = 1$, $\nu_2(3^{10L_2+7L_3}W_2 - 19Q_{L_3}) = 7L_3 + 10L_4$ and $\nu_2(3^{10L_2+7m_1}U_1 + 19R_{L_2}) = 10L_2 + 7L_3$. In the short-run regime ($19Q_{L_3} < 2^{7L_3+10L_4}$, which holds for all $L_3 \le 4$ and whenever $3^{7L_3} \le 108 \cdot 2^{7L_3+10L_4}$; and $19R_{L_2} < 2^{10L_2+7L_3}$, which holds for all $L_2 \le 3$ and whenever $3^{10L_2} \le 3053 \cdot 2^{10L_2+7L_3}$), Euclidean division yields the unwrapped 2-adic quotients and remainders:

$$\left\lfloor \frac{3^{10L_2+7L_3} \cdot W_2}{2^{7L_3+10L_4}} \right\rfloor = W_4, \qquad \big(3^{10L_2+7L_3} \cdot W_2\big) \bmod 2^{7L_3+10L_4} = 19 \cdot Q_{L_3},$$

$$\left\lfloor \frac{3^{10L_2+7m_1} \cdot U_1}{2^{10L_2+7L_3}} \right\rfloor = W_3 - 1, \qquad \big(3^{10L_2+7m_1} \cdot U_1\big) \bmod 2^{10L_2+7L_3} = 2^{10L_2+7L_3} - 19 \cdot R_{L_2}.$$

2. **Symmetric $(2^{S_N}, 3^{S_N})$ Multi-Wave Master Identity & Boundary-Run Independence:** Across $N+1$ consecutive waves $(0 \dots N)$, the interior wave step-sum $S_N = \text{waveInteriorExp}(L_2, L_1, N)$ and interior drift cocycle $\text{Acc}_N = \text{waveInteriorAcc}(L_2, L_1, N)$ depend **only on the interior** $(2N+1)$ **runs** $(L_1(0 \dots N), L_2(1 \dots N))$—with **zero dependence on** $L_2(0)$ **or** $L_2(N+1)$—and satisfy the symmetric master equation:

$$2^{S_N} \cdot \Big(2^{10L_2(N+1)} \cdot W(N+1)\Big) \;+\; 19 \cdot \text{Acc}_N \;=\; 3^{S_N} \cdot \Big(3^{10L_2(0)} \cdot W(0)\Big).$$

3. **Multi-Wave Interior-Match Cancellation, Right-Special Valuation Locks & $275 \cdot 2^{S_N+10L_{\text{end}}+4}$ Super-Divisibility:** Whenever two $(N+1)$-wave blocks share their interior $(2N+1)$ runs, $19 \cdot \text{Acc}_N$ cancels identically. When $L_2(0) = L_2'(0)$ and $W(0) \le W'(0)$, right-special branching $L_2(N+1) < L_2'(N+1)$ locks $\nu_2(W'(0) - W(0)) = S_N + 10L_2(N+1)$, whereas matching terminal runs $L_2(N+1) = L_2'(N+1) = L_{\text{end}}$ (where $4400 = 16 \times 275$ divides $W'(N+1) - W(N+1)$ by Section 51) force:

$$\Big(275 \cdot 2^{S_N+10L_{\text{end}}+4}\Big) \;\Big|\; \big(W'(0) - W(0)\big) \qquad \big(\text{and } W(0) + 275 \cdot 2^{S_N+10L_{\text{end}}+4} \le W'(0) \text{ when } W(0) < W'(0)\big).$$

4. **The $+12$-Bit Macro-Wave Archimedean Repetition Cutoff:** Combining $2^{S_N+10L_{\text{end}}+12} < 275 \cdot 2^{S_N+10L_{\text{end}}+4} + W_a \le W(M+1)$ (since $275 \times 16 = 4400 > 4096 = 2^{12}$) with the $(M+1)$-wave origin bound $2^{S_M+10L_2(M+1)}W(M+1) < 3^{S_M+10L_2(0)}2^{W_0}$ and raising to $q \ge 1$ ($3^q \le 2^{p+q}$) **cancels** $q \cdot S_M$ **identically** and proves:

$$q \cdot \big(S_N + 10L_{\text{end}} + 10L_2(M+1) + 12\big) \;<\; p \cdot S_M + 10(p+q)L_2(0) + qW_0,$$

yielding
$17(17N + 19 + 10L_{\text{end}} + 10L_2(M+1)) \le 17(S_N + 10L_{\text{end}} + 10L_2(M+1) + 12) < 10S_M + 270L_2(0) + 17W_0$
at $(p,q) = (10,17)$! **- Telescopic Cross-Defect Laws ($M_W = 16{,}243{,}286{,}400$ and $M_U = 124{,}499{,}856{,}384$), Path-Independent Mirror Collapses of `waveInteriorAcc` and `dualWaveInteriorAcc`, Universal Same-Run-Length CRT Moduli (8.12 Billion$\times$ Wave Descent), the Upgraded $58025 \cdot 2^{S_N+10L_{\text{end}}+7}$ ($+22$-Bit) Super-Divisibility Cutoff, Multi-Wave Half-Step ($W(0) \to U(N+1)$) Elimination & the Dual $(2^{\tilde{S}_N}, 3^{\tilde{S}_N})$ Multi-Wave Master Identity**

(`dense_core_waveInteriorAcc_telescopic_cross_defect_and_mirror_collapses`, `dense_core_same_run_length_crt_moduli_and_billion_step_descent`, `dense_core_multi_wave_58025_super_divisibility_and_22_bit_repetition_cutoff`, `dense_core_dual_one_run_multi_wave_master_identity_and_2059_super_divisibility`, `dense_core_multi_wave_half_step_elimination_and_7427200_sieve`, `dense_core_dualWaveInteriorAcc_telescopic_cross_defect_and_mirror_collapses`): 1. **Telescopic Cross-Defect Identities & Path-Independent Mirror Collapses (Sections 57 & 62):** Defining `waveCrossDefect` (CD_N) and `dualWaveCrossDefect` ($\widetilde{\text{CD}}_N$), we prove that for every $N \in \mathbb{N}$ (with $1 \le L_1(t)$ and $1 \le L_2(t)$):

$$2059 \cdot \text{Acc}_N + 2^{S_N} + 16{,}243{,}286{,}400 \cdot \text{CD}_N = 3^{S_N}, \qquad 58025 \cdot \tilde{\text{Acc}}_N + 2^{\tilde{S}_N} + 124{,}499{,}856{,}384 \cdot \widetilde{\text{CD}}_N = 3^{\tilde{S}_N},$$

where $M_W = 128 \times 2187 \times 58025 = 16{,}243{,}286{,}400$ and $M_U = 1024 \times 59049 \times 2059 = 124{,}499{,}856{,}384$. Consequently, all path dependence in Acc_N and $\tilde{\mathrm{Acc}}_N$ vanishes modulo $(16{,}243{,}286{,}400, 58025, 128, 2187)$ and $(124{,}499{,}856{,}384, 2059, 1024, 59049)$:
$(2187 \cdot \mathrm{Acc}_N) \bmod 128 = 3^{S_N} \bmod 128$, $(128 \cdot \mathrm{Acc}_N) \bmod 2187 = 2^{S_N} \bmod 2187$,
$(59049 \cdot \tilde{\mathrm{Acc}}_N) \bmod 1024 = 3^{\tilde{S}_N} \bmod 1024$, $(1024 \cdot \tilde{\mathrm{Acc}}_N) \bmod 59049 = 2^{\tilde{S}_N} \bmod 59049$,
$\gcd(\mathrm{Acc}_N, 6) = \gcd(\tilde{\mathrm{Acc}}_N, 6) = 1$, and any two macro-wave anagrams with $S_N = S'_M$ (or $\tilde{S}_N = \tilde{S}'_M$) satisfy $16{,}243{,}286{,}400 \mid (\mathrm{Acc}'_M - \mathrm{Acc}_N)$ (and $124{,}499{,}856{,}384 \mid (\tilde{\mathrm{Acc}}'_M - \tilde{\mathrm{Acc}}_N)$). 2. **Universal Same-Run-Length CRT Moduli M_W and M_U & 8.12 Billion$\times$ Wave Descent (Section 58):** Any two 2-run states W, W' of the same length L_2 flanked by 1-runs satisfy $7{,}427{,}200 = 128 \times 58025 \mid (W' - W)$ (from the outgoing interface alone), $2187 \mid (W' - W)$ (from the incoming interface), and $16{,}243{,}286{,}400 \mid (W' - W)$ (with unique residues $W \bmod 16{,}243{,}286{,}400 \in \{9{,}344{,}804{,}209,\ 3{,}539{,}170{,}441\}$ for $L_2 = 1, 2$). Dually, any two 1-run states W_3, W'_3 of the same length L_3 flanked by 2-runs satisfy $2{,}108{,}416 = 1024 \times 2059 \mid (W'_3 - W_3)$ (outgoing), $59049 \mid (W'_3 - W_3)$ (incoming), and $124{,}499{,}856{,}384 \mid (W'_3 - W_3)$ (with unique residues $W_3 \bmod 124{,}499{,}856{,}384 \in \{9{,}506{,}086{,}031,\ 29{,}917{,}871{,}245\}$ for $L_3 = 1, 2$), accelerating same-length wave descent/growth to $W + 16{,}243{,}286{,}400 \le W'$ and $W_3 + 124{,}499{,}856{,}384 \le W'_3$! 3. **Upgraded $58025 \cdot 2^{S_N+10L_{\text{end}}+7}$ Super-Divisibility & $+22$-Bit Repetition Cutoff (Section 59):** Because matching terminal 2-runs $L_2(N+1) = L'_2(N+1) = L_{\text{end}}$ with outgoing 1-runs automatically satisfy $7{,}427{,}200 = 58025 \times 2^7 \mid (W'(N+1) - W(N+1))$, coprimality with $3^{S_N+10L_2(0)}$ transfers the full factor across the $(N+1)$-wave tower:

$$\left(58025 \cdot 2^{S_N+10L_{\text{end}}+7}\right) \,\Big|\, \big(W'(0) - W(0)\big),$$

forcing $2^{S_N+10L_{\text{end}}+22} < 58025 \cdot 2^{S_N+10L_{\text{end}}+7} + W_a \le W(M+1)$ (since $7{,}427{,}200 > 4{,}194{,}304 = 2^{22}$) and upgrading Section 56's macro-wave repetition cutoff by $+10$ **additional bits** (to $+22$ **bits**):

$$q \cdot \big(S_N + 10L_{\text{end}} + 10L_2(M+1) + 22\big) \;<\; p \cdot S_M + 10(p+q)L_2(0) + qW_0.$$

4. **Dual $(2^{\tilde{S}_N}, 3^{\tilde{S}_N})$ Multi-Wave Master Identity, Opposite-Sign Drift Monotonicity & Multi-Wave Half-Step Elimination (Sections 60 & 61):** Across $N+1$ consecutive $1 \to 2 \to 1$ waves, `dualWaveInteriorExp` ($\tilde{S}_N$) and `dualWaveInteriorAcc` ($\tilde{\mathrm{Acc}}_N$) are 100% independent of $L_1(0)$ and $L_1(N+1)$ and satisfy:

$$3^{\tilde{S}_N} \cdot \left(3^{7L_1(0)} \cdot U(0)\right) \;+\; 19 \cdot \tilde{\mathrm{Acc}}_N \;=\; 2^{\tilde{S}_N} \cdot \left(2^{7L_1(N+1)} \cdot U(N+1)\right),$$

forcing $3^{\tilde{S}_N+7L_1(0)}U(0) < 2^{\tilde{S}_N+7L_1(N+1)}U(N+1)$ universally and $(2059 \cdot 2^{\tilde{S}_N+7L_{\text{end}}+10}) \mid (U'(0) - U(0))$ ($+21$ bits, since $2{,}108{,}416 > 2^{21}$) on interior-matching blocks. Finally, coupling the W-wave master identity with the outgoing interface at $W(N+1)$ yields the half-step identity $58025 \cdot 2^{S_{N+1}}U_{\text{end}} + 19(2^{S_N+10L_2(N+1)} + 2059 \cdot 3^{10L_2(N+1)}\mathrm{Acc}_N) = 2059 \cdot 3^{S_N+10L_2(0)+10L_2(N+1)}W(0)$, locking $W(0)$ modulo $7{,}427{,}200 \cdot 2^{S_N+10L_2(N+1)}$! - **Exact Boundary $(2,3)$-Adic Valuations of `waveCrossDefect` ($\nu_2(\mathrm{CD}_{m+1}) = 7L_1(0) - 7$ and $\nu_3(\mathrm{CD}_{m+1}) = 7L_1(m+1) - 7$), Prefix-Sharing Cross-Defect Filtration ($126{,}900{,}675 \cdot 2^{S_K}$ Anagram Divisibility), Incoming-Interface Multi-Wave Elimination ($126{,}900{,}675 \cdot 3^{S_N+10L_2(0)}$ Terminal-State Sieve) & Two-Way $(16{,}243{,}286{,}400 \cdot 2^{S_N+10L_{\text{end}}},\ 16{,}243{,}286{,}400 \cdot 3^{S_N+10L_2(0)})$ Tower Super-Divisibility with $+33$-Bit / $+36$-Bit Repetition Cutoffs (`dense_core_waveCrossDefect_boundary_valuations_and_anagram_locks`, `dense_core_waveCrossDefect_prefix_filtration_and_anagram_divisibility`, `dense_core_multi_wave_incoming_elimination_and_126900675_sieve`, `dense_core_two_way_16243286400_tower_super_divisibility_and_33_bit_cutoff`):** 1. **Exact Boundary $(2,3)$-Adic Valuations of `waveCrossDefect` (Section 63):** Because $R_k = (3^{10k} - 2^{10k})/58025$ is universally coprime to 6 for all $k \ge 1$ ($R_k \bmod 2 = 1$ and $3 \nmid R_k$), for every $m \ge 0$ ($N = m+1 \ge 1$) the initial 1-run $L_1(0)$ uniquely locks the 2-adic valuation $\nu_2(\mathrm{CD}_{m+1}) = 7L_1(0) - 7$ (via a ≥ 17-bit spectral gap to later waves) while the terminal 1-run $L_1(m+1)$ uniquely locks the 3-adic valuation $\nu_3(\mathrm{CD}_{m+1}) = 7L_1(m+1) - 7$, so $(2^{7L_1(0)-7} \cdot 3^{7L_1(m+1)-7}) \mid \mathrm{CD}_{m+1}$. 2. **Prefix-Sharing Cross-Defect Filtration & Higher-Power $126{,}900{,}675 \cdot 2^{S_K}$ Anagram Divisibility (Section 64):** For any $K \le N$, $3^{S_N-S_K}\mathrm{CD}_K \le \mathrm{CD}_N$ and $2^{S_K-7} \mid (\mathrm{CD}_N - 3^{S_N-S_K}\mathrm{CD}_K)$. Consequently, if two macro-wave anagrams ($S_N = S'_M$) share their first K interior waves, $3^{S_N-S_K}\mathrm{CD}_K$ cancels identically from $\mathrm{CD}_N - \mathrm{CD}'_M$, forcing $2^{S_K-7} \mid (\mathrm{CD}_N - \mathrm{CD}'_M)$ and $(126{,}900{,}675 \cdot 2^{S_K}) \mid (\mathrm{Acc}'_M - \mathrm{Acc}_N)$ (where $S_K \ge 17K + 7$). 3. **Incoming-Interface Multi-Wave Elimination & the $126{,}900{,}675 \cdot 3^{S_N+10L_2(0)}$ Terminal-State Sieve (Section 65):** Coupling the $(N+1)$-wave master identity with the incoming interface at $W(0)$ ($58025 \cdot 3^{7L_{1,\text{start}}}U_{\text{start}} + 19 = 2059 \cdot 2^{10L_2(0)}W(0)$) yields $2059 \cdot 2^{S_N+10L_2(0)+10L_2(N+1)}W(N+1) + 19 \cdot 2059 \cdot 2^{10L_2(0)}\mathrm{Acc}_N = 58025 \cdot 3^{S_N+10L_2(0)+7L_{1,\text{start}}}U_{\text{start}} + 19 \cdot 3^{S_N+10L_2(0)}$, locking $W(N+1)$ modulo $126{,}900{,}675 \cdot 3^{S_N+10L_2(0)}$ independently of $W(0)$, U_{start}, and $L_{1,\text{start}}$. 4. **Two-Way $16{,}243{,}286{,}400$ and $124{,}499{,}856{,}384$ Tower Super-Divisibility & $+33$-Bit / $+36$-Bit Repetition Cutoffs (Section 66):** Combining incoming ($2187 \mid (W'(0) - W(0))$) and outgoing ($7{,}427{,}200 \mid (W'(N+1) - W(N+1))$) interfaces across the $(N+1)$-wave tower proves simultaneously:

$$\left(16{,}243{,}286{,}400 \cdot 2^{S_N+10L_{\text{end}}}\right) \,\Big|\, \big(W'(0) - W(0)\big) \qquad \text{and} \qquad \left(16{,}243{,}286{,}400 \cdot 3^{S_N+10L_2(0)}\right) \,\Big|\, \big(W'(N+1) - W(N+1)\big),$$

and dually on U-waves:

$$\Big(124{,}499{,}856{,}384\cdot 2^{\tilde{S}_N+7L_{\text{end}}}\Big)\;\Big|\;\big(U'(0)-U(0)\big)\qquad\text{and}\qquad\Big(124{,}499{,}856{,}384\cdot 3^{\tilde{S}_N+7L_1(0)}\Big)\;\Big|\;\big(U'(N+1)-U(N+1)\big).$$

Since $16{,}243{,}286{,}400 > 2^{33}$ and $124{,}499{,}856{,}384 > 2^{36}$, this forces $2^{S_N+10L_{\text{end}}+33} < W(M+1)$ and $2^{\tilde{S}_N+7L_{\text{end}}+36} < U'(0)$, boosting our macro-wave repetition cutoff to $+33$ **bits** (nearly two full waves of extra divisibility):

$$q\cdot\big(S_N+10L_{\text{end}}+10L_2(M+1)+33\big)\;<\;p\cdot S_M+10(p+q)L_2(0)+qW_0,$$

and at $(p,q)=(10,17)$:
$17(17N+40+10L_{\text{end}}+10L_2(M+1))\le 17(S_N+10L_{\text{end}}+10L_2(M+1)+33)<10S_M+270L_2(0)+17W_0$
! - Automatic Self-Propagation of $(2059,58025)$ Across Macro-Run Interfaces, the Universal Multi-Wave Constant of Motion ($W(0)-1$ and $U(0)$) & Full-Power 3-Adic Remainders for All Horizons $N\ge 0$, Exact Cross-Defect–Dual-Accumulator Factorization ($\mathrm{CD}_{m+1}=2^{7L_1(0)-7}3^{7L_1(m+1)-7}\tilde{\mathrm{Acc}}_m$) & the W–U Complementarity Law, and Two-Sided Prefix+Suffix $(58025\cdot 2^{S_K}\cdot 3^{S_{\text{end}}-S_J+7})$ Anagram Super-Divisibility (`dense_core_interface_2059_and_58025_automatic_self_propagation`, `dense_core_universal_multi_wave_constant_of_motion_and_three_adic_remainders`, `dense_core_cross_defect_dual_accumulator_factorization_and_complementarity`, `dense_core_two_sided_prefix_suffix_anagram_super_divisibility`): 1. **Automatic Self-Propagation of $(2059,58025)$ Across Macro-Run Interfaces (Section 67):** Whenever $58025\cdot 2^{7L_1}U_1+19=2059\cdot 3^{10L_2}W_2$, the next interface numerator satisfies $2^{7L_1}(58025\cdot 3^{7L_1}U_1+19)+19(2059Q_{L_1})=2059\cdot 3^{10L_2+7L_1}W_2$, forcing $2059\mid(58025\cdot 3^{7L_1}U_1+19)$ automatically for all $L_1\in\mathbb{N}$. Dually, whenever $58025\cdot 3^{7L_1}U_1+19=2059\cdot 2^{10L_2}W_2$, $58025\mid(2059\cdot 3^{10L_2}W_2-19)$ holds automatically for all $L_2\in\mathbb{N}$. 2. **Universal Multi-Wave Constant of Motion & Full-Power 3-Adic Remainders for All $N\ge 0$ (Section 68):** Because $2059\cdot\mathrm{Acc}_N<3^{S_N}$ and $58025\cdot\tilde{\mathrm{Acc}}_N<3^{\tilde{S}_N}$ hold unconditionally at every horizon $N\in\mathbb{N}$, the multi-wave integer quotients and 3-adic remainders are locked across **all horizons $N\ge 0$ simultaneously**:

$$\left\lfloor\frac{2^{S_N+10L_2(N+1)}\cdot W(N+1)}{3^{S_N+10L_2(0)}}\right\rfloor=W(0)-1,\qquad\Big(2^{S_N+10L_2(N+1)}\cdot W(N+1)\Big)\bmod 3^{S_N+10L_2(0)}=3^{S_N+10L_2(0)}-19\cdot\mathrm{Acc}_N,$$

$$\left\lfloor\frac{2^{\tilde{S}_N+7L_1(N+1)}\cdot U(N+1)}{3^{\tilde{S}_N+7L_1(0)}}\right\rfloor=U(0),\qquad\Big(2^{\tilde{S}_N+7L_1(N+1)}\cdot U(N+1)\Big)\bmod 3^{\tilde{S}_N+7L_1(0)}=19\cdot\tilde{\mathrm{Acc}}_N.$$

3. **Exact Cross-Defect–Dual-Accumulator Factorization & the W–U Complementarity Law (Section 69):** For every $m\in\mathbb{N}$, `waveCrossDefect L_2 L_1 (m + 1)` is literally $2^{7L_1(0)-7}\cdot 3^{7L_1(m+1)-7}$ times the shifted dual accumulator `dualWaveInteriorAcc L_1 (fun t => L_2 (t + 1)) m` (and dually for `dualWaveCrossDefect`), proving that the W-wave drift and the interleaved U-wave drift are **exact complementary halves of $3^{S_{m+1}}-2^{S_{m+1}}$** with zero path-dependent remainder:

$$2059\cdot\mathrm{Acc}_{m+1}\;+\;58025\cdot 2^{7L_1(0)}\cdot 3^{7L_1(m+1)}\cdot\tilde{\mathrm{Acc}}_m\;+\;2^{S_{m+1}}\;=\;3^{S_{m+1}},$$

$$58025\cdot\tilde{\mathrm{Acc}}_{m+1}\;+\;2059\cdot 2^{10L_2(0)}\cdot 3^{10L_2(m+1)}\cdot\mathrm{Acc}_m\;+\;2^{\tilde{S}_{m+1}}\;=\;3^{\tilde{S}_{m+1}}.$$

4. **Right-Suffix 3-Adic Cancellation & Two-Sided Prefix+Suffix Anagram Super-Divisibility (Section 70):** Whenever two macro-wave anagrams ($S_{J+D}=S'_{J'+D}$) share their last D interior waves, $S_J=S'_{J'}$ holds automatically and the suffix drift cancels identically: $\mathrm{Acc}'_{J'+D}-\mathrm{Acc}_{J+D}=3^{S_{J+D}-S_J}(\mathrm{Acc}'_{J'}-\mathrm{Acc}_J)$. If they also share their first K interior waves ($K\le J,J'$), then:

$$\Big(58025\cdot 2^{S_K}\cdot 3^{S_{J+D}-S_J+7}\Big)\;\Big|\;\big(\mathrm{Acc}'_{J'+D}-\mathrm{Acc}_{J+D}\big)!$$

- Multi-Wave $+19$-Cancellation Homogeneous Difference Laws, the Global All-Horizon Two-Sided Wave Corridors ($3^{10L_2(0)}W(0)<2^{10L_2(N+1)}W(N+1)$ and $3^{7L_1(0)}U(0)<2^{7L_1(N+1)}U(N+1)$), Complete Impossibility of Same-Run-Length Multi-Wave Descent ($+16{,}243{,}286{,}400$ and $+124{,}499{,}856{,}384$ Monotone Level Growth), Mandatory All-Horizon Run Expansion ($3L(0)<2L(N+1)$) & the Three-Point Multi-Horizon Homogeneous Wronskian Syzygy (`dense_core_multi_wave_homogeneous_difference_and_all_horizon_corridor`, `dense_core_all_horizon_same_run_strict_growth_and_mandatory_expansion`, `dense_core_dual_multi_wave_homogeneous_difference_and_124499856384_growth`, `dense_core_multi_horizon_shift_composition_and_three_point_wronskian_syzygy`): 1. **Multi-Wave Homogeneous Difference Laws & Global All-Horizon Two-Sided Wave Corridors (Sections 71 & 73):** Although the single-wave bounds $3^{10L_2(n)}W(n)<2^{10L_2(n+1)}W(n+1)$ cannot be chained by transitivity because the base switches from 2 to 3 at each step, eliminating $+19$ between the $(N+1)$-wave master identity and the initial/terminal macro-run interfaces via the cross-defect identity proves for **every horizon $N\in\mathbb{N}$**:

$$2^{S_N}\cdot\Big(2^{10L_2(N+1)}W(N+1)\Big)=2^{S_N}\cdot\Big(3^{10L_2(0)}W(0)\Big)+16{,}243{,}286{,}400\cdot 3^{10L_2(0)}\mathrm{CD}_N W(0)+58025\cdot 2^{7L_1(0)}\mathrm{Acc}_N U(0),$$

$$2^{\tilde{S}_N}\cdot\Big(2^{7L_1(N+1)}U(N+1)\Big)=2^{\tilde{S}_N}\cdot\Big(3^{7L_1(0)}U(0)\Big)+124{,}499{,}856{,}384\cdot 3^{7L_1(0)}\widetilde{\mathrm{CD}}_N U(0)+2059\cdot 2^{10L_2(0)}\tilde{\mathrm{Acc}}_N W_{\text{first}},$$

establishing $58025 \mid (2^{10L_2(N+1)}W(N+1) - 3^{10L_2(0)}W(0))$, $2059 \mid (2^{7L_1(N+1)}U(N+1) - 3^{7L_1(0)}U(0))$, and canceling 2^{S_N} and $2^{\tilde{S}_N}$ to yield the **Global All-Horizon Lower Bounds**:

$$3^{10L_2(0)} \cdot W(0) \;<\; 2^{10L_2(N+1)} \cdot W(N+1), \qquad 3^{7L_1(0)} \cdot U(0) \;<\; 2^{7L_1(N+1)} \cdot U(N+1) \qquad (\forall N \geq 0).$$

2. **Complete Impossibility of Same-Run-Length Multi-Wave Descent,** $+16.24\text{B}/+124.5\text{B}$ **Monotone Level Growth & All-Horizon Mandatory Run Expansion (Sections 72 & 73):** Across **any** number $N+1$ of waves: - Whenever $2^{10L_2(N+1)} \leq 3^{10L_2(0)}$ (including $L_2(N+1) \leq L_2(0)$, or $2 \leq L_2(0)$ with $L_2(N+1) \leq 3$), $W(0) < W(N+1)$ strictly (and dually $U(0) < U(N+1)$ whenever $L_1(N+1) \leq L_1(0)$). - Whenever $L_2(N+1) = L_2(0)$ (or $L_1(N+1) = L_1(0)$), multi-wave descent is **100% impossible**, and Section 58's CRT moduli force strict monotone growth by billions along every constant run-length level:

$$W(0) + 16{,}243{,}286{,}400 \;\leq\; W(N+1), \qquad U(0) + 124{,}499{,}856{,}384 \;\leq\; U(N+1).$$

- Whenever $W(N+1) \leq W(0)$ (or $U(N+1) \leq U(0)$) across any $(N+1)$-wave span, the terminal run length must expand via $3L_2(0) < 2L_2(N+1)$ and $19L_2(0) < 12L_2(N+1)$ (and $3L_1(0) < 2L_1(N+1)$). In particular, in any $\{1,2\}$-confined short-run regime, once $L = 2$ is visited, the state can **never** descend at any future wave in $\{1,2\}$! 3. **Multi-Horizon Shift-Composition & the Three-Point Homogeneous Wronskian Syzygy (Section 74):** Across any three nested horizons $0 < K+1 < K+2+d$, `waveInteriorExp` and `waveInteriorAcc` satisfy $S_{K+1+d} = S_K + 10L_2(K+1) + S_{\text{tail}}$ and $\text{Acc}_{K+1+d} = 3^{10L_2(K+1)+S_{\text{tail}}}\text{Acc}_K + 2^{S_K+10L_2(K+1)}\text{Acc}_{\text{tail}}$, and eliminating $+19$ yields the $+19$**-free Three-Point Wronskian Syzygy**:

$$\text{Acc}_{K+1+d} \cdot W(K+1) \;=\; 3^{S_K+10L_2(0)} \cdot \text{Acc}_{\text{tail}} \cdot W(0) \;+\; 2^{S_{\text{tail}}+10L_2(K+2+d)} \cdot \text{Acc}_K \cdot W(K+2+d).$$

- **Cumulative K-Visit Billion-Step Level Growth, Exponential Run-Length Explosion (** $3^K L(0) < 2^K L(K)$ **and** $19^K L(0) < 12^K L(K)$**), Exclusion of Bounded-Run Non-Increasing Chains, Prime-Factor** $(29, 71)$ **Homoclinic Bijections (** $\bmod 29, 71, 464, 1392$**), Exact Three-Way Multi-Wave GCD Equalities & Multi-Horizon Wronskian Congruence Locks (** `dense_core_cumulative_level_growth_and_exponential_run_explosion` **,** `dense_core_one_run_mod_29_71_464_1392_bijections` **,** `dense_core_multi_wave_three_way_gcd_equality_and_corridor_bound` **,** `dense_core_multi_horizon_wronskian_archimedean_and_divisibility_bounds` **):** 1. **Cumulative K-Visit Level Growth, Exponential Run Explosion & Exclusion of Bounded-Run Non-Increasing Chains (Section 75):** Along any K visits to a constant run-length level, $W(m_0) + 16{,}243{,}286{,}400K \leq W(m_K)$ and $U(m_0) + 124{,}499{,}856{,}384K \leq U(m_K)$. Along any $K \geq 1$ non-increasing W-epochs ($W(m_{k+1}) \leq W(m_k)$), run lengths explode exponentially via $3^K L_2(m_0) < 2^K L_2(m_K)$ and $19^K L_2(m_0) < 12^K L_2(m_K)$ ($L_2(m_0) + K \leq L_2(m_K)$), ruling out all infinite non-increasing W-chains under bounded run lengths ($L_2(n) \leq B$) as well as all strictly decreasing W-chains of length $> W(m_0)/88$ ($W(m_K) + 88K \leq W(m_0)$). 2. **Prime-Factor** 29 **and** 71 **Homoclinic Sieves on** 1**-Run Quotients (**$2059 = 29 \times 71$**, Section 76):** Because $2^7 \equiv 12 \pmod{29}$ has multiplicative order 4 and $2^7 \equiv 57 \pmod{71}$ has multiplicative order 5, every 1-run state W_3 satisfies a **4-way bijection** between $L_3 \bmod 4$ and $W_3 \bmod 29 \in \{1, 17, 28, 12\}$, a **5-way bijection** between $L_3 \bmod 5$ and $W_3 \bmod 71 \in \{46, 17, 14, 70, 66\}$, and combined **4-way bijections modulo** $464 = 16 \times 29$ ($\{175, 365, 231, 389\}$) and **modulo** $1392 = 16 \times 3 \times 29$ ($\{1103, 829, 695, 853\}$). 3. **Exact Three-Way Multi-Wave GCD Equalities &** $1/6{,}377{,}292$ **and** $1/6{,}676{,}911$ **GCD Bounds (Section 77):** Because $\gcd(W(0), 2) = \gcd(W(N+1), 3) = 1$ and $\gcd(U(0), 6) = \gcd(19\tilde{\text{Acc}}_N, 3) = 1$, across every horizon $N \in \mathbb{N}$:

$$\gcd\big(W(0), W(N+1)\big) = \gcd\big(W(0), 19\text{Acc}_N\big) = \gcd\big(W(N+1), 19\text{Acc}_N\big) < \frac{3^{S_N+10L_2(0)}}{6{,}377{,}292},$$

$$\gcd\big(U(0), U(N+1)\big) = \gcd\big(U(0), 19\tilde{\text{Acc}}_N\big) = \gcd\big(U(N+1), 19\tilde{\text{Acc}}_N\big) < \frac{3^{\tilde{S}_N+7L_1(0)}}{6{,}676{,}911}.$$

4. **Multi-Horizon Wronskian Archimedean Sandwich & Two-Way Congruence Locks (Section 78):** Across any three nested horizons $0 < K+1 < K+2+d$, $\text{Acc}_{\text{full}}W(K+1)$ strictly dominates both $3^{S_K+10L_2(0)}\text{Acc}_{\text{tail}}W(0)$ and $2^{S_{\text{tail}}+10L_2(K+2+d)}\text{Acc}_K W(K+2+d)$ and satisfies the exact two-way congruences:

$$\big(\text{Acc}_{\text{full}} \cdot W(K+1)\big) \bmod 3^{S_K+10L_2(0)} = \big(2^{S_{\text{tail}}+10L_2(K+2+d)} \cdot \text{Acc}_K \cdot W(K+2+d)\big) \bmod 3^{S_K+10L_2(0)},$$

$$\big(\text{Acc}_{\text{full}} \cdot W(K+1)\big) \bmod 2^{S_{\text{tail}}+10L_2(K+2+d)} = \big(3^{S_K+10L_2(0)} \cdot \text{Acc}_{\text{tail}} \cdot W(0)\big) \bmod 2^{S_{\text{tail}}+10L_2(K+2+d)}.$$

- **Universal Exponential Bands on** `waveInteriorAcc` **(**$[2059/2187, 1) \cdot 3^{S_N}$**) and** `dualWaveInteriorAcc` **(**$[58025/59049, 1) \cdot 3^{\tilde{S}_N}$**), Two-Sided Cauchy Pinch Across Nested Horizons, Full-Exponent Cumulative Step-Sum Run-Expansion Laws (**$7S_N + 190L_2(0) < 120L_2(N+1) + 7$ **and** $7\tilde{S}_N + 133L_1(0) < 84L_1(N+1)$**) & Unconditional** $631\times$ **Two-Wave** W**-Growth and** $7^{N+1}\times$ U**-Growth in** $\{1,2\}$ **(** `dense_core_accumulator_exponential_bands_and_cauchy_pinch` **,** `dense_core_full_exponent_archimedean_wave_bound_and_step_sum_expansion` **,** `dense_core_short_run_two_wave_631_growth_and_isolated_dip_law` **,** `dense_core_dual_one_run_full_exponent_expansion_and_seven_pow_growth` **):** 1. **Universal Exponential**

Accumulator Bands & Two-Sided Cauchy Pinch (Section 79): For all $N \in \mathbb{N}$, $3^{S_N-7} \le \mathrm{Acc}_N$ and $3^{\tilde{S}_N-10} \le \tilde{\mathrm{Acc}}_N$, trapping $2059 \cdot \mathrm{Acc}_N \in [2059 \cdot 3^{S_N-7},\ 3^{S_N})$ ($[0.94147, 1) \cdot 3^{S_N}$) and $58025 \cdot \tilde{\mathrm{Acc}}_N \in [58025 \cdot 3^{\tilde{S}_N-10},\ 3^{\tilde{S}_N})$ ($[0.98266, 1) \cdot 3^{\tilde{S}_N}$), and pinching $\mathrm{Acc}_{K+1+d} - 3^{10L_2(K+1)+S_{\text{tail}}}\mathrm{Acc}_K$ between $2^{S_K+10L_2(K+1)} \cdot 3^{S_{\text{tail}}-7}$ and $2^{S_K+10L_2(K+1)} \cdot 3^{S_{\text{tail}}}/2059$. 2. **Full-Exponent Archimedean Wave Bound & Cumulative Step-Sum Run-Expansion Law (Section 80):** Preserving the full exponential factor 3^{S_N} in the master identity proves $2 \cdot 3^{S_N+10L_2(0)-1} \cdot W(0) < 2^{S_N+10L_2(N+1)} \cdot W(N+1)$ for all $N \ge 0$. Consequently, whenever $W(N+1) \le W(0)$, the **single terminal run length $L_2(N+1)$ must beat the entire cumulative step-sum S_N of all $N+1$ waves**:

$$p' \cdot S_N + 10(p'+q')L_2(0) \ < \ 10q' \cdot L_2(N+1) + p' \qquad \big(7S_N + 190L_2(0) < 120L_2(N+1) + 7, \ \ 119N + 190L_2(0) + 42 < 120L_2(N$$

3. **Unconditional $631\times$ Two-Wave W-Growth & Isolated Dip Law in $\{1,2\}$ (Section 81):** Whenever $L_2(N+1) \le 2$ and $1 \le L_2(0)$, $9 \le S_N$ (including a single wave $N=0$ with $2 \le L_1(0)$) forces $W(0) < W(N+1)$ strictly, isolating $(L_2(0)=1, L_1(0)=1, L_2(1)=2)$ as the **only** transition in $\{1,2\}$ where W can dip. Across **any $N+1 \ge 2$ waves ($N \ge 1$)**, $24 \le S_N$ and $631 \cdot 2^{43} < 3^{33}$ force $631 \cdot W(0) < W(N+1)$ ($512 \cdot W(0) < W(N+1)$)! 4. **Dual U-Wave Full-Exponent Expansion & Unconditional $7^{N+1}\times$ Growth in $\{1,2\}$ (Section 82):** On U-waves, $U(N+1) \le U(0)$ forces $p'\tilde{S}_N + 7(p'+q')L_1(0) < 7q'L_1(N+1)$ ($7\tilde{S}_N + 133L_1(0) < 84L_1(N+1)$, so $3 \le L_1(N+1)$). In $\{1,2\}$ ($L_1(N+1) \le 2$), $U(n)$ **never dips at all—not even for a single wave ($N=0$)**—satisfying $7 \cdot U(0) < U(N+1)$ at every wave and $7^K \cdot U(0) \le U(K)$ across any K waves! - **Accumulator-Eliminated Three-Point Wronskian Power Inequalities ($2059:2187$ S_K-Independent Sandwich), Cumulative 631^K-Fold Even/Odd-Wave Exponential Growth ($631^K W(0) \le W(2K)$) & the Balanced Macro-Wave Subcritical Complexity Cutoff at the Full $(q:p)$ Ratio (`dense_core_accumulator_eliminated_three_point_wronskian_inequalities`, `dense_core_short_run_even_wave_631_pow_and_joint_exponential_tower`, `dense_core_balanced_macro_wave_subcritical_complexity_cutoff`):** 1. **Accumulator-Eliminated Three-Point Wronskian Power Inequalities (Section 83):** Substituting Section 79's $[2059/2187, 1)$ exponential bounds into Section 74's Three-Point Wronskian Syzygy eliminates Acc_K, $\mathrm{Acc}_{\text{tail}}$, and $\mathrm{Acc}_{\text{full}}$ and cancels 3^{S_K-7} from all three terms, trapping $3^{10L_2(K+1)+S_{\text{tail}}}W(K+1)$ and $3^{10L_2(0)+S_{\text{tail}}}W(0) + 2^{S_{\text{tail}}+10L_2(K+2+d)}W(K+2+d)$ within a $2059:2187$ ratio **independently of S_K**:

$$2059 \cdot \Big(3^{10L_2(0)+S_{\text{tail}}}W(0) + 2^{S_{\text{tail}}+10L_2(K+2+d)}W(K+2+d)\Big) \ < \ 2187 \cdot 3^{10L_2(K+1)+S_{\text{tail}}}W(K+1),$$

$$2059 \cdot 3^{10L_2(K+1)+S_{\text{tail}}}W(K+1) \ < \ 2187 \cdot \Big(3^{10L_2(0)+S_{\text{tail}}}W(0) + 2^{S_{\text{tail}}+10L_2(K+2+d)}W(K+2+d)\Big).$$

2. **Cumulative 631^K-Fold Even-Wave and Odd-Wave Exponential Growth in $\{1,2\}$ (Section 84):** Iterating $631 \cdot W(n) < W(n+2)$ across K two-wave blocks proves $631^K \cdot W(0) \le W(2K)$, $512^K \cdot W(0) = 2^{9K} \cdot W(0) \le W(2K)$, $631^K \cdot W(1) \le W(2K+1)$, and $512^K \cdot W(1) \le W(2K+1)$ for all $K \in \mathbb{N}$. 3. **Balanced Macro-Wave Repetition Cutoff at the Full $(q:p)$ Critical Ratio (Section 85):** Because Section 54's symmetric master identity has the **exact same exponent S_N** on 2^{S_N} and 3^{S_N}, whenever a macro-wave sequence has balanced step-sum rate μ ($\mu N \le S_N + D_{\text{low}}$ and $S_M \le \mu M + D_{\text{high}}$), Section 66's $+33$-bit repetition cutoff forces:

$$q \cdot \mu \cdot N + q \cdot \big(10L_{\text{end}} + 10L_2(M+1) + 33\big) \ < \ p \cdot \mu \cdot M + p \cdot D_{\text{high}} + q \cdot D_{\text{low}} + 10(p+q)L_2(0) + qW_0,$$

recovering the **full critical Mahler slope** $q/p \to \log 2/\log(3/2) \approx 1.709511$ on macro-waves (with zero $7/10$ loss)! - **Macro-Wave Pair Encoding (`macroWaveCode`), Automatic Same-Start Strict Growth $W(r_1) < W(r_2)$, Unconditional Exclusion of Balanced Subcritical Macro-Wave Orbits Up to the Full Critical Threshold $\kappa < \frac{\log 2}{\log(3/2)} \approx 1.709511$, Two-Way Anagram Prefix-to-Start & Suffix-to-End State-Difference Super-Divisibility & the $(p+q)$-Weighted Suffix-Anagram 3-Adic Cutoff (`dense_core_macroWaveCode_pigeonhole_and_automatic_growth_cutoff`, `no_subcritical_macroWaveCode_dense_core`, `dense_core_anagram_prefix_start_and_suffix_end_state_super_divisibility`, `dense_core_suffix_sharing_anagram_three_adic_cutoff`):** 1. **Macro-Wave Pair Encoding, Automatic Strict Growth $W(r_1) < W(r_2)$ & Pigeonhole `prefixFactorCount` Cutoff (Section 86):** Defining `macroWaveCode L_2 L_1 n = Nat.pair (L_2 n) (L_1 n)`, whenever an $(N+2)$-wave subword repeats at $r_1 < r_2 \le C$, equality of the initial runs $L_2(r_1) = L_2(r_2)$ **automatically forces** $W(r_1) < W(r_2)$ by Section 72 (with zero monotonicity assumption on W) and $\big(58025 \cdot 2^{S_N^{(r_1)}+10L_2(r_1+N+1)+7}\big) \mid \big(W(r_2) - W(r_1)\big)$, yielding the Pigeonhole cutoff $q(17N+49) \le q(S_N^{(r_1)} + 42) < pS_C + 10(p+q)L_2(0) + qW_0$ whenever `prefixFactorCount (macroWaveCode L_2 L_1) (N + 2) C ≤ C`. 2. **Unconditional Exclusion of Subcritical Macro-Wave Orbits Up to the Full Critical Threshold $\kappa < \frac{\log 2}{\log(3/2)} \approx 1.709511$ (Section 87):** Combining Section 86 with `exists_pq_of_lt_log_threshold` excludes: (i) all macro-wave orbits with density-weighted complexity $\mu \cdot \kappa \cdot \log(3/2) < 17 \log 2$, and (ii) **all balanced macro-wave orbits with subcritical complexity $\kappa < \frac{\log 2}{\log(3/2)} \approx 1.709511$** ($\kappa \cdot \log(3/2) < \log 2$), completely closing the $[1.1967,\ 1.709511)$ complexity gap for balanced macro-waves! 3. **Two-Way Anagram Prefix-to-Start & Suffix-to-End State-Difference Super-Divisibility (Section 88):** Across any two interior macro-wave anagrams ($S_{J+D} = S'_{J'+D}$) with matching boundary 2-runs, sharing K prefix waves forces $\big(58025 \cdot 2^{S_K}\big) \mid \big(W'(0) - W(0)\big)$, while sharing D suffix waves forces

$\big(7{,}427{,}200\cdot 3^{S_{J+D}-S_J}\big)\mid\big(W'(J'+D+1)-W(J+D+1)\big)$. 4. **The $(p+q)$-Weighted Suffix-Anagram 3-Adic Archimedean Cutoff (Section 89):** Because $7{,}427{,}200\cdot 3^{E_{\text{suf}}}>2^{22}\cdot 3^{E_{\text{suf}}}$ is already a base-3 power, canceling $3^{E_{\text{suf}}}$ directly against $3^{S_M+10L_2(0)}$ gives suffix steps a $(p+q)$ **weight (27 vs. 10 at $(10,17)$, $1.588\times$ stronger than a 2-adic prefix bit)**:

$$(p+q)\cdot E_{\text{suf}}+q\cdot\big(10L_2(M+1)+22\big)\;<\;p\cdot S_M+10(p+q)L_2(0)+qW_0\qquad\big(27E_{\text{suf}}+17(10L_2(M+1)+22)<10S_M+270L_2(0$$

- First-Divergence Cauchy-Pinch Separation (`54569 * 2^E_div * 3^(S_full - E_div - 10)`), the `28`-Step Middle-Anagram Floor, Unconditional Two-Sided Anagram Elimination, Universal `2`-Run Spike Upper Bounds & the Mod-`336` Exact State-Collision Jump Lock (`dense_core_first_divergence_cauchy_pinch_and_54569_exponential_gap`, `dense_core_unconditional_anagram_divisibility_and_joint_prefix_suffix_cutoff`, `dense_core_universal_two_run_spike_upper_bound_and_dip_corridor`, `dense_core_exact_state_collision_syzygy_and_81648_run_jump_lock`): 1. **First-Divergence Cauchy-Pinch Separation & the `28`-Step Middle-Anagram Floor (Section 90):** Whenever two macro-wave anagrams ($S_{K+1+d}=S'_{K+1+d'}=S_{\text{full}}$) share their first K interior waves and diverge at $L_2(K+1)<L'_2(K+1)$, setting $E_{\text{div}}:=S_K+10L_2(K+1)$ forces $\text{Acc}'_{K+1+d'}<\text{Acc}_{K+1+d}$ strictly and traps their difference inside the razor-thin $[54569/59049,\,1)\approx[0.9241,\,1)$ band:

$$54{,}569\cdot 2^{E_{\text{div}}}\cdot 3^{S_{\text{full}}-E_{\text{div}}-10}\;<\;2059\cdot\big(\text{Acc}_{K+1+d}-\text{Acc}'_{K+1+d'}\big)\;<\;2^{E_{\text{div}}}\cdot 3^{S_{\text{full}}-E_{\text{div}}}.$$

When they also share a D-wave suffix ($E_{\text{suf}}=S_{\text{full}}-S_J$), canceling 2^{S_K} and $3^{E_{\text{suf}}}$ against Section 70's two-sided divisibility forces $261{,}288{,}489{,}825<1024\cdot 3^{E_{\text{mid}}-10}$ and $28\le E_{\text{mid}}=S_J-S_K$ unconditionally—eliminating all short-middle anagrams! 2. **Unconditional Two-Way Anagram Divisibility & Joint Prefix+Suffix Elimination (Section 91):** By `Nat.le_total`, both $(58025\cdot 2^{S_K})\mid(W'(0)-W(0))$ and $(7{,}427{,}200\cdot 3^{S_{\text{end}}-S_J})\mid(W'(\text{end})-W(\text{end}))$ hold unconditionally (with no ordering assumption on Acc_{end} vs. Acc'_{end}), and combining the prefix and suffix cutoffs along an orbit eliminates **both S_K and E_{suf} from the right-hand side**:

$$(q-p)\big(S_K+10L_2(r_2)\big)+q\big(E_{\text{suf}}+10L_2(\text{end})\big)+37q\;<\;2pS_{\text{pre}}+pE_{\text{mid}}+20(p+q)L_2(0)+2qW_0.$$

3. **Universal Upper Bound on 2-Run Spikes, Two-Sided Dip Corridor & Mandatory $N\ge 2B$ Strict Growth (Section 92):** Every horizon $N\in\mathbb{N}$ satisfies $10qL_2(N+1)<pS_N+10(p+q)L_2(0)+qW_0$ ($170L_2(N+1)<10S_N+270L_2(0)+17W_0$), trapping any dip $W(N+1)\le W(0)$ inside a two-sided linear corridor and proving that whenever $L_2(N+1)\le B$ and $2B\le N$, $W(0)<W(N+1)$ **strictly**. 4. **Exact State-Collision Syzygy $W(0)=W(N+1)$ & the Mod-`42 / 16 / 336` Run-Length Jump Lock (Section 93):** Any exact state collision $W(0)=W(N+1)$ satisfies $\big(3^{S_N+10L_2(0)}-2^{S_N+10L_2(N+1)}\big)W(0)=19\text{Acc}_N$ and forces $42\mid(L_2(N+1)-L_2(0))\pmod{58{,}025}$ and $16\mid(L_2(N+1)-L_2(0))\pmod{128}$, hence $336\mid(L_2(N+1)-L_2(0))$, $L_2(0)+336\le L_2(N+1)$, and $57{,}120<10S_N+100L_2(0)+17W_0$. **- Interior State-Collision Mod-`729` & Mod-`81,648` Jump Lock (`+816,480` Binary Bits), Dual `1`-Run First-Divergence Cauchy Pinch, Global Injectivity of `(waveInteriorExp, waveInteriorAcc)` on Macro-Wave Blocks & Unconditional Divergence $W(n)\to\infty$ Under Bounded `2`-Run Lengths (`dense_core_interior_state_collision_mod_729_and_81648_jump_lock`, `dense_core_one_run_first_divergence_cauchy_pinch_and_54569_gap`, `dense_core_waveInteriorAcc_global_anagram_injectivity_and_31_step_floor`, `dense_core_bounded_run_mandatory_mod88_growth_and_unconditional_divergence`):** 1. **Interior State-Collision Mod-`729` & Mod-`81,648` Run-Length Jump Lock (Section 94):** Coupling the incoming interfaces at $W(0)$ and $W(N+1)=W(0)$ modulo 81 and $2{,}187=3^7$ ($\text{ord}_{2187}(2^{10})=729$) forces $729\mid(L_2(N+1)-L_2(0))$. Combining with Section 93's $336\mid(L_2(N+1)-L_2(0))$ ($\gcd(112,729)=1$) proves:

$$81{,}648\;\Big|\;\big(L_2(N+1)-L_2(0)\big),\qquad L_2(0)+81{,}648\;\le\;L_2(N+1),\qquad 13{,}880{,}160\;<\;10S_N+100L_2(0)+17W_0.$$

2. **Dual `1`-Run First-Divergence Cauchy-Pinch Separation (`L_1(0) < L'_1(0)`) & the `54,569` Gap (Section 95):** Whenever two macro-wave anagrams ($S_{1+d}=S'_M=S_{\text{full}}$) diverge at their initial 1-run $L_1(0)<L'_1(0)$, the identity $58{,}025-27\times 128=54{,}569$ forces $\text{Acc}_{1+d}<\text{Acc}'_M$ strictly and traps their difference inside:

$$54{,}569\cdot 2^{S_0}\cdot 3^{S_{\text{full}}-S_0-10}\;<\;2059\cdot\big(\text{Acc}'_M-\text{Acc}_{1+d}\big)\;<\;2^{S_0}\cdot 3^{S_{\text{full}}-S_0}.$$

3. **Global Injectivity of `(waveInteriorExp, waveInteriorAcc)` on Macro-Wave Blocks & the `31`-Step Universal Anagram Floor (Section 96):** Combining Section 90 ($L_2(1)\ne L'_2(1)$) and Section 95 ($L_1(0)\ne L'_1(0)$) with shift-composition (`waveInteriorExp_shift_comp`, `waveInteriorAcc_shift_comp`) by induction on N proves that **any two macro-wave blocks with $S_N=S'_M$ and $\text{Acc}_N=\text{Acc}'_M$ are 100% identical** ($N=M$, $\forall t\le N$, $L_1(t)=L'_1(t)$, and $\forall t<N$, $L_2(t+1)=L'_2(t+1)$), and any two non-identical anagrams satisfy $16{,}243{,}286{,}400\le|\text{Acc}'_M-\text{Acc}_N|$, $261{,}288{,}489{,}825<3^{S_N-7}$, and $31\le S_N$. 4. **Mandatory `+88` Growth Every $2B+1$ Waves & Unconditional Exclusion of Bounded-State Orbits Under Bounded 2-Run Lengths (Section 97):** Whenever $L_2(n)\le B$ for all $n\in\mathbb{N}$, $W(n)<W(n+2B+1)$ holds strictly for all n, $W(n_0)+88K\le W\big(n_0+K(2B+1)\big)$ under mod-88 invariance, and **no global upper bound $\forall n,\,W(n)\le M_{\max}$ can exist**. **- General Sublevel Step-Sum Run-Expansion (`W(M+1) ≤ 2^B * W(0)`), the `63 : 40` (`1,890 : 1,200`) All-Prefix Run-Dominance**

Law for Non-Monotone Sawtooth Visits, Unconditional Exclusion of Bounded-Run Recurrent Sublevel Trajectories & Immediate `59,049^(dL)` Post-Spike Exponential Rebound (`dense_core_general_sublevel_step_sum_expansion_and_finite_horizon_cutoff`, `dense_core_non_monotone_sawtooth_all_prefix_run_dominance`, `dense_core_non_monotone_sawtooth_exponential_explosion_and_exclusion`, `dense_core_post_spike_next_wave_exponential_rebound`): 1. **General Sublevel Step-Sum Run-Expansion & Explicit Finite-Horizon Cutoff (Section 98):** Whenever $W(M+1) \le 2^B \cdot W(0)$, canceling $2W(0)$ from Section 80's full-exponent wave bound forces $3^{S_M+10L_2(0)-1} < 2^{S_M+10L_2(M+1)+B-1}$ and, for every $2^{p'+q'} \le 3^{q'}$ ($q' \ge 1$):

$$p' \cdot S_M + 10(p'+q')L_2(0) \;<\; 10q' \cdot L_2(M+1) + q' \cdot B + p' \qquad \big(7S_M + 190L_2(0) < 120L_2(M+1) + 12B + 7\big),$$

yielding $119M + 190L_2(0) + 42 < 120L_2(M+1) + 12B$ and proving that whenever $L_2(M+1) \le L_{\max}$ and $2L_{\max} + B \le M$, $2^B \cdot W(0) < W(M+1)$ **holds strictly**. 2. **The `63 : 40` (`1,890 : 1,200`) All-Prefix Run-Dominance Law for Non-Monotone Sawtooth Visits (Section 99):** Whenever $W(N+2+d) \le 2^B \cdot W(0)$ along an orbit starting from $W(0) < 2^{W_0}$, eliminating S_N between Section 92 at $N+1$, shift-composition $S_{N+1+d} = S_N + 10L_2(N+1) + S_{\text{tail}}$, and Section 98 at $N+2+d$ proves—with **zero monotonicity assumption on** $W(N+1)$—that $L_2(N+2+d)$ dominates **every earlier 2-run** $L_2(N+1)$ **in the entire prefix of the orbit** by a factor of $63/40 = 1.575\times$:

$$1{,}890 \cdot L_2(N+1) + 70 \cdot S_{\text{tail}} + 10 \cdot L_2(0) \;<\; 1{,}200 \cdot L_2(N+2+d) + 120 \cdot B + 119 \cdot W_0 + 70.$$

3. **Cumulative `63 : 40` Sawtooth Run Growth & Unconditional Exclusion of Bounded-Run Recurrent Sublevel Visits (Section 100):** Iterating Section 99's $63:40$ inequality forces strict growth $L_{\text{seq}}(0) + K \le L_{\text{seq}}(K)$ along any sublevel visit chain above the initial threshold, and proves unconditionally that **no infinite strictly increasing sequence of horizons** $m_0 < m_1 < m_2 < \dots$ **can satisfy both** $W(m_k+1) \le 2^B \cdot W(0)$ **and** $L_2(m_k+1) \le L_{\max}$. 4. **Immediate Post-Spike Next-Wave Exponential Rebound (Section 101):** Across wave $N+1$ immediately following a 2-run spike $L_2(N+1)$, $2 \cdot 3^{7L_1(N+1)+10L_2(N+1)-1}W(N+1) < 2^{7L_1(N+1)+10L_2(N+2)}W(N+2)$; whenever $L_2(N+2) \le L_2(N+1)$, $59{,}049^{L_2(N+1)-L_2(N+2)} \cdot W(N+1) < W(N+2)$ (an immediate exponential rebound in a single wave), and whenever $W(N+2) \le W(N+1)$, $49L_1(N+1) + 190L_2(N+1) < 120L_2(N+2) + 7$. - **Boundary-Free Scaled-State Coordinates (`X(n) = 3^(10*L_2(n)) * W(n)`, `Y(n) = 2^(10*L_2(n)) * W(n)`), Universal Strict Interleaving (`Y(0) < X(0) < Y(N+1) < X(N+1)`), Boundary-Free Interior Subword & Single-`1`-Run Collision Cutoffs & Boundary-Free Two-Way Anagram Super-Divisibility (`dense_core_scaled_wave_strict_interleaving_and_universal_crt_locks`, `dense_core_boundary_free_interior_repeat_two_way_super_divisibility_and_cutoff`, `dense_core_single_one_run_collision_boundary_free_sieve_and_cutoff`, `dense_core_boundary_free_anagram_scaled_state_super_divisibility`):** 1. **Universal Strict Interleaving & Run-Length-Independent CRT Locks (Section 102):** Defining `scaledWaveX W L_2 n = 3^(10*L_2(n)) * W(n)` and `scaledWaveY W L_2 n = 2^(10*L_2(n)) * W(n)` absorbs the boundary powers of Section 54's master identity (`2^(S_N) * Y(N+1) + 19 * Acc_N = 3^(S_N) * X(0)`). Across **every** multi-wave horizon $N \in \mathbb{N}$—with **zero** restriction on $L_2(0)$ vs. $L_2(N+1)$ and even when $W(N+1) < W(0)$—we have $Y(0) < X(0) < Y(N+1) < X(N+1)$ (eliminating all Sawtooth dips in (X, Y) coordinates!) and universal divisibilities $7{,}427{,}200 \mid (X' - X)$ and $126{,}900{,}675 \mid (Y' - Y)$ with no same-run-length hypothesis. 2. **Boundary-Free Interior Subword Repetition Two-Way Super-Divisibility & Terminal $3^{S_{\text{sub}}}$-Cancellation Cutoff (Section 103):** Any two $(N+1)$-wave blocks matching only on their interior runs $(L_1(0 \dots N), L_2(1 \dots N))$—with **no** condition on $L_2(0)$ or $L_2(N+1)$—satisfy $2^{S_N}(Y'(N+1) - Y(N+1)) = 3^{S_N}(X'(0) - X(0))$, $(58025 \cdot 2^{S_N}) \mid (X'(0) - X(0))$, and $(58025 \cdot 3^{S_N}) \mid (Y'(N+1) - Y(N+1))$, forcing:

$$(p+q) \cdot S_{\text{sub}} + 15q \;<\; p \cdot S_M + 10(p+q)L_2(0) + qW_0 \qquad \big(27S_{\text{sub}} + 255 < 10S_M + 270L_2(0) + 17W_0\big).$$

3. **Single-`1`-Run Collision (`L_1(r_1) = L_1(r_2)`) Boundary-Free Sieve & Cutoff (Section 104):** Specializing Section 103 to $N = 0$ (whose interior is solely the single 1-run $L_1(r)$!) proves that whenever **any** two waves $r_1 < r_2$ have $L_1(r_1) = L_1(r_2)$ (regardless of all surrounding 2-runs!), $(58025 \cdot 2^{7L_1(r_1)}) \mid (X(r_2) - X(r_1))$, $(58025 \cdot 3^{7L_1(r_1)}) \mid (Y(r_2+1) - Y(r_1+1))$, and:

$$7(p+q) \cdot L_1(r_1) + 15q \;<\; p \cdot S_{r_2} + 10(p+q)L_2(0) + qW_0 \qquad \big(189L_1(r_1) + 255 < 10S_{r_2} + 270L_2(0) + 17W_0\big).$$

4. **Boundary-Free Two-Way Anagram Scaled-State Super-Divisibility (Section 105):** Removing both boundary-run hypotheses `L_2(0) = L'_2(0)` and `L_2(end) = L'_2(end)` from Section 91, any two macro-wave anagrams ($S_{J+D} = S'_{J'+D}$) with $X(0) \le X'(0)$ and $Y(\text{end}) \le Y'(\text{end})$ satisfy $(58025 \cdot 2^{S_K}) \mid (X'(0) - X(0))$ (when sharing K prefix waves) and $(58025 \cdot 3^{S_{\text{end}} - S_J}) \mid (Y'(\text{end}) - Y(\text{end}))$ (when sharing D suffix waves). - **The Simultaneously `(58,025 * 2^S_K * 3^E_suf)`-Reduced Anagram Syzygy, Mutual Defect-Absorption Cross-Bounds, Bounded-Middle Anagram Palette Pigeonhole Cutoff & Dual Boundary-Free Scaled `U`-Wave Theory (`dense_core_two_sided_reduced_anagram_syzygy_and_mutual_defect_absorption`, `dense_core_bounded_middle_palette_pigeonhole_exact_repeat_and_cutoff`, `dense_core_scaled_dual_wave_strict_interleaving_and_universal_crt_locks`, `dense_core_boundary_free_dual_repeat_and_single_two_run_collision_sieve`):** 1. **Simultaneously `(58,025 * 2^S_K * 3^E_suf)`-Reduced Anagram Syzygy & Mutual Defect Absorption (Section 106):**

Dividing all three terms of the (X, Y) master identity difference by $58{,}025 \cdot 2^{S_K} \cdot 3^{S_{\text{end}}-S_J}$ yields positive quotients $q_X, q_Y \geq 1$ and q_A satisfying $2^{S_{\text{end}}-S_K} q_Y \pm 41{,}553 q_A = 3^{S_J} q_X$, and proves—**regardless of the sign of** $\text{Acc}'_{\text{end}} - \text{Acc}_{\text{end}}$—the unconditional cross-bounds:

$$58{,}025 \cdot 2^{S_{\text{end}}} \;<\; 3^{S_J} \cdot X'(0) \qquad \text{and} \qquad 58{,}024 \cdot 3^{S_{\text{end}}} \;<\; 2^{S_{\text{end}}-S_K} \cdot Y'(\text{end}).$$

2. **Multi-Window Pigeonhole Reduction of Bounded-Middle Anagram Palettes to Exact Interior Repeats (Section 107):** Whenever $C+1$ windows at $r(0) < r(1) < \cdots < r(C)$ draw their $(N+1)$-wave interior configurations from a palette of size $\leq C$, Pigeonhole forces two windows $r(i) < r(j)$ to match on all interior runs ($q_A = 0$), forcing $(p+q)S_{\text{sub}} + 15q < pS_{r(j)+N} + 10(p+q)L_2(0) + qW_0$. 3. **Dual Boundary-Free Scaled `U`-Wave Coordinates ($\tilde{X}(n) = 3^{7L_1(n)}U(n)$, $\tilde{Y}(n) = 2^{7L_1(n)}U(n)$) & Universal CRT Locks (Section 108):** Across every horizon $N \in \mathbb{N}$, $\tilde{Y}(0) < \tilde{X}(0) < \tilde{Y}(N+1) < \tilde{X}(N+1)$ strictly, with universal divisibilities $2{,}108{,}416 \mid (\tilde{X}' - \tilde{X})$ and $121{,}581{,}891 \mid (\tilde{Y}' - \tilde{Y})$ (independent of L_1). 4. **Dual Boundary-Free Interior Repeat & Single-`2`-Run Collision Sieve (Section 109):** Any two $(N+1)$-wave U-blocks matching on their interior runs $(L_2(0 \ldots N), L_1(1 \ldots N))$ (with no condition on $L_1(0)$ or $L_1(N+1)$) satisfy $(2059 \cdot 2^{\tilde{S}_N}) \mid (\tilde{X}'(0) - \tilde{X}(0))$ and $(2059 \cdot 3^{\tilde{S}_N}) \mid (\tilde{Y}'(N+1) - \tilde{Y}(N+1))$; at $N = 0$, any single repeated 2-run $L_2(r_1) = L_2(r_2)$ (independent of all 1-runs!) forces $(2059 \cdot 2^{10L_2(r_1)}) \mid (\tilde{X}' - \tilde{X})$ and $(2059 \cdot 3^{10L_2(r_1)}) \mid (\tilde{Y}' - \tilde{Y})$.

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8. Architecture of the 142 Lean 4 Files & Verification

Every theorem in this repository is machine-checked in Lean 4.19.0 (`mahler_demand_a/`) without a single `sorry` . Each core module `Foo.lean` is paired with a verification module `FooChecks.lean` that runs `#check` and `#print axioms` on every exported theorem.

Key Module Groups

- **Foundations & Odd-Block Reduction (Modules 1–36):** `MahlerDemandA.lean` through `TwoSidedSqueeze.lean` — basic Mahler step/orbit properties, companion intervals, odd-to-odd block reduction, `coprimeRemainder` , `integerSlack` , and initial two-sided squeezes.
- **1D Dynamical Collapse & Converse Reconstruction (Modules 37–52):** `SharpenedSqueezes.lean` , `AperiodicSlackCocycle.lean` , `MinSlackDynamicalSystem.lean` , `GlobalStepRateBounds.lean` , `BaseTwoDivModLock.lean` , `CeilingSaturationSynthesis.lean` , `TelescopingSeriesDuality.lean` , `ConverseCompanionReconstruction.lean` .
- **2-Adic LTE, Cauchy Convergence & Time-Independent Bounds (Modules 53–80):** `ExplicitHorizonCutoffs.lean` , `CeilingDeficitAmplification.lean` , `UniversalTwoVisitSeparation.lean` , `MonotonicStepSumSeparation.lean` , `CeilingSaturatedModTraps.lean` , `TwoAdicLiftingExponent.lean` , `FullStepSumValuation.lean` , `CoprimeRemainderAdditionLaw.lean` , `TwoAdicCauchyConvergence.lean` , `AllHistoryQuotientRecovery.lean` , `ParametricRemainderCorridor.lean` , `AperiodicBoundedBlockSqueeze.lean` , `TimeIndependentDeficiencySqueeze.lean` , `SlackDeadEndsAndWitnesses.lean` .
- **Record-Holder, Sieves, Aperiodic Exclusions & Finite-Seed Isolation (Modules 81–122):**
- `MahlerGrandSynthesis.lean` : Equivalence `validBlockChain_iff_coprime_recurrence` and 113-step survival / 114-step death of `recordHolderA0 = 29871555608238677` .
- `Mod256CompleteSieve.lean` , `CoprimeUnitRigidity.lean` , `DeterminantFactorization.lean` : Mod-256 two-block sieve, unit-sum rigidity, and rank-1 determinant collapse (Dead End 12).
- `BakerMatveevFramework.lean` : Consecutive coprimality, 3-logarithm `Real.log` squeezes, and Weil height growth (Dead End 13).
- `RidoutSubspaceFramework.lean` : Intrinsic $\gcd(\Delta_m, 6) = 1$, $2^{T(m)}$-cancellation law $2^{s(m)+1}u(m)\Delta_m < Q_m$, and Q_m^{-2} Ridout bound (Dead End 14).
- `MultiOriginSubspaceSyzygy.lean` : Additive separation of variables, shift-free `minSlack` equivalence, and 3-origin Subspace syzygy (Dead End 15).
- `RepeatedSubwordComplexity.lean` : Strict quotient aperiodicity $u(r_1) < u(r_2)$, parity-enhanced common multiplier, and linear subword-complexity cutoffs.
- `SimultaneousBaseDigitRigidity.lean` : Carryless Base-3 prefix equivalence and dual Base-2 / Base-3 remainder complementarity (Dead End 16).
- `MultiStepQuotientGCD.lean` : Exact three-way multi-step GCD equality $\gcd(u(r), u(r+d)) = \gcd(u(r), R_d^{(r)})$ and reduced-denominator exponential growth (Dead End 17).
- `MahlerSeriesHankelFramework.lean` : Power-series functional equation and two-point Padé–Hankel determinant factorization (Dead End 18).
- `ThreeAdicAndSeedGCDLocks.lean` , `CoupledFractionalPartDynamics.lean` , `Mod2304CombinedSieve.lean` , `TwoSidedCRTAndBackwardObstructions.lean` , `ThreeAdicValuationConservation.lean` , `Mod216TwoSidedQuotientFloor.lean` : Seed/slack GCD locks, 3-adic orbit factorization $b_{k+1} = 3^{s(k)-1}u(k)$, μ_2 interval map, disjoint Lüroth–Markov partition, Mod-2304 / Mod-1728 / Mod-216 sieves, $u(k) \neq 1$ ($k \geq 1$), and 3-adic valuation conservation.
- `InfiniteZNumberBridgeAndRationalBase.lean` : Ceiling-vs-floor odd-step proof, direct `Int.fract` finite/infinite companion equivalences, upstream `Int.fract` 113-step theorem, and rational-base $T_{3/2}$ deterministic subword exclusions.

- `SturmianAndBalancedComplexityExclusion.lean`: $5:3$ and $17:10$ weighted repetition cutoffs, Finite Pigeonhole Principle on pattern classifiers, and exclusion of aperiodic ceiling-saturated $\{3,4\}$ / $[3,5]$ and weighted-balanced ($\delta \geq 1$, slope $\leq (17-\delta)/10 \leq 1.6$) block chains.
- `FiniteSeedCarryAndBackwardDescent.lean`: $G_{3,6}$ false-positive cycle $(5,5) \to (4,2) \to (6,3) \to (5,5)$ vs. $8^m \mid (19a_0+9)$ positive-seed cutoff, arithmetic block-parameter uniqueness, and backward-lifting descent dichotomy (`a < b`, `r_a = 1/3 + (2/3)^c r_b < 13/27` for $c \geq 3$).
- `SubstitutionMahlerMethodAndGrandSynthesis.lean`: Dual-place partial-sum bounds (`0 < \Sigma_m < 1/3` vs. `-(6a_0+2)/3 \le -2/3 < 0`), closed-form uniform macro-block summation (`2^{Dm} \mid ((3^D-2^D)(3a_0+1) + 2^{s(0)} R_0)`), 3-point macro-block remainder syzygy, and `mahler_grand_122_module_synthesis`.
- `EndpointCompactnessAndBalancedExclusion.lean`: Stage 0 perpetual finite survival supremum `supCompanion a_0 = ⊔ n, lowerBound a_0 n`, strict-vs-dyadic-endpoint shift alternative (`perpetual_finite_survival_endpoint_alternative`), and positive-real forward shift (`exists_positive_seed_ge_one_of_pos_companion`); standard prefix factor complexity equivalence `prefixFactorCount s L C ≤ C ↔ ∃ r1 < r2 ≤ C, ...` (`prefixFactorCount_le_iff_exists_repeated_subword`) and global factor complexity `globalFactorComplexity s L`; parameterized (p,q) rational power bounds $3^q < 2^{p+q}$ ($qS+1 \leq pT+qW+p$); the **real logarithmic threshold selection bridge** (`exists_pq_of_lt_log_threshold` for every $\kappa \log(3/2) < \log 2$); the **asymptotic-mean ($T(n)/n \to \lambda > 0$), normalized maximum-discrepancy convergence ($E_H/H \to 0$), uniform-window-mean, and liminf-complexity exclusions** (`maxPrefixDiscrepancy_div_tendsto_zero`, `no_infinite_asymptotic_mean_liminf_subcritical_block_chain`, `no_infinite_tendsto_mean_log_threshold_block_chain`, `no_infinite_global_complexity_log_threshold_block_chain`, `tendsto_mean_of_uniformWindowMean`, `no_infinite_uniformWindowMean_global_complexity_block_chain`); the **scale-local balance exclusion without a global limiting mean (`no_infinite_scale_local_balanced_block_chain`) and unconditional exclusion of the concrete non-stationary oscillating-epoch family `nonStationaryEpochSeq a b` (`nonStationaryEpochSeq_no_limiting_mean`, `no_infinite_nonStationaryEpochSeq_block_chain`)**; and the **derived phase-threshold factor complexity bound $\leq L+1$** (`mechTwoLetterSeq_finset_factor_card_le`, `mechTwoLetterSeq_prefixFactorCount_le`) yielding the **assumption-free exclusion of arbitrary two-letter $\{a,b\}$ ($3 \leq a \leq b$) mechanical/Sturmian block chains** (`unconditional_mechanical_two_letter_finite_cutoff`, `no_infinite_mechanical_two_letter_block_chain`).
- `ExactSeedResidueTreeAndStabilization.lean`: Stage 2 bidirectional parity-prefix bijection `(∀ n < N, orbit a n % 2 = orbit b n % 2) ↔ a % 2^N = b % 2^N`, constructive `seedResidue ε N < 2^N` and bit extension $r_{N+1} = r_N + \eta_N 2^N$, exact characterization of ordinary nonnegative/positive integer seeds by residue stabilization, canonical `FeasibleSeedNode` (`FeasibleSeedNode.eq_canonical`), full companion-set membership (`validCompanionUpTo_add_offset_iff_mem_ratInterval`), **denominator-cleared signed integer margin $D_{N+1} = \min(2^{N+1}, 3D_N - 2^{N+1}(\varepsilon_N \bmod 2))$ (`signedMargin_cast_rat`, `validCompanionUpTo_iff_signedMargin_pos`), coupled quotient-residue-margin transition (`coupled_seedQuotient_and_margin_transition`), the 5-way general Mahler conjecture equivalence (`general_mahler_conjecture_iff_positive_stabilized_branch_escape`, `mahler_conjecture_iff_leastPositiveSurvivor_unbounded`), rank-certificate escape interface (`no_perpetual_feasible_stabilized_branch_of_rank_certificate`), scalar-margin `11`/`10101` contraction vs. `100` ceiling recovery (`signedMargin_10101_neg`, `signedMargin_100_ceiling_recovery`), Lean-certified step-28 survival / step-29 failure of seed `149` and global step-29 lower bound $m \geq 4096$ ($B(29) \geq 4096$) (`seed_149_survives_28_and_fails_29`, `positive_survivor_step29_ge_4096`, `leastPositiveSurvivor_certified_bounds`)**, single-block spike cutoff (`no_infinite_superLacunary_block_chain`), and the **well-founded odd-quotient descent ranking $u(m+1)+2 \leq u(m)$ and $u(m)=1$ mod-8 trap** (`no_infinite_rapid_growth_block_chain`) excluding all rapid-growth chains with $3 \leq s(m+1)$ and $3^{s(m)} < 2^{s(m+1)}$.
- `ThueMorseMahlerFunctionalEquation.lean`: Stage 3 formalization of the Thue–Morse sequence (`tmBit (2k) + tmBit (2k+1) = 1`), its 3-step encoded Mahler stream (`0 ≤ fractionalTail ≤ 9/19 < 1/2`, 1 odd step per 6-step aligned window), closed finite and infinite Mahler functional equations $F(z) = (1-z)F(z^2) + \frac{z}{1-z^2}$ (`tmSeries_mahler_functional_equation`), 2-adic seed Mahler bridge `64^M ∣ (729^M (665 a_0 + 72) + 113715 * tmIntCarry 64 729 M)`, **unconditional non-eventual-periodicity** (`tmBit_not_eventually_periodic`, `tmEncodedParity_not_eventually_periodic`, `pdBit_not_eventually_periodic`, `pdEncodedParity_not_eventually_periodic`), **reusable dyadic-inflation repetition cutoff (`dyadic_inflated_repetition_cutoff`), finite seed cutoffs ($< 30 \cdot 2^k$ and $< 60W$ for `tmEncodedParity`; $< 21 \cdot 2^k$ and $< 42W$ for Period-Doubling `pdEncodedParity_cutoff_and_exclusion`), quantitative seed-residue lower bounds** $\mathrm{seedResidue}(\varepsilon, 30 \cdot 2^k) \geq 2^{2^k}$ **and bounded waiting window $[N_0, 60N_0)$ for a forced nonzero extension bit (`seedResidue_tmEncodedParity_ge_two_pow`, `tmEncodedParity_forced_seedBit_in_sixty_mul_window`, `seedResidue_pdEncodedParity_ge_two_pow`), and unconditional exclusion of two-letter Thue–Morse block-length chains** $s(n) = a + (b-a)t_n$ **($3 \leq a \leq b$) (`tmBlockSeq_finite_seed_cutoff`, `no_infinite_tmBlockSeq_block_chain`)**.

- `SparseDefectCocycleAndValuationLock.lean`: Stage 4 signed single-block defect identity $2^{s(k)}I(b_{k+1}) = 3^{s(k)}I(b_k) + (9\cdot 2^{s(k)} - 8\cdot 3^{s(k)-1})$, constant 2-adic valuation $v_2(E(c)) = 3$ for all $c \geq 4$, iterated multi-block defect cocycle and its **proved explicit `Finset.sum` expansion (`defectCocycleSum_eq_finset_sum`)**, the **short local prefix derivation (`local_prefix_first_defect_valuation_lock`)** showing that the finite prefix up to the first defect $s(j_0) \geq 4$ after j_0 blocks of 3 already forces $2^{3j_0+3} \mid (19a_0 + 9)$ and $2^{3j_0+4} \nmid (19a_0 + 9)$, multi-block valuation non-cancellation (`first_defect_valuation_lock_and_non_cancellation`), and the growing boundary invariant identity (`defectCocycle_residual_eq_growing_boundary_affInv`).
- `DubickasConfinementAndComplexityExtension.lean`: Quantitative extensions of Dubickas (2008 / 2009) on shifted carry words $s_n = qx_{n+1} - px_n$ (`dubBaseCarry_eq_fract_linear`, `dubBaseY_multi_step_carry_identity`, `dubBaseX_repeated_window_exp_and_power_cutoff`), universal two-letter alphabet $s_n \in \{\lfloor (p-q)s \rfloor, \lfloor (p-q)s \rfloor + 1\}$ (`dubBaseCarry_mem_two_letters`), general-base no-bispecial-pair property (`dubBaseCarry_no_bispecial_pair`), **finite bispecial-free binary complexity bound** $\mathrm{prefixFactorCount}(w, L, C) \leq L + 1$ **(`prefixFactorCount_le_add_one_of_no_bispecial`, `dubBaseCarryNat_prefixFactorCount_le`) establishing the full general-base Dubickas (2009) Theorem 1 for all coprime** $1 < q < p < q^2$ **across all** $s \in [0, 1 - 1/p]$ **(`dubickas2009_general_base_explicit_and_infinite_escape`), its strictly linear-in-W bound** $N \leq 2bW + 2a + 2$ **(`dubickas2009_general_base_linear_in_W_escape`), and the sharp** $5W + 14$ **escape bound at base** $3/2$ **(`shifted_third_unconditional_explicit_escape`), general-base floor-residue non-stabilization** $\lfloor \xi(p/q)^n \rfloor \bmod m$ **for all coprime** $1 < q < p < q^2$ **and** $m \geq p$ **(`general_base_floor_mod_non_stabilization`, `floor_mod_non_stabilization_of_three_le`), universal infinite-family single-cell theorem covering the open regime** $p > q^2$ **(**$p \geq 2q$**, `general_base_non_straddling_cell_escape`), upgrade of Dubickas's Sturmian carry-word endgame (**$\kappa = 1$**) to the general-base critical threshold** $\kappa < \log q / \log(p/q)$ **(`exists_ab_of_lt_general_log_threshold`, `no_subcritical_general_log_threshold_dubBaseCarry`, `no_subcritical_log_threshold_dubCarry`)**, and **explicit** $O(W) = O(\log \xi)$ **finite-horizon escape bounds and infinite-horizon eventual escape** for the nearest-integer arc $\|\xi(3/2)^n\| < 1/5$ ($n \leq W + 5$), the beyond-$1/3$ single window $[1/6, 13/24)$ of length $3/8 > 1/3$ at base $3/2$ ($n \leq W + 15$), **the beyond-$1/4$ single window** $[1/3, 5/8)$ **of length** $7/24 > 1/4 = 1/p$ **at base** $4/3$ **(**$n \leq W + 8$**, `base_four_three_beyond_quarter_explicit_and_infinite_escape`), Dubickas's 2008 closed union** $[8/39, 18/39] \cup [21/39, 31/39]$ **of total length** $20/39 > 1/2$ ($n \leq W + 8$ on the open union, $n \leq 3W + 19$ on the closed union, and direct $2^m \mid (13x_r + v_r)$ divisibility on the rational 4-cycle), and **multi-interval unions of length** $7/12 \approx 0.5833$ **(**$n \leq W + 12$**),** $2/3 \approx 0.6667$ **(**$n \leq W + 14$**, `union_two_thirds_explicit_and_infinite_escape`), and** $25/36 \approx 0.6944 > 2/3 > 1/2$ **(**$n \leq W + 16$**, `union_twenty_five_thirty_sixths_explicit_and_infinite_escape`) via verified backward–forward integer funnel pruning (`inRatIntervalList_terminal_of_funnelChainOk`, `inRatIntervalList_next_of_forwardStepOk`)**, plus the **Thue–Morse period-doubling Cantor cascade on** $[c, 1-c]$ **(`tmCascadeCarryAcc_mahlerProd_identity`, `tmCascadeEntry_closed_form`, `tmCascadeEntry_step_ratio_and_strictAnti`, `tmCascadeEntry_explicit_values_and_two_sevenths_cutoff`, `tmCascade_periodic_orbit_two_pow_dvd`, `cantorTwoSevenths_matches_mahlerBinaryPartition_up_to_15_and_splits_at_16`)** and the **general gap-g window ladder (**$1/3 < \alpha_g < \beta_g < 1/2$**) and g-rung first-defect valuation lock** $v_2((3^g - 2^g)a_0 + 3^{g-1}) = g(j_0 + 1)$ **(`rungAlpha_lt_rungBeta_lt_half`, `gap_ladder_short_gap_cost_and_cap`, `general_rung_first_defect_valuation_lock`)**, together with the **punctured** $[0, 1/2)$ **homoclinic catapult trap on** $[8/57, 1/2)$ **(**$n \leq 3W + 20$**, `punctured_eight_fifty_sevenths_explicit_and_infinite_escape`), the pure period-3 lock on** $[4/27, 1/2)$ **(**$n \leq W + 8$**, `punctured_four_twenty_sevenths_period_three_lock_and_escape`), the** $\varepsilon^* = 8/57$ **homoclinic phase transition (`homoclinicCycleMin_closed_form_and_strictMono`, `cantorPuncturedHalf_phase_transition_counts`), and the universal rung-g catapult staircase** $\varepsilon_g^* = \frac{2^g}{3(3^g - 2^g)} \to 0$**, leak-free inductive elevator** $s_{m+1} \leq s_m$**, self-cleaning death-band decomposition** $D_{g,m}(\varepsilon)$ **(**$18/19 \approx 94.74\%$ **of** L_3 **cleaned at** $8/57$**), 7-step catapult-into-death-band trap below** $8/57$**, higher-order gap-to-trigger bijection** $\mathrm{Gap}_{g,m}(\varepsilon) \xrightarrow{T_g^m} [\varepsilon, b_g)$**,** $2^{7m} \mid (2059x_r + 280)$ **leak-threading survivor lock, universal homoclinic cycle lock family** $2^{(3k+4)m} \mid (D_k x_r + N_k)$**, general rung-$g$ defect cocycle identity, mutually incompatible multi-defect valuation locks, the two-island punctured-**$[0, 1/2)$ **set** $\mathcal{W}_{280/2187}$ **(**73.43% **of** $[0, 1/2)$**,** $n \leq 3W + 27$**), the self-refining funnel engine and** $17/24 > 70\%$ **7-interval union (**$n \leq W + 17$**), the arbitrary K-defect simultaneous valuation lock tower, and the k-block minimum defect spacing law and** $(3, 3k + 4)$ **Perron–Fibonacci hierarchy inside the Cantor dust below** $8/57$ **(`rungCatapult_ordering_and_corridor_overlap`, `universal_rung_catapult_and_upward_expansion`, `leak_free_inductive_elevator_monotone_rung_descent`, `rungDeathBand_step_and_measure_decomposition`, `catapult_into_death_band_seven_step_escape`, `rungDeathGap_maps_onto_catapult_zone`, `homoclinic_seven_cycle_in_gap_and_divisibility_lock`, `homoclinic_family_cycle_divisibility_lock`, `rungAffInv_general_defect_step_identity`, `exactTwoPowLock_incompatible_of_ne`, `general_rung_two_defect_simultaneous_lock`, `two_island_punctured_half_explicit_and_infinite_escape`, `inRatIntervalList_terminal_of_selfFunnelChainOk`,**

`union_seventeen_twenty_fourths_explicit_and_infinite_escape`, `rungDefectCocycle_tower_identity`, `general_rung_multi_defect_valuation_tower`, `cantor_dust_minimum_defect_spacing_law`, `punctured_eps1_forced_seven_step_macro_block`, `cantorDust_perron_fibonacci_counts`, `perronCharPoly_sign_change_and_root_squeeze`, `universal_catapult_death_band_ladder`, `punctured_eight_eighty_firsts_forbids_four_zeros_and_short_gaps`, `cantor_dust_defect_to_defect_quotient_recurrence`, `cantor_dust_mod_eight_lock_and_sparse_descent`, `no_infinite_sparse_defect_cantor_dust_chain`, `three_pow_ne_two_pow_add_one_of_four_le`, `cantor_dust_three_adic_valuation_and_normalized_recurrence`, `cantor_dust_multi_step_telescoping_and_epoch_descent`, `no_infinite_cumulative_sparse_epoch_cantor_dust_chain`, `defectNormalizedQuotient_exact_two_adic_valuation_and_forward_uniqueness`, `defectNormalizedQuotient_mod_sixteen_spacing_bijection`, `defectNormalizedQuotient_orbit_unique_of_V0`, `defectNormalizedQuotient_binary_crt_parametrization`, `defectNormalizedQuotient_cooldown_mod_128_and_three_adic_lock`, `defectNormalizedQuotient_renormalized_k_recurrence`, `defectNormalizedQuotient_level2_crt_parametrization`, `defectNormalizedQuotient_level3_renormalized_l_recurrence_j1_eq_one`, `defectNormalizedQuotient_level3_renormalized_l_recurrence_j1_eq_two`, `defectNormalizedQuotient_multi_step_homogeneous_tower`, `renormalized_three_adic_carry_digit_identity`, `backward_two_adic_residue_lift_of_stepOk`, `binary_cantor_dust_min_seed_of_certifyBinaryDustDepth`, `binary_cantor_dust_certified_horizon_bounds_up_to_depth_ten`, `one_step_quotient_and_remainder_conservation`, `universal_two_three_adic_quotient_duality`, `binary_cantor_dust_certified_depth_11_and_12`, `defectNormalizedQuotient_ternary_cooldown_mod_128_sieve`, `ternary_cantor_dust_min_seed_of_certifyTernaryDustDepth`, `ternary_cantor_dust_certified_horizon_bounds_up_to_depth_six`, `defectStepSum_shift_cancellation_and_single_exponent_identity`, `defectNormalizedQuotient_archimedean_floor_invariant`, `defectNormalizedQuotient_cyclotomic_mod_7_13_19_locks`, `mahler_min_seed_of_mahlerLeafOk`, `mahler_binary_cantor_dust_min_seed_of_certifyMahlerBinaryDustDepth`, `mahler_binary_cantor_dust_certified_bounds_up_to_depth_12`, `dense_core_strict_doubling_and_monotonicity`, `dense_core_repeated_subword_divisibility_and_exp_bound`, `dense_core_repeated_subword_cutoff_and_complexity_exclusion`, `dense_core_all_V_odd_and_mod_four_divergence`, `dense_core_right_special_exact_two_adic_valuation_and_spectral_gap`, `dense_core_symmetric_bispecial_resonance`, `dense_core_parameterized_pq_cutoff_and_convergents`, `no_subcritical_log_threshold_dense_core`, `no_density_weighted_subcritical_log_threshold_dense_core`, `bottom_part_remainder_invariance_and_floor_conservation`, `dense_core_shifted_horizon_floor_and_ninety_eight_bit_corridor`, `dense_core_repeated_window_top_part_isolation_and_squeeze`, `dense_core_three_occurrence_ultrametric_pigeonhole`, `normalizedDefectAcc_master_identity_and_infinite_sieve_tower`, `dense_core_depth2_four_in_16384_and_depth3_eight_in_2097152_sieves`, `dense_core_exact_prefix_collision_biconditional_and_mod_cutoff`, `normalizedDefectAcc_cocycle_composition_law`, `dense_core_frozen_remainder_and_exact_valuation_conspiracy_law`, `dense_core_return_loop_homoclinic_denominator_syzygy`, `dense_core_global_origin_return_loop_syzygy`, `dense_core_seed_free_two_return_loop_syzygy_and_congruence`, `dense_core_every_three_step_window_spawns_maximal_branching_return_loop`, `dense_core_seed_free_two_loop_bit_flip_and_spectral_gap`, `dense_core_constant_run_length_2059_and_58025_valuation_locks`, `dense_core_alternating_run_129009091_valuation_locks`, `dense_core_one_run_compression_and_macro_transition`, `dense_core_two_run_compression_and_macro_transition`, `dense_core_anagram_weight_decomposition_and_universal_pigeonhole`, `dense_core_anagram_collision_syzygy_and_drift_cancellation_cutoff`, `dense_core_one_to_two_run_interface_syzygy_and_ten_mul_valuation_lock`, `dense_core_two_to_one_run_interface_and_wave_elimination`, `dense_core_two_one_two_wave_elimination_and_strict_W_descent`, `dense_core_multi_wave_telescoping_and_epoch_W_descent`, `dense_core_same_middle_run_wave_cancellation_and_divisibility`, `dense_core_non_descent_run_growth_speed_limit_and_212_dichotomy`, `dense_core_macro_run_mod_16_128_1024_bijections_and_step_8_descent`, `dense_core_singleton_bridge_wave_identities_and_cross_wave_power_balance`,

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`dense_core_macro_wave_archimedean_floor_invariant_and_corridor`,
`dense_core_three_adic_mirror_sieves_and_mod_24_144_bijections`,
`dense_core_dual_wave_full_three_adic_remainders_and_one_run_floor_invariant`,
`dense_core_plus_19_cancellation_homogeneous_difference_and_run_expansion_laws`,
`dense_core_fermat_homoclinic_sieves_and_mod_4400_39600_bijections`,
`dense_core_four_run_W2_free_cross_wave_syzygy`,
`dense_core_wave_exact_two_adic_valuations_and_short_run_euclidean_laws`,
`dense_core_symmetric_multi_wave_master_identity_and_boundary_independence`,
`dense_core_multi_wave_interior_match_cancellation_and_275_super_divisibility`,
`dense_core_macro_wave_archimedean_repetition_cutoff`,
`dense_core_waveInteriorAcc_telescopic_cross_defect_and_mirror_collapses`,
`dense_core_same_run_length_crt_moduli_and_billion_step_descent`,
`dense_core_multi_wave_58025_super_divisibility_and_22_bit_repetition_cutoff`,
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`dense_core_all_horizon_same_run_strict_growth_and_mandatory_expansion`,
`dense_core_dual_multi_wave_homogeneous_difference_and_124499856384_growth`,
`dense_core_multi_horizon_shift_composition_and_three_point_wronskian_syzygy`,
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`dense_core_multi_wave_three_way_gcd_equality_and_corridor_bound`,
`dense_core_multi_horizon_wronskian_archimedean_and_divisibility_bounds`,
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`dense_core_full_exponent_archimedean_wave_bound_and_step_sum_expansion`,
`dense_core_short_run_two_wave_631_growth_and_isolated_dip_law`,
`dense_core_dual_one_run_full_exponent_expansion_and_seven_pow_growth`,
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`dense_core_short_run_even_wave_631_pow_and_joint_exponential_tower`,
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`no_subcritical_macroWaveCode_dense_core`,
`dense_core_anagram_prefix_start_and_suffix_end_state_super_divisibility`,
`dense_core_suffix_sharing_anagram_three_adic_cutoff`,
`dense_core_first_divergence_cauchy_pinch_and_54569_exponential_gap`,
`dense_core_unconditional_anagram_divisibility_and_joint_prefix_suffix_cutoff`,
`dense_core_universal_two_run_spike_upper_bound_and_dip_corridor`,
`dense_core_exact_state_collision_syzygy_and_81648_run_jump_lock`,
`dense_core_interior_state_collision_mod_729_and_81648_jump_lock`,
`dense_core_one_run_first_divergence_cauchy_pinch_and_54569_gap`,
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`dense_core_general_sublevel_step_sum_expansion_and_finite_horizon_cutoff`,
`dense_core_non_monotone_sawtooth_all_prefix_run_dominance`,
`dense_core_non_monotone_sawtooth_exponential_explosion_and_exclusion`,
`dense_core_post_spike_next_wave_exponential_rebound`,
`dense_core_scaled_wave_strict_interleaving_and_universal_crt_locks`,
`dense_core_boundary_free_interior_repeat_two_way_super_divisibility_and_cutoff`,
`dense_core_single_one_run_collision_boundary_free_sieve_and_cutoff`,
`dense_core_boundary_free_anagram_scaled_state_super_divisibility`,
`dense_core_two_sided_reduced_anagram_syzygy_and_mutual_defect_absorption`,
`dense_core_bounded_middle_palette_pigeonhole_exact_repeat_and_cutoff`,
`dense_core_scaled_dual_wave_strict_interleaving_and_universal_crt_locks`,
`dense_core_boundary_free_dual_repeat_and_single_two_run_collision_sieve`).

Reproducing the Verification Build

```
 cd mahler_demand_a
lake build
lake env lean -DwarningAsError=true EndpointCompactnessAndBalancedExclusionChecks.lean
lake env lean -DwarningAsError=true ExactSeedResidueTreeAndStabilizationChecks.lean
```

```
lake env lean -DwarningAsError=true ThueMorseMahlerFunctionalEquationChecks.lean
lake env lean -DwarningAsError=true SparseDefectCocycleAndValuationLockChecks.lean
lake env lean -DwarningAsError=true DubickasConfinementAndComplexityExtensionChecks.lean
```

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References

1. **K. Mahler (1968),** *"An unsolved problem on the powers of* $3/2$*,"* Journal of the Australian Mathematical Society **8**, 313–321.
2. **D. Ridout (1957),** *"Rational approximations to algebraic numbers,"* Mathematika **4**, 125–131.
3. **T. Vijayaraghavan (1940),** *"On the fractional parts of the powers of a number (I),"* Journal of the London Mathematical Society **15**, 159–160.
4. **A. D. Pollington (1979),** *"On the density of sequence* $\{\theta^k\}$*,"* Illinois Journal of Mathematics **23**, 511–515.
5. **G. Choquet (1980),** *"θ-fermés et dimension de Hausdorff. Conjectures de travail. Arithmétique des θ-cycles (où* $\theta = 3/2$*),"* Comptes Rendus de l'Académie des Sciences Paris Sér. A–B **290**, A863–A868.
6. **L. Flatto, J. C. Lagarias, and A. D. Pollington (1995),** *"On the range of fractional parts* $\{\xi(p/q)^n\}$*,"* Acta Arithmetica **70**, 197–227.
7. **H. P. Schlickewei (1977),** *"The* $\mathfrak{p}$*-adic Thue–Siegel–Roth–Schmidt theorem,"* Archiv der Mathematik **29**, 267–270.
8. **C. L. Stewart (1980),** *"On the representation of an integer in two different bases,"* Journal für die reine und angewandte Mathematik **319**, 63–72.
9. **A. Baker and G. Wüstholz (1993),** *"Logarithmic forms and group varieties,"* Journal für die reine und angewandte Mathematik **442**, 19–62.
10. **E. M. Matveev (2000),** *"An explicit lower bound for a homogeneous rational linear form in the logarithms of algebraic numbers. II,"* Izvestiya: Mathematics **64**, 1217–1269.
11. **P. Corvaja and U. Zannier (2002),** *"Some new applications of the Subspace Theorem,"* Compositio Mathematica **131**, 319–340.
12. **P. Corvaja and U. Zannier (2003),** *"On the greatest prime factor of* $(ab+1)(ac+1)$*,"* Proceedings of the American Mathematical Society **131**, 1705–1709.
13. **B. Adamczewski and Y. Bugeaud (2007),** *"On the complexity of algebraic numbers I. Expansions in integer bases,"* Annals of Mathematics **165**, 547–565.
14. **A. Dubickas (2006),** *"On the distance from a rational power to the nearest integer,"* Journal of Number Theory **117**, 222–239.
15. **A. Dubickas (2008),** *"On the powers of* $3/2$ *and other rational numbers,"* Mathematische Nachrichten **281**, 951–958.
16. **A. Dubickas (2009),** *"Powers of a rational number modulo* 1 *cannot lie in a small interval,"* Acta Arithmetica **137**, 233–239.
17. **The Formal Conjectures Authors (Google DeepMind, 2025–2026),** *"Formal Conjectures in Lean 4:* `Mahler32.lean`, `Problem10_9.lean`, *and* `ZNumber.lean`*,"* `https://github.com/google-deepmind/formal-conjectures`.
18. **S. Akiyama, C. Frougny, and J. Sakarovitch (2008),** *"Powers of rationals modulo 1 and rational base number systems,"* Israel Journal of Mathematics **168**, 53–91.
19. **M. Andrieu, S. Eliahou, and L. Vivion (2025),** *"A Normality Conjecture on Rational Base Number Systems,"* `arXiv:2510.11723`.